場の量子論 I　　福来 正孝

NO. 1

Apr 14

Dirac 理論

spin 0

boson anti particle

particle number を変える

対象	fundamental	Higgs	composition
spin 0		$\phi^0\ \phi^\pm$	$\pi^{0,\pm}$
1/2	$e^-\ \nu_e$	$u_1\ u_2\ u_3$ 自由度	p, n, Λ, Σ
		$d_1\ d_2\ d_3$	
1	A_μ (photon) gluon	$W_\mu^\pm\ Z_\mu$	$\rho^{\pm,0}, \omega, \cdots$
3/2			Δ
2	graviton		$A_2^{\pm,0}, f$

Abel

QED　γ, e

Weinberg-Salam　(e, ν) (d, u)　γ, W, Z　$\phi^0\ \phi^+$

QCD　gluon, quark　SU(3)

gauge 理論
vector field が charge に universal に couple する

場

流体　$x_i(t)$　Lagrange

$\phi(x,t)$　Euler

近接相互作用　$\phi\ \dot\phi\ \partial\phi$　$\partial_\mu\phi$

$$I = \int \mathcal{L}(\phi, \partial_\mu\phi)\, d^4x$$

Hamilton の原理 $\delta I = 0$

Euler eq → eq. of motion

対称性　　nether の定理

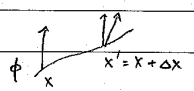
不変変分論　$x \to x'$　　大局的変換 → 保存則

$\phi(x) \to \phi'(x')$　　局所的 〃 → gauge 理論

$I = I'$

gauge 理論　$A_\mu \to A_\mu - \partial_\mu\lambda(x)$

$\psi \to e^{-ie\lambda(x)}\psi(x)$



$\psi'(x+\Delta x) = (1+ie\Delta x^\mu A_\mu)\psi(x)$

$= \psi(x+\Delta x) - \psi'(x+\Delta x)$

Eichinvariance

Weyl (1918)　接続係数 connection

(1929)　$\psi(x) \to \psi'(x) = e^{-ie\lambda(x)}\psi(x)$

$A_\mu(x) \to A_\mu(x) - \partial_\mu\lambda(x) = A_\mu'$

$(\partial_\mu - ieA_\mu')\psi'(x) = (\partial_\mu - ieA_\mu)\psi(x)$

coop Filing Leaf 6mm×41行

量子論

$\varphi(x) \to q_i$

$\pi(x) \leftarrow p_i$

$[\varphi(x,t), \pi(x',t)] = i\delta_{xx'}$ 第二量子化

φ $a_{p,\sigma}\ldots$ $a^+_{p,\sigma}\ldots$

1 particle $a^+_{p\sigma}|0\rangle = |p,\sigma\rangle$ Φ: 不変群の規約表現

symmetry transformation

linear → Unitary operator $U^{-1} = U^+$

antilinear → (antiunitary op)

$U = 1$ Hermite — observable

$U = 1 + i\epsilon(t)$ real parameter

正準量子化

経路積分

$\langle o|o\rangle = \int e^{i\int \mathcal{L}\, d^4x}\, d\varphi$

$\hbar \to 0$ $\frac{\delta\mathcal{L}}{\delta\varphi} = 0$ Euler eq

統計力学

$Z = \sum_{\{\phi_i\}} e^{-H(\phi_i)/kT}$

Minkovski Euclid

$t \to i\tau$

$kT \to \hbar g$

二次相転移 ＝ 連続理論

S. Weinberg : The Quantum Theory of Field

vol I (1995)

II (1996)

Chmbridge U.P.

- 自由場
- 相互作用のある場合 摂動展開

$\langle o|\phi(x)\phi(y)|o\rangle$ { propergator, Green function, correlation function

—○—○—

metric $ds^2 = \eta_{\mu\nu}\, dx^\mu dx^\nu$ $\eta = \begin{bmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}$ $x^\mu = (x^0, x^1, x^2, x^3) = (t, x, y, z)$

$x'_\mu = (x_0, x_1, x_2, x_3) = (t, -x, -y, -z)$

$p^\mu = (E, \vec{p})$ $p^2 = p^\mu p_\mu = E^2 - \vec{p}^2 = m^2$ $p_\mu = i\partial_\mu$

units $c = \hbar = 1$ natural unit

全ての次元量はエネルギー単位 fine structure constant $\frac{e^2}{4\pi\hbar c} = \frac{e^2}{4\pi} = \frac{1}{137}$

$\hbar c = 197.35\ \text{MeV fm}$

$\lambda_{m_e}{}^2 = (\hbar c)^2/(0.511\,\text{MeV})^2 = 1.5 \times 10^{-21}\ \text{cm}^2$

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作用原理

$L(q_i, \dot{q}_i)$　Hamilton の原理　$I = \int_{t_1}^{t_2} L\,dt$　$\frac{\partial L}{\partial q} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) = 0$　Euler eq

正準理論　$p_i = \frac{\partial L}{\partial \dot{q}_i}$　Hamilton 関数　$H = \sum \dot{q}_i p_i - L$

$\{A, B\} = \sum \left(\frac{\partial A}{\partial q_i}\frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i}\frac{\partial B}{\partial q_i} \right)$　$\dot{q} = \{q, H\} = \frac{\partial H}{\partial p}$

$\dot{p} = \{p, H\} = -\frac{\partial H}{\partial q}$

量子化

$\{A, B\} \leftarrow \frac{1}{i\hbar}[A, B]$　要請　$[q_i, p_i] = i\delta_{ij}$　$\{q_i, p_j\} = \delta_{ij}$

$\begin{cases} \dot{q}_i = i[H, q_i] \\ \dot{p}_i = i[H, p_i] \end{cases}$　Heisenberg picture

無限自由度への拡張

$q_i \to \varphi_i \to \varphi(x)$　$I = \int dt\, L = \int dt \int d^3x\, \mathcal{L}[\varphi(x), \partial_\mu \varphi(x)]$

$\delta I = 0$

$\pi(x) = \frac{\delta \mathcal{L}}{\delta \dot{\varphi}(x)}$

$\sum_j [q_i, p_j] f_j = i f_i$

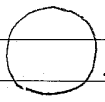
$\sum_j \to \int$　$\int d^3x' [\varphi(x), \pi(x')] f(x') = i f(x)$

$[\varphi(x), \pi(x')] = i\delta^3(x - x')$

$\mathcal{H} = \dot{\varphi}(x)\pi(x) - \mathcal{L}$

$H = \int d^3x\, \mathcal{H}$

変分原理　$\delta \int_\Omega d^4x\, \mathcal{L}[\phi(x), \partial_\mu \phi(x)] = 0$　$\delta\phi = 0$ on $\partial\Omega$



$\int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi}\delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\delta(\partial_\mu \phi) \right]$　δ と ∂_μ は可換

$\int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right] \delta\phi + \int d\sigma_\mu \frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)}\delta\phi$　（第2項 = 0）

Gauss' theorem

$\int d^4x \frac{\partial A^\mu}{\partial x^\mu} = \int d\sigma_\mu A^\mu$

$d\sigma_\mu = \frac{d^4x}{dx^\mu}$

$\frac{\partial \mathcal{L}}{\partial \phi_\alpha} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_\alpha)} = 0$　Euler eq

$\phi_1 \cdots \phi_N$

$\frac{\partial}{\partial x^\mu}\frac{\partial}{\partial x_\mu} = \partial_t^2 - \nabla^2 \equiv \Box$

example　neutral scalar field　$\mathcal{L}(\phi) = \frac{1}{2}\left(\frac{\partial \phi}{\partial x^\mu}\frac{\partial \phi}{\partial x_\mu} - m^2\phi^2 \right)$

$m^2\phi - \frac{\partial}{\partial x^\mu}\frac{\partial}{\partial x_\mu}\phi = (m^2 - \Box)\phi = 0$

$\pi(x) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}(x)} = \dot{\phi}(x)$

$\mathcal{H} = \dot{\phi}\pi - \mathcal{L} = \frac{1}{2}(\pi^2 + (\nabla\phi)^2 + m^2\phi^2)$

Euler → EOM
Hamiltonian → EOM
）等価？　等しいとき canonical theory

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不変変分論 Apr 24

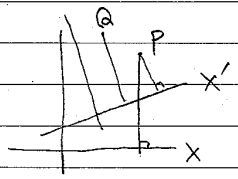
$$I = \int d^4x\, \mathcal{L}\left[\varphi(x), \frac{\partial}{\partial x}\varphi(x)\right]$$

$x \to x^{\mu\prime} = x^\mu + \delta x^\mu$

$\varphi(x) \to \varphi'(x') = \varphi(x) + \delta\varphi(x)$

$I \to I'$ $\quad \delta'\varphi = \varphi'(x) - \varphi(x)$ Lie 微分

$$\delta\varphi = \underbrace{\varphi'(x')}_{P} - \underbrace{\varphi(x)}_{Q}$$



$\partial_\mu \delta'\varphi(x) = \delta\varphi_{,\mu}(x)$ $\quad \partial_\mu, \delta'$ が交換, δ は ∂_μ と交換しない

$$\delta\varphi = \varphi'(x') - \varphi'(x) + \underbrace{\varphi'(x) - \varphi(x)}_{\delta'\varphi}$$

$$\frac{\partial\varphi'}{\partial x'^\mu}(x'-x)^\mu \quad \delta x^\mu$$

$$\delta\varphi = \delta'\varphi + \varphi_{,\mu}\delta x^\mu$$

$$\frac{\partial}{\partial x^\mu} = \frac{\partial x'^\nu}{\partial x^\mu}\frac{\partial}{\partial x'^\nu} = \frac{\partial}{\partial x'^\mu}(\delta x^\nu)$$

$$\frac{\partial}{\partial x^\mu}\delta\varphi = \frac{\partial}{\partial x'^\mu}\varphi'(x') - \frac{\partial}{\partial x'^\mu}\varphi'(x') + \frac{\partial}{\partial x'^\mu}\varphi'(x') - \frac{\partial}{\partial x^\mu}\varphi(x)$$

$\downarrow$ $\frac{\partial}{\partial x'^\mu}(\delta x')\varphi'_{,\mu}$

$$\delta\varphi_{,\mu} = \delta'\varphi_{,\mu} + \varphi_{,\mu\nu}\delta x^\nu$$

$$\delta I = \int_{\Omega'} d^4x\, \mathcal{L}\left[\varphi(x'), \partial_\mu\varphi(x')\right] - \int_\Omega d^4x\, \mathcal{L}\left[\varphi(x), \partial_\mu\varphi(x)\right] \quad \leftarrow I'-I$$

$$J = \frac{\partial^4 x'}{\partial^4 x} = \frac{\partial(x^{0\prime}x'^{\cdots}x^{3\prime})}{\partial(x^0\cdots x^3)} = 1 + \frac{\partial}{\partial x^\nu}(\delta x^\nu)$$

$\downarrow$

$$= \int d^4x \left\{\frac{\partial\mathcal{L}}{\partial\varphi}\delta\varphi + \frac{\partial\mathcal{L}}{\partial\varphi_{,\mu}}\delta\varphi_{,\mu} + \mathcal{L}\frac{\partial}{\partial x^\mu}(\delta x^\mu)\right\}$$

$$\delta\varphi \to \delta'\varphi : \delta I = \int_\Omega d^4x \left\{\frac{\partial\mathcal{L}}{\partial\varphi}(\delta'\varphi + \varphi_{,\mu}\delta x^\mu) + \underbrace{\frac{\partial\mathcal{L}}{\partial\varphi_{,\mu}}(\delta'\varphi_{,\mu} + \varphi_{,\mu\nu}\delta x^\nu)}_{=} + \mathcal{L}\frac{\partial(\delta x^\mu)}{\partial x^\mu}\right\}$$

$$\partial_\mu\left[\frac{\partial\mathcal{L}}{\partial\varphi_{,\mu}}\delta'\varphi\right] - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial\varphi_{,\mu}}\right)\delta'\varphi$$

$$\delta I = \int d^4x \left\{[\mathcal{L}]\delta'\varphi + \partial_\mu\left[\frac{\partial\mathcal{L}}{\partial\varphi_{,\mu}}\delta'\varphi + \mathcal{L}\delta x^\mu\right]\right\} \qquad \delta\varphi = \delta'\varphi + \varphi_{,\mu}\delta x^\mu$$

$\downarrow$

$[\mathcal{L}] =$

$$= \int_\Omega d^4x \left\{[\mathcal{L}]\delta\varphi' + \partial_\mu\left[\frac{\partial\mathcal{L}}{\partial\varphi_{,\mu}}\delta\varphi - \left(\frac{\partial\mathcal{L}}{\partial\varphi_{,\mu}}\varphi_{,\nu} - \mathcal{L}\right)\delta x^\nu\right\}\right\} = 0$$

$$\underbrace{[\mathcal{L}]\delta'\varphi}_{\text{field}} + \partial_\mu\left[\frac{\partial\mathcal{L}}{\partial\varphi_{,\mu}}\delta\varphi - \underbrace{T^\mu{}_\nu\delta x^\nu}_{\text{coordinate}}\right] = 0 \qquad T^\mu{}_\nu = \frac{\partial\mathcal{L}}{\partial\varphi_{,\mu}}\varphi_{,\nu} - \mathcal{L}$$

φ が field eqn を満す $[\mathcal{L}]_\varphi = 0$ $\qquad T^0{}_0 = \frac{\partial\mathcal{L}}{\partial\dot\varphi}\dot\varphi - \mathcal{L} = \mathcal{H}$

$\partial_\mu J^\mu = 0 \quad J^\mu = \frac{\partial\mathcal{L}}{\partial\varphi_{,\mu}}\delta\varphi_\alpha - T^\mu{}_\nu\delta x^\nu$ Noether の定理

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global r_3 symmetry $\delta x^\mu = x'^\mu - x^\mu = \sum \varepsilon_r f_r^\mu(x)$

n-parameter の群 $r = 1, \cdots, n$

$\delta\varphi_\alpha = \varphi'_\alpha(x') - \varphi_\alpha(x) = \sum \varepsilon_r F_{\alpha r}(\varphi)$

$J^\mu = \frac{\partial \mathcal{L}}{\partial \varphi_{\alpha,\mu}} F_{\alpha r}(\varphi) - T^\mu{}_\nu f_r^\nu(x)$ $\quad \alpha = 1, \cdots, N$ $\quad r = 1, \cdots, n$

n 個の保存則

$\partial_\mu J^\mu = 0$, $\partial_0 \int d^3x\, J^0 = \partial_k \int d^3x\, J^k = \int d^3\sigma_k J^k = 0$

孤立系

$Q = \int d^3x\, J^0$

$\dot{Q} = 0$

Lie 群

$\delta\varphi_\alpha = F(\vec{\varepsilon})_{\alpha\beta}\varphi_\beta$

$= \sum \varepsilon_r (F_r)_{\alpha\beta}\varphi_\beta$

$F_r = \frac{\partial F(\vec{\varepsilon})}{\partial \varepsilon_r}$

$[F_r, F_s] = C_{rs}^t F_t$

Structure constant

Lorentz 群

special theory of relativity ｜ 2つの慣性の間で物理法則は同等 ／ 光速不変

$x^\mu \to x'^\mu$

$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu = \eta_{\rho\sigma} dx'^\rho dx'^\sigma$

$\eta_{\mu\nu} \frac{\partial x^\mu}{\partial x'^\rho} \frac{\partial x^\nu}{\partial x'^\sigma} = \eta_{\rho\sigma}$

無限小変換

｜ $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu$

$a^\mu = \varepsilon^\mu$

位) translation

$\delta x^\mu = \epsilon^\mu$

$\delta\varphi = 0$

$\partial_\mu T^\mu{}_\nu = 0$

$\partial_0 \int d^3x\, T^0{}_\nu = P_\nu$ $\quad \dot{P}_\nu = 0$

位相変換

$\varphi_\alpha, \varphi_\alpha^*$

$\varphi(x) \to e^{i\lambda}\varphi(x)$ $\quad \delta\varphi = i\lambda\varphi$

$\varphi^*(x) \to e^{-i\lambda}\varphi(x)$ $\quad \delta\varphi^* = -i\lambda\varphi^*$

$\delta x = 0$

$\partial_\mu \underbrace{\left[\frac{\partial \mathcal{L}}{\partial \varphi_{,\mu}} \varphi - \frac{\partial \mathcal{L}}{\partial \varphi^*_{,\mu}} \varphi^* \right]}_{J^\mu \text{ EM current}} = 0$

EM $\partial_\mu J^\mu = 0$

interacting $\partial_\mu F^{\mu\nu} = J^\mu$

$\partial_\mu \partial_\nu F^{\mu\nu} = 0$

(ex 1)

$U(\theta) = e^{i\theta t}$

$U(\theta)U(\bar{\theta}) =$

Baker-Hausdorff's Formula $e^A e^B = e^C$ $\quad C = A + B + [A,B]$

$= \frac{1}{12}([A,[A,B]] + [[A,B],B])$

$+ \frac{1}{48}([A,[[A,B]B]] + \cdots$

$+ \frac{1}{120}$

$- \frac{1}{360}$

$\eta_{\mu\nu}\Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma = \eta_{\rho\sigma}$ Lorentz condition

$x^\mu \to x'^\mu$ $\quad x'^\mu = \Lambda^\mu{}_\nu x^\nu + a^\mu$ $\quad (\det\Lambda)^2 = 1$ $\quad \exists \Lambda \quad (\Lambda^{-1})^\mu{}_\nu = \Lambda_\nu{}^\mu$

$a = 0$ homogeneous

$a \neq 0$ in " or Poincaré group

$\det\Lambda = +1$ proper

-1 improper P.T.

$(\Lambda^0{}_0)^2 = 1 + (\Lambda^1{}_0)^2 + (\Lambda^2{}_0)^2 + (\Lambda^3{}_0)^2 \geq 1$

$\Lambda^0{}_0 \geq 1$ orthochronous

≤ -1

3 rotation 3 L-boost + 4 = 10

$\eta_{\mu\nu}\Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma = \eta_{\rho\sigma}$

$\eta_{\mu\nu}(\delta^\mu{}_\rho + \omega^\mu{}_\rho)(\delta^\nu{}_\sigma + \omega^\nu{}_\sigma) = \eta_{\rho\sigma}$

$\omega^\nu{}_\rho + \omega^\rho{}_\nu = 0$ $\quad \omega_{\mu\nu} = -\omega_{\nu\mu}$

$T \to U(T)|P_1 P_2 \cdots\rangle$　　$U(T)$ は operator の変換を induce する

$a_{p_i}^+ \cdots U^{-1}U$

$U(1+\omega) = 1 - \frac{i}{2}\omega_{\mu\nu}J^{\mu\nu} - i\varepsilon_\mu P^\mu + \cdots$

unitality $\begin{cases} J^{\mu\nu\dagger} = J^{\mu\nu} \\ P^{\mu\dagger} = P^\mu \end{cases}$

$J^{\mu\nu} = J^{\nu\mu}$　　$\epsilon = 0$ とする　$U(\Lambda)U(1+\omega)U(\Lambda)^{-1} = U(\Lambda(1+\omega)\Lambda^{-1})$

$1 - \frac{i}{2}\omega_{\mu\nu}U(\Lambda)J^{\mu\nu}U(\Lambda)^{-1} = 1 - \frac{i}{2}(\Lambda\omega\Lambda^{-1})_{\rho\sigma}J^{\rho\sigma}$

$U(\Lambda)J^{\mu\nu}U(\Lambda)^{-1} = \Lambda_\rho{}^\mu\Lambda_\sigma{}^\nu J^{\rho\sigma}$　　$J^{\mu\nu}$ Lorentz tensor　　$\Lambda_\rho{}^\mu\omega_{\mu\nu}(\Lambda^{-1})^\nu{}_\sigma$ ($\Lambda_\sigma{}^\nu$)

$\Lambda = 1+\omega$

$(1 - \frac{i}{2}\omega_{\alpha\beta}J^{\alpha\beta})J^{\mu\nu}(1 + \frac{i}{2}\omega_{\alpha\beta}J^{\alpha\beta}) = (\delta_\rho{}^\mu + \omega_\rho{}^\mu)(\delta_\sigma + \omega_\sigma{}^\nu)J^{\rho\sigma}$

$-\frac{i}{2}\omega_{\alpha\beta}[J^{\alpha\beta}, J^{\mu\nu}] = (\delta_\rho{}^\mu\omega_\sigma{}^\nu + \delta_\sigma{}^\nu\omega_\rho{}^\mu)J^{\rho\sigma} = \omega_\sigma{}^\nu J^{\mu\sigma} + \omega_\rho{}^\mu J^{\rho\nu}$

$= \omega_{\sigma\lambda}\eta^{\lambda\nu}J^{\mu\sigma} + \omega_{\rho\lambda}\eta^{\lambda\mu}J^{\rho\nu}$

$= \frac{1}{2}\left(\begin{array}{l}\omega_{\alpha\beta}\eta^{\beta\nu}J^{\mu\alpha} + \omega_{\alpha\rho}\eta^{\beta\mu}J^{\alpha\nu} \\ -\omega_{\beta\rho}\eta^{\alpha\nu}J^{\mu\beta} - \omega_{\alpha\beta}\eta^{\alpha\mu}J^{\beta\nu}\end{array}\right)$

$[J^{\alpha\beta}, J^{\mu\nu}] = i(\eta^{\beta\nu}J^{\mu\alpha} - \eta^{\alpha\nu}J^{\mu\beta} + \eta^{\beta\mu}J^{\alpha\nu} - \eta^{\alpha\mu}J^{\beta\nu}) = iC^{\alpha\beta\mu\nu}{}_{\rho\sigma}J^{\rho\sigma}$

$\vec{J} = (J^{23}, J^{31}, J^{12}) = (J^1\ J^2\ J^3)$　　$[J^1, J^2] = iJ^3$　　$[J^i J^j] = i\varepsilon_{ijk}J^k$

$\vec{K} = (J^{10}, J^{20}, J^{30}) = (K^1\ K^2\ K^3)$　　$[J^i\ K^j] = i\varepsilon_{ijk}K^k$

$[\ ,\ ] =$　　$[K^i K^j] = -i\varepsilon_{ijk}J^k$

(ex 2)　$\omega^\mu{}_\nu$, ε_μ があるとき

Poincaré 群の Lie algebra を導け

$U(\Lambda, a')U(\Lambda, a) = U(\Lambda'\Lambda, \Lambda a + a')$

$U(\Lambda, a)U(1+\omega, \epsilon)U^{-1}(\Lambda, a)$

$\vec{P} = \{P^1 P^2 P^3\}$　$H = P^0$

Ans.
$[J_i, P_j] = i\epsilon_{ijk}P_k$
$[K_i P_j] = i\delta_{ij}H$
$[K_i, H] = iP_i$
$[J_i, H] = [P_i, H] = [H, H] = 0$

$U(\Lambda, a)P^\mu U(\Lambda, a)^{-1} = \Lambda_\rho{}^\mu P^\rho$

P^μ vector の変換性

回転 (ex3)

$J_1 = i\begin{bmatrix} 0 & & \\ & 0 & 0 \\ & 0 & 0 \\ & 0 & -1 \end{bmatrix}$　　0 0 0 / 0 0 1 / 0 −1 0

$J_2 =$

$J_3 =$

$U(R_\theta) = e^{i\vec{\theta}\vec{J}}$

$U(\Lambda) = e^{-\frac{i}{2}\omega_{\mu\nu}J^{\mu\nu}}$

を matrix で書け

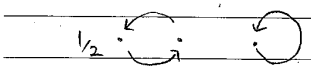
$K_1 =$

$K_2 =$

$K_3 =$

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Spin と統計の関係



$J = 1/2 \quad \theta = 2\pi \quad U(2\pi) = -1$

$$U(R_3(\theta)) = \begin{pmatrix} 1 & & & \\ & \cos\theta & \sin\theta & \\ & -\sin\theta & \cos\theta & \\ & & & 1 \end{pmatrix} \qquad B_3(p) = \begin{pmatrix} \cosh\varphi & 0 & 0 & \sinh\varphi \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \sinh\varphi & 0 & 0 & \cosh\varphi \end{pmatrix}$$

$\cosh\varphi = \gamma \quad v = \beta$

$\gamma = 1/\sqrt{1-\beta^2}$

Noether's theorem

Lorentz transform $x^\mu \to x^\mu + \omega^\mu{}_\nu x^\nu$

$$\varphi_\alpha(x) \to \varphi_\alpha(x) - \frac{i}{2}\omega_{\mu\nu}[S^{\mu\nu}]_{\alpha\beta}\varphi_\beta(x)$$

$$\partial_\mu\left[-i\frac{\partial\mathcal{L}}{\partial\varphi_{\alpha,\mu}}\frac{1}{2}\omega_{\rho\sigma}[S^{\rho\sigma}]_{\alpha\beta}\varphi_\beta - T^\mu{}_\nu\,\omega^\nu{}_\lambda x^\lambda\right] = 0$$

$T^\mu{}_\nu\,\omega^{\nu\lambda}x_\lambda$

$\frac{1}{2}(T^\mu{}_\nu\,\omega_\nu{}^\lambda x_\lambda - T^{\mu\lambda}\omega_{\cdot\lambda}^{\nu}x_\nu)$

$\omega^{\rho\sigma}$

$$\partial_\mu\left\{i\frac{\partial\mathcal{L}}{\partial\varphi_{\alpha,\mu}}[S^{\rho\sigma}]_{\alpha\beta}\varphi_\beta + (T^{\mu\rho}x^\sigma - T^{\mu\sigma}x^\rho)\right\} = 0$$

$$\partial_\mu M^{\mu\rho\sigma} = 0 \qquad M^{\mu\rho\sigma} = \underbrace{i\frac{\partial\mathcal{L}}{\partial\varphi_{\alpha,\mu}}[S^{\rho\sigma}]_{\alpha\beta}\varphi_\beta}_{-S^{\mu\rho\sigma}} + (T^{\mu\rho}x^\sigma - T^{\mu\sigma}x^\rho)$$

$\partial_\nu T^{\nu\rho} = 0$

$$\partial_\mu[T^{\mu\rho}x^\sigma - T^{\mu\sigma}x^\rho = S^{\mu\rho\sigma}]$$

$T^{\sigma\rho} - T^{\mu\rho} = \partial_\mu S^{\mu\rho\sigma}$

$\Theta^{\rho\sigma} = T^{\rho\sigma} + \partial_\mu g^{\mu\rho\sigma}$

$g^{\mu\rho\sigma} = \frac{1}{2}\{S^{\mu\rho\sigma} - S^{\rho\mu\sigma} - S^{\sigma\mu\rho}\}$ と置く

$\partial_\rho\Theta^{\rho\sigma} = \underbrace{\partial_\rho T^{\rho\sigma}}_{0} + \underbrace{\partial_\rho\partial_\mu g^{\mu\rho\sigma}}_{0} = 0$ 保存則

$$\int d^3x\,\Theta^{0\sigma} = \int d^3x\,(T^{0\sigma} + \partial_k g^{k0\sigma}) = \int d^3x\,T^{0\sigma} = P^\sigma$$

$$\Theta^{\rho\sigma} - \Theta^{\sigma\rho} = T^{\rho\sigma} - T^{\sigma\rho} + \partial_\mu(g^{\mu\rho\sigma} - g^{\mu\sigma\rho})$$

$$= \partial_\mu S^{\mu\sigma\rho} + \partial_\mu\frac{1}{2}\{S^{\mu\rho\sigma} - S^{\rho\mu\sigma} - S^{\mu\sigma\rho} + S^{\sigma\mu\rho} + S^{\rho\mu\sigma} - S^{\sigma\mu\rho}\}$$

(×) (×) (×) (×)

Θ Belinfante tensor

$$\int d^3x\,\Theta^{0\nu} = P^\nu$$

$$\partial_\mu\Theta^{\mu\nu} = 0$$

$$\Theta^{\mu\nu} = \Theta^{\nu\mu}$$

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$\bar{M}^{\lambda\mu\nu} = x^\mu \Theta^{\lambda\nu} - x^\nu \Theta^{\lambda\mu}$

$\partial_\lambda \bar{M}^{\lambda\mu\nu} = 0$

$J^{\mu\nu} = \int d^3x\, \bar{M}^{0\mu\nu} = \int d^3x\, (x^\mu \Theta^{0\nu} - x^\nu \Theta^{0\mu})$ $\quad J^k \equiv \epsilon_{ijk} \frac{1}{2} J^{ij}$ $\quad \dot{J} = i[H, J] = 0$

$[P_j, J^k] = \frac{1}{2}\epsilon_{ilk}[P_j, J^{il}]$ $\quad [\vec{P}, f(x)] = i\vec{\nabla} f(x)$

$= \frac{1}{2}\epsilon_{ilk}\int d^3x\, i\frac{\partial}{\partial x^j}(x^i \Theta^{0l} - x^l \Theta^{0i}) = i\epsilon_{jlk}\int d^3x\, \Theta^{0l}$ $\quad [\vec{P}_i, J^j] = -i\epsilon_{ijk}P_k$

ex) $[P_j, K_k] = -iH\delta_{jk}$

⋮

▷ Apr 28

Noether 第二定理 $\quad r = 1, \cdots, n$

大局的 $\delta x^\mu = x'^\mu - x^\mu = f^\mu(\varepsilon, x) = \sum \varepsilon_r f_r^\mu(x)$

第一定理 $\delta\varphi_\alpha = \varphi'_\alpha(x) - \varphi_\alpha(x) = F_\alpha(\varepsilon, \varphi) = \sum \varepsilon_r F_{\alpha r}(\varphi)$

局所的 $\delta x^\mu = f^\mu(\lambda(x), x) = \sum \lambda_r f_r^\mu(x)$ $\quad F_{\alpha r} = \left.\frac{\partial F}{\partial \lambda}\right|_{\lambda=0}$

$\delta\varphi_\alpha = F_\alpha(\lambda(x), \partial_\mu\lambda(x), \cdots \varphi, x) = \sum \lambda_\mu F_{\alpha r}(\varphi, x) + \lambda_{r,\mu} F_{\alpha r}^\mu(\varphi, x)$ $\quad F_{\alpha r}^\mu = \left.\frac{\partial F}{\partial \lambda_{,\mu}}\right|_{\lambda=0}$

$\delta I = \int d^4x \left\{ [\mathcal{L}]_\alpha \delta'\varphi_\alpha + \partial_\mu\left[\frac{\partial \mathcal{L}}{\partial \varphi_{\alpha,\mu}}\delta\varphi_\alpha - T^\mu{}_\nu \delta x^\nu\right]\right\}$

$\delta\varphi - \varphi_{\alpha,\mu}\delta x^\mu$

$\partial_\mu([\mathcal{L}]_\alpha \lambda_r F_{\alpha r}^\mu) - \lambda_r \partial_\mu([\mathcal{L}]_\alpha F_{\alpha r}^\mu)$

$= \int d^4x \left\{ [\mathcal{L}]_\alpha (\lambda_r F_{\alpha r} + \lambda_{r,\mu} F_{\alpha r}^\mu - \varphi_{\alpha,\mu}\lambda_r f_r^\mu) + \partial_\mu\left[\frac{\partial \mathcal{L}}{\partial \varphi_{\alpha,\mu}}(\lambda_\nu F_{\alpha r} + \lambda_{r,\nu} F_{\alpha r}^\nu) - T^\mu{}_\nu \delta x^\nu\right]\right.$

$= \int d^4x \left\{ \lambda_r \left[[\mathcal{L}]_\alpha F_{\alpha r} - \varphi_{\alpha,\mu} f_r^\mu - \partial_\mu([\mathcal{L}]_\alpha F_{\alpha r}^\mu)\right] + \partial_\mu\left[\frac{\partial \mathcal{L}}{\partial \varphi_{\alpha,\mu}}\lambda_r F_{\alpha r} - T^\mu{}_\nu \lambda_r f_r^\nu + [\mathcal{L}]_\alpha \lambda_r F_{\alpha r}^\mu\right]\right.$

$+ \partial_\mu\left(\frac{\partial \mathcal{L}}{\partial \varphi_{\alpha,\mu}}\lambda_{r,\nu} F_{\alpha r}^\nu\right)$

$= \int d^4x \left\{ [\mathcal{L}]_\alpha (F_{\alpha r} - \varphi_{\alpha,\mu} f_r^\mu) - \partial_\mu([\mathcal{L}]_\alpha F_{\alpha r}^\mu)\right\} + \partial_\mu(B_r^\mu \lambda_r + C_r^{\mu\nu}\lambda_{r,\nu}) = 0$

$B_r^\mu = \frac{\partial \mathcal{L}}{\partial \varphi_{\alpha,\mu}} F_{\alpha r} - T^\mu{}_\nu f_r^\nu + [\mathcal{L}] F_{\alpha r}^\mu$

$\lambda_r(x) = 0 \quad \partial\Omega \qquad (\alpha = 1 \cdots N) \qquad C_r^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial \varphi_{\alpha,\mu}} F_{\alpha r}^\nu$

① $[\mathcal{L}]_\alpha (F_{\alpha r} - \varphi_{\alpha,\mu} f_r^\mu) - \partial_\mu([\mathcal{L}]_\alpha F_{\alpha r}^\mu) = 0$ $\quad (r = 1 \cdots n)$ $\quad \longrightarrow$ $N-n$ 個の Euler eqn のみ独立

$\partial_\mu(\cdots) = 0$ $\quad$ n 個の任意関数

$(\partial_\mu B_r^\mu)\lambda_r + B_r^\nu \lambda_{r,\nu} + (\partial_\mu C_r^{\mu\nu})\lambda_{r,\nu} + C_r^{\mu\nu}\lambda_{r,\nu\mu} = 0$

② $\partial_\mu B_r^\mu = 0 \rightarrow \partial_\mu\left(\frac{\partial \mathcal{L}}{\partial \varphi_{\alpha,\mu}} F_{\alpha r} - T^\mu{}_\nu f_r^\nu + [\mathcal{L}] F_{\alpha r}^\mu\right) = 0 \rightarrow$ Euler eq $[\mathcal{L}] = 0$ $\quad \partial_\mu J_r^\mu = 0$

$J_r^\mu = \frac{\partial \mathcal{L}}{\partial \varphi_{\alpha,\mu}} F_{\alpha r} - T^\mu{}_\nu f_r^\nu$

∂_μ の中 $= [\mathcal{L}]_\alpha (F_{\alpha r} - \varphi_{\alpha,\mu} f_r^\mu)$

global transfer の Neother current と同じ

③ $B_r^\nu + \partial_\mu C_r^{\mu\nu} = 0$

④ $C_r^{\nu\mu} + C_r^{\mu\nu} = 0$ $\quad Q = \int d^3x\, J^0 = -\int d^3x\, \partial_k C_r^{k0}$

$= -\int_{\partial\Omega} d\sigma_k C_r^{k0}$ $\quad \infty$ の場の生成 $\rightarrow$ charge が計算可

coop Filing Leaf 6mm×41行

- Field eqn
- gauge inv (Neother) } Maxwell eqn

$\varphi \to \varphi' = e^{+ie\lambda(x)}\varphi(x) = (1+ie\lambda)\varphi$

$\varphi^* \to \varphi^{*\prime} = e^{-ie\lambda(x)}\varphi^*(x) = (1-ie\lambda)\varphi^*$

$A_\mu \to A_\mu - \partial_\mu \lambda(x)$

$f^\mu_\nu = 0$

① → $[\mathcal{L}]_\varphi ie\varphi + [\mathcal{L}]_{\varphi^*}(-ie\varphi) - \partial_\mu [\mathcal{L}]_{A_\mu} = 0$

② → $B^\mu = \frac{\partial \mathcal{L}}{\partial \varphi_{,\mu}} ie\varphi - \frac{\partial \mathcal{L}}{\partial \varphi^*_{,\mu}} ie\varphi^* + [\mathcal{L}]_{A_\nu}(-\delta^\mu_\nu) = 0$

③ → $C^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial A_{\lambda,\mu}}(-)\delta^\nu_\lambda = -\frac{\partial \mathcal{L}}{\partial A_{\nu,\mu}} \to \frac{\partial \mathcal{L}}{\partial A_\mu} - \partial_\lambda\left(\frac{\partial \mathcal{L}}{\partial A_{\mu,\lambda}}\right)$

④ → $\frac{\partial \mathcal{L}}{\partial A_{\mu,\nu}} + \frac{\partial \mathcal{L}}{\partial A_{\nu,\mu}} = 0$ $\quad \mathcal{L}$ の中に $A_{\mu,\nu}$

ex

$\varphi \to \varphi' = e^{iT^\alpha \theta^\alpha}\varphi$

$A^\alpha_\mu \to A^{\alpha\prime}_\mu = A^\alpha_\mu - \frac{1}{g}\partial_\mu \theta^\alpha + i\varepsilon_{\alpha\beta\gamma} A^\beta_\mu \theta^\gamma$

$SU(2)$ $[T^\alpha, T^\beta] = i\varepsilon^{\alpha\beta\gamma}T^\gamma$

$\alpha,\beta = 1,2,3$

$F_{\nu\mu} = \partial_\nu A_\mu - \partial_\mu A_\nu$

$ie\left(\frac{\partial \mathcal{L}}{\partial \varphi_{,\mu}}\varphi - \frac{\partial \mathcal{L}}{\partial \varphi^*_{,\mu}}\varphi^*\right) + \frac{\partial \mathcal{L}}{\partial A_\mu} = 0$

$(\partial_\mu - ieA_\mu)\varphi$

$(\partial_\mu + ieA_\mu)\varphi^*$ minimum sub

①～④ で導かれる物理は何か？

(Yang-Mills 場)

自由場の量子化 Real Scalar Field (neutral, spin 0)

$\varphi(x) = \int d^3p\, a(p) f_p(x)$

$(\Box + m^2) f_p(x) = 0$ $\quad f_p \sim e^{-ipx}$ $\quad p^{0^2} = \vec{p}^{\,2} + m^2$ mass shell 条件

$\int d^3p\, a(p) e^{-ipx} + a'(p) e^{+ipx}$

$\delta(f) = \frac{\delta(x)}{|f'|}$

$\varphi^* = \int d^3p\, a^\dagger(p) e^{+ipx} + a'^\dagger(p) e^{-ipx}$ $\quad \varphi = \varphi^*$ $\quad a' = a^\dagger$ $\quad \to \varphi(x) \propto \int d^3p\, a(p) e^{-ipx} + a^\dagger(p) e^{+ipx}$

$\int d^4p\, \delta(p^2 - m^2)\theta(p^0) e^{-ipx}$

$= \int d^4p\, \delta(p^{0^2} - \vec{p}^{\,2} - m^2)\theta(p^0) e^{-ipx}$

$= \int d^3p \frac{e^{-ipx}}{2\sqrt{\vec{p}^{\,2}+m^2}} \leftarrow f(\sqrt{\vec{p}^{\,2}+m^2}, \vec{p})$

$\omega_p = +\sqrt{\vec{p}^{\,2}+m^2}$

invariant volume element $\frac{d^3p}{2\sqrt{\vec{p}^{\,2}+m^2}}$

$f_p(x) = \frac{1}{(2\pi)^{3/2}}\frac{1}{\sqrt{2\omega_p}} e^{-ipx}$

$\varphi(x) = \int d^3p \left(a_p f_p(x) + a_p^\dagger f_p^*(x)\right)$

逆 $\int d^3x\, f^*(\vec{x},t)\varphi(\vec{x},t) = \frac{1}{2\omega_p} a_p + a^\dagger(p) e^{2i\omega_p t}$

$\int d^3x\, f^*(\vec{x},t)\dot{\varphi}(\vec{x},t) = -\frac{i}{2}\left(a_p - a^\dagger(p) e^{2i\omega_p t}\right)$

$a_p = \int f^*(x,t)\left\{\omega_p \varphi(x,t) + i\dot{\varphi}(x,t)\right\} d^3x$

$= i\int f^*(x,t) \overleftrightarrow{\partial_0} \varphi(x,t)$

$A \overleftrightarrow{\partial_0} B = A\dot{B} - \dot{A}B$

$a_p^\dagger = -i\int f(x,t) \overleftrightarrow{\partial_0} \varphi(x,t)$

f_p に対する orthonormality

$$\int f_p^* \overset{\leftrightarrow}{\partial_0} f_{p'} \, d^3x \equiv \delta^3(p-p')$$

$$\int f_p \overset{\leftrightarrow}{\partial_0} f_p \, d^3x = 0$$

canonical quantisation

$$\pi(x) = \frac{\partial \mathcal{L}}{\partial \dot{\varphi}} = \dot{\varphi}(x)$$

$$[\varphi(x,t), \pi(x',t)] = i\delta^3(\vec{x}-\vec{x}')$$

$$[a(p), a^\dagger(p')] = \int d^3x\, d^3y \, [f_p^* \overset{\leftrightarrow}{\partial_0} \varphi, f_{p'} \overset{\leftrightarrow}{\partial_0} \varphi]$$

$$= i\int d^3x \, f_p^* \overset{\leftrightarrow}{\partial_0} f_{p'} = \delta^3(\vec{p}-\vec{p}')$$

$a(p)$ positive energy part, annihilation op

$a^\dagger(p)$ negative 〃 , creation op

Hamiltonian $H = \int d^3x \, \mathcal{H} = \int d^3x \, (\pi\dot{\varphi} - \mathcal{L}) = \int d^3x \, \frac{1}{2}(\pi^2(x) + (\nabla\varphi)^2 + m^2\varphi^2)$

$$= \int d^3p \, [a^\dagger(p)a(p) + a(p)a^\dagger(p)]\,\omega_p$$

$$\vec{P} = -\int d^3x \, \pi(x) \underbrace{\nabla}_{\partial_k} \varphi(x) = \int d^3p \, [\cdots]\vec{p}$$

number op

$n(p) = a^\dagger(p)a(p)$ $\quad n(p)|p^{n_p}\rangle = n_p|p^{n_p}\rangle$ $\quad \langle 0|\varphi(x)|p\rangle = \frac{1}{(2\pi)^{3/2}} \frac{e^{-ipx}}{\sqrt{2\omega_p}}$

$a(p)|0\rangle = 0$ $\quad |p^{n_p}\rangle = \frac{1}{\sqrt{n_p!}}(a^\dagger)^{n_p}|0\rangle$

$a^\dagger(p)|0\rangle = |p\rangle$

$n(p)|p\rangle = |p\rangle$ $\quad$ Fock space

$\langle p^{n_p-1}|a_p|p^{n_p}\rangle = \sqrt{n_p}$

Invariant function

$$[\varphi(x), \varphi(y)] = \int d^3p\, d^3q \, \frac{1}{(2\pi)^3\sqrt{2p^0 2q^0}} [a_p e^{-ipx} + a_p^\dagger e^{ipx}, a_q e^{-iqy} + a_q^\dagger e^{iqy}]$$

$$\delta^3(p-q)(e^{-ip(x-y)} - e^{+ip(x-y)})$$

$i\Delta(x-y)$

〃 ゼロ次

$$= \int d^3p \, \frac{1}{(2\pi)^3 2p_0} \{e^{-ip(x-y)} - e^{+ip(x-y)}\}$$

$$= \frac{1}{(2\pi)^3}\int d^4p \, \delta(p^2-m^2)[\theta(p^0) - \theta(-p^0)]e^{-ip(x-y)}$$

$$= \frac{1}{(2\pi)^3}\int d^4p \, \delta(p^2-m^2)\,\delta(p^0)\, e^{-ip(x-y)}$$

性質 ① Lorentz invariant

② $\Delta(-x) = \Delta(x)$

③ $\Delta(x) = 0$ for space like $x^2 < 0$

$x^2 = x^{0\,2} - \vec{x}^2 = -\vec{x}^2$ $\quad \Delta(x)$ は 180° rot に対して inv

$x^0 = 0$

④ $(\Box + m^2)\Delta(x) = 0$ homogeneous inv func $\quad x\delta(x) = 0$

⑤ $x_0 = 0$ $\quad \Delta(x) = 0$

$\frac{\partial}{\partial x_0}\Delta(x) = -\delta^3(x)$

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⑥ ex $i\Delta(x) = \frac{1}{4\pi r}\frac{\partial}{\partial r}$ $\begin{cases} J_0(m\sqrt{t^2-r^2}) & t>r \\ 0 & -r<t<r \\ -J_0(m\sqrt{t^2-r^2}) & t<-r<0 \end{cases}$

$r=\sqrt{|\vec{x}|^2}$

$i\Delta^+(x) = \int \frac{d^3p}{(2\pi)^3} e^{-ipx}$

$i\Delta^-(x) = \int \frac{d^3p}{(2\pi)^3} e^{ipx}$

$\pm i\pi H^{(i)}(z) = \int dx\, e^{\pm iz\cosh x}$

$\Delta(x) = \Delta^+(x) + \Delta^-(x)$

⑦ $(\Box + m^2) G(x) = 0$

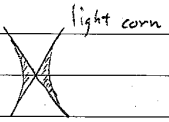
$G(x) : \Delta(x), \quad \Delta_1(x) = \int \frac{d^3p}{(2\pi)^3}(e^{ipx} - e^{-ipx}) = i(\Delta^+ - \Delta^-)$

$\Delta_1(x) = \frac{1}{4\pi r}\frac{\partial}{\partial r} N_0(m\sqrt{t^2-r^2}) \quad |t|>r$

$-\frac{1}{2\pi^2}\frac{1}{r}\frac{\partial}{\partial r} K_0(m\sqrt{r^2-t^2}) \quad |t|<r$

$\sim \frac{e^{-m\sqrt{r^2-t^2}}}{r^2-t^2}$

non causal



⑧ $m \to 0$

$\Delta(x) = -\frac{1}{4\pi r}[\delta(r-t) - \delta(r+t)]$

$\Delta_1(x) = \frac{1}{4\pi r}\left[\frac{P}{r-t} + \frac{P}{r+t}\right]$

Spin stastistics connection microscopic causality

observable Θ

$\Theta^\alpha = \int d^3x\, \Theta^\alpha(x) \quad [\Theta^\alpha(x), \Theta^\beta(y)] = 0 \quad (x-y)^2 < 0$

$\Theta(x) = \varphi_a \varphi_b$

$[\varphi_a(x)\varphi_b(x), \varphi_c(y)\varphi_d(y)] = 0 \quad (x-y)^2<0$

comitator $[\varphi_a(x), \varphi_c(y)] = 0$ Boson

anticomitator $\{\varphi_a(x), \varphi_c(y)\} = 0$

Fierz Helv Phys Acta 12 3 (1939)

Pauli Phys Rev 58 716 (1940)

Liichers & Zumino Phys Rev 110 1950 (1958)

K-G field と anticomitator 2次量子化 $\longrightarrow \{\varphi_a(x), \varphi_c(y)\} = \Delta_1(x-y)$

$-\{\varphi_a\varphi_a\}\varphi_c\varphi_b + \varphi_c\varphi_a\varphi_d\varphi_b$

$+\{\varphi_c\varphi_a\}\varphi_a\varphi_b + \varphi_a\varphi_c\varphi_a\varphi_b + \cdots$

$\sum \Delta_1(x-y)\varphi_c\varphi_d$

microscopic causality を満す

$\varphi(x)$ causal field ▷ May 12

$i\Delta(x-y) = [\varphi(x), \varphi(y)] = 0 \quad (x-y)^2<0$

$= \int \frac{1}{(2\pi)^3}\frac{d^3p}{2p^0}(e^{-ipx} + e^{+ipx}) = i(\Delta^+(x-y) - \Delta^-(x-y))$

$\Delta_1 = \langle 0|\{\varphi(x), \varphi(y)\}|0\rangle = i(\Delta^+ + \Delta^-)$

$i\Delta^+ = \langle 0|\varphi(x)\varphi(y)|0\rangle$

$i\Delta^- = \langle 0|\varphi(y)\varphi(x)|0\rangle$

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Feynmann propagator

$$\Delta_F(x-y) = \theta(x^0-x^{0\prime})\,\Delta^+(x-x') + \theta(x^{0\prime}-x^0)\,\Delta^-(x-x')$$

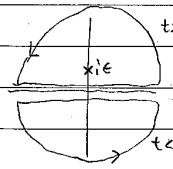
$(-i)\langle 0|\varphi(x)\varphi(x')|0\rangle$

$x^0 > x^{0\prime}$ $\quad e^{-ip(x-x')}$ 平面波 $x' \to x$ 伝播

$P(x' \to x) \geq 0$

$P(x \to x') \geq 0$

$$\theta(t) = \frac{1}{2\pi i}\int_{-\infty}^{\infty} d\tau\, \frac{e^{it\tau}}{\tau - i\epsilon}$$

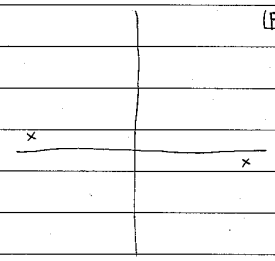


$$\Delta_F(x) = \frac{1}{2\pi i}\int_{-\infty}^{\infty} d\tau\, \frac{e^{it\tau}}{\tau - i\epsilon}\, \frac{-i}{(2\pi)^3}\int d^4p\, \theta(p^0)\,\delta(p^2-m^2)\, e^{-ipx}$$

$$+ \frac{1}{2\pi i}\int_{-\infty}^{\infty} d\tau\, \frac{e^{-it\tau}}{\tau - i\epsilon}\, \frac{-i}{(2\pi)^3}\int d^4p\, \theta(-p^0)\,\delta(p^2-m^2)\, e^{-ipx}$$

$$= \frac{-1}{(2\pi)^4}\int_{-\infty}^{\infty} d\tau\, \frac{e^{it\tau - ipx}}{\tau - i\epsilon}\, \theta(p^0)\,\delta(p^2-m^2)$$

$$\frac{e^{-ip'x}}{\tau - i\epsilon}\,\theta(p^0+\tau)\,\delta[(p^{0\prime}+\tau)^2 - m^2 - \vec{p}^{\,2}]$$



$$= \frac{-1}{(2\pi)^4}\left\{\int \frac{e^{-ipx}}{-p^0+\sqrt{\vec{p}^2+m^2}-i\epsilon}\right\} d^4p$$

$$\int \frac{e^{-ipx}}{+p^0+\sqrt{\quad}-i\epsilon}\Bigg\} \times \frac{1}{\sqrt{p^2+m^2}}$$

$$= \frac{1}{(2\pi)^4}\int d^4p\, \frac{e^{-ipx}}{p^2+m^2+i\epsilon}$$

$$\langle 0|T(\varphi(x)\varphi(x'))|0\rangle$$
$$= \theta(x^0-x^{0\prime})\langle 0|\varphi(x)\varphi(x')|0\rangle$$
$$+ \theta(x^{0\prime}-x^0)\langle 0|\varphi(x')\varphi(x)|0\rangle$$
$$= i\,\Delta_F(x-x')$$

$(\Box + m^2)\Delta = 0$ homogenious

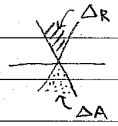
$\Delta(x),\ \Delta^+(x),\ \Delta^-(x)$

$\Delta_1(x)$

$\Delta_F(x)$

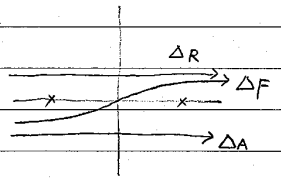
$\Delta_R(x) = -\theta(x^0)\,\Delta(x)$

$\Delta_A(x) = \theta(-x^0)\,\Delta(x)$

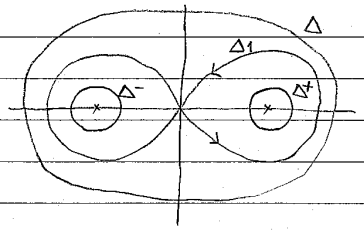


$(\Box + m^2)\Delta_F = -\delta^4(x)$ inhomogenious

$$\Delta_F(x) = \theta(x^0)\underbrace{\Delta^+}_{-\Delta^-} + \theta(-x^0)\underbrace{\Delta^-}_{\theta(x^0)+\Delta^-} = \Delta_R(x) + \Delta^-(x)$$



$$\Delta(x) = \frac{1}{(2\pi)^4}\int d^4p\, \frac{e^{-ipx}}{p^2-m^2}$$



cf Euclidian

$$\int \frac{e^{ikx}}{\underbrace{k^2+m^2}_{0}}\, d^4x$$

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Complex scalar field　charge のある場合

$\mathcal{L} = \frac{\partial \varphi^+}{\partial x}\frac{\partial \varphi}{\partial x} - m^2\varphi^+\varphi$　$\left\{ \begin{array}{l} \varphi \to e^{i\lambda}\varphi \\ \varphi^+ \to e^{-i\lambda}\varphi^+ \end{array} \right.$

$J_\mu = -i\left[(\partial_\mu \varphi^+)\varphi - \varphi^+(\partial_\mu \varphi)\right]$

$\pi = \frac{\partial \mathcal{L}}{\partial \dot{\varphi}} = \dot{\varphi}^+$

$\pi^+ \equiv \dot{\varphi}$

$\frac{\partial}{\partial x^\mu}\left[\frac{\partial \mathcal{L}}{\partial \varphi_{,\mu}}\varphi - \frac{\partial \mathcal{L}}{\partial \varphi^+_{,\mu}}\varphi^+\right] = 0$

$N_Q = \int d^3x\, J^0 = -i\int d^3x\,(\pi\varphi - \pi^+\varphi^+)$

$\varphi = \frac{1}{\sqrt{2}}(\varphi_1 + i\varphi_2)$

$\varphi^+ = \frac{1}{\sqrt{2}}(\varphi_1 - i\varphi_2)$　$\mathcal{L} = \frac{1}{2}\left(\sum_{i=1}^{2}\frac{\partial \varphi_i}{\partial x^\mu}\frac{\partial \varphi_i}{\partial x_\mu} - m^2\varphi_i^2\right)$

φ_1, φ_2 real　　Complex scalor field → 2つの real scalar field

$[\varphi_i, \dot{\varphi}_j] = i\delta_{ij}\delta^3(x-x')$

$[\varphi_i, \varphi_j] = 0$

$[\varphi_i, \varphi_j) = i\Delta(x-y)$

$[\varphi(x), \varphi^+(y)] = i\Delta(x-y)$

$[\varphi(x), \varphi(y)] = 0$

$[\varphi^+ \quad \varphi^+ \quad] = 0$

$\varphi(x) = \frac{1}{(2\pi)^3}\int \frac{d^3p}{2p^0}\left[a(p)e^{-ipx} + b^+(p)e^{ipx}\right]$

φ^+　　$\left[a^+ e^{ipx} + b\ e^{-ipx}\right]$

$a = \frac{a_1 + ia_2}{\sqrt{2}}$　　$a(p)$　φ 場の + freq. annihilate

$b = \frac{a_1 - ia_2}{\sqrt{2}}$　　$b^+(p)$　φ^+　　create

$N_Q = \int d^3p\,(n_+(p) - n_-(p))$

$n_+(p) = a^+(p)a(p)$

$N_-(p) = b^+(p)b(p)$

charge conjugation 荷電共役

charge conjugation operator

$\mathcal{C}\,a^+\,\mathcal{C}^{-1} = \xi\, b^+$　　$\mathcal{C}\,a\,\mathcal{C}^{-1} = \xi^* b$　（ξ: phase）

$\mathcal{C}\,b^+\,\mathcal{C}^{-1} = \xi^* a^+$　　$\mathcal{C}\,b\,\mathcal{C}^{-1} = \xi\, a$

real field　$\varphi = \varphi^+$

$\xi^* = \xi$　$\xi = \pm 1$

$\mathcal{C}\varphi\mathcal{C}^{-1} = \xi^*\varphi^+$

$\mathcal{C}\varphi^+\mathcal{C}^{-1} = \xi\varphi^-$

$\mathcal{C}P_\mu\mathcal{C}^{-1} = P_\mu$

$[H, \mathcal{C}] = 0$　　$\mathcal{C}|0\rangle = |0\rangle$

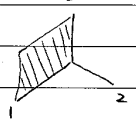
$\underbrace{+\ -\ 0}_{1\pm i2}$

$\varphi, \varphi^+ \to \varphi_1, \varphi_2$

$\mathcal{C}\varphi_1\mathcal{C}^{-1} = \varphi_1$

$\mathcal{C}\varphi_2\mathcal{C}^{-1} = -\varphi_2$

$\mathcal{C}\varphi_3\mathcal{C}^{-1} = \varphi_3$



(1,3)平面の鏡映

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Dirac field spin 1/2

$\mathcal{L} = \bar{\psi}(i\partial\!\!\!/ - m)\psi$ $\quad \partial\!\!\!/ = \partial_\mu \gamma^\mu$

Dirac matrices

$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ Clifford algebra $\quad \vec{\gamma} = \beta\vec{\alpha}$ $\quad \beta = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$ $\quad \vec{\alpha} = \begin{pmatrix} 0 & \vec{\sigma} \\ \vec{\sigma} & 0 \end{pmatrix}$

$\gamma^0 = \beta$ $\quad \gamma^5 = \gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

ex) γ^μ (Weyl) $= P\gamma^\mu(D-P)P^{-1}$ $\quad \beta$ を diagonal $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

Pを求めよ

Dirac-Pauli 表現

例えば $\gamma^0 = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$, $\gamma^5 = \begin{pmatrix} & 1 \\ 1 & \end{pmatrix}$ Weyl 表現

Majorana 表現 $\gamma^0 = \begin{pmatrix} 0 & \sigma_2 \\ \sigma_2 & 0 \end{pmatrix}$.

運動方程式

$[\mathcal{L}]_{\bar{\psi}} = (i\partial\!\!\!/ - m)\psi = 0$

$[\mathcal{L}]_{\psi} = \bar{\psi}(-i\overleftarrow{\partial\!\!\!/} - m) = 0$ $\quad A\overleftarrow{\partial} = -(\partial A)$

● $\psi, \psi^\dagger$ ではなくて $\psi, \bar{\psi}$ をとる理由？ → P15 頭

$\bar{\psi} = \psi^\dagger\gamma^0$

$\mathcal{H} = \pi\dot{\psi} - \mathcal{L}$

$= i\bar{\psi}\gamma^0\dot{\psi} - \bar{\psi}(i\partial\!\!\!/ - m)\psi = \psi^\dagger(-i\vec{\alpha}\nabla + \beta m)\psi$

$H = \int d^3x\, \mathcal{H}$

$\vec{P} = \int d^3x\,(-\pi\vec{\nabla}\psi) = \int d^3x\, \psi^\dagger(-i\nabla)\psi$

Lorentz transformation の表現

(mass less spin ↑↓ が decouple → Weyl)

$\Lambda \quad D(\Lambda)$

$\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$ $\quad \sigma^{ij} = \begin{pmatrix} \sigma^k & 0 \\ 0 & \sigma^k \end{pmatrix}$ (ijk) cyclic

$\sigma^{0i} = i\alpha^i = i\begin{pmatrix} 0 & \sigma^i \\ \sigma^i & 0 \end{pmatrix}$

$[\sigma^{\mu\nu}, \gamma^\sigma]$

$[\sigma^{\mu\nu}, \sigma^{\rho\sigma}]$

$\gamma^\mu\gamma^\nu\gamma^\sigma = -\gamma^\mu\gamma^\sigma\gamma^\nu + 2\eta^{\nu\sigma}\gamma^\mu = \gamma^\sigma\gamma^\mu\gamma^\nu + 2\eta^{\nu\sigma}\gamma^\mu - 2\eta^{\mu\sigma}\gamma^\nu$

$[\sigma^{\mu\nu}, \gamma^\sigma] = 2i(\eta^{\nu\sigma}\gamma^\mu - \eta^{\mu\sigma}\gamma^\nu)$

$[\sigma^{\mu\nu}, \sigma^{\rho\sigma}] = 2i\{\eta^{\nu\sigma}\sigma^{\mu\rho} + \eta^{\mu\rho}\sigma^{\nu\sigma} - \eta^{\mu\sigma}\sigma^{\nu\rho} - \eta^{\nu\rho}\sigma^{\mu\sigma}\} = 2i\,C^{\mu\nu\rho\sigma}{}_{\alpha\beta}\,\sigma^{\alpha\beta}$

$\frac{\sigma^{\mu\nu}}{2} = j^{\mu\nu}$ 同一 $\begin{cases} [J^{\mu\nu}, P^\sigma] & [j^{\mu\nu}, j^{\rho\sigma}] = iC^{\mu\nu\rho\sigma}{}_{\alpha\beta}\, j^{\alpha\beta} \\ [j^{\mu\nu}, \gamma^\sigma] & [J^{\mu\nu}, J^{\rho\sigma}] = iC^{\mu\nu\rho\sigma}{}_{\alpha\beta}\, J^{\alpha\beta} \end{cases}$

γ^σ: vector

$\Lambda \cdot D(\Lambda) = 1 - \frac{i}{2}\omega_{\mu\nu}J^{\mu\nu}$

$D(\Lambda)\gamma^\mu D^{-1}(\Lambda) = \Lambda_\rho{}^\mu\gamma^\rho$ vector

$D(\Lambda)j^{\mu\nu}D^{-1}(\Lambda) = \Lambda_\rho{}^\mu\Lambda_\sigma{}^\nu j^{\rho\sigma}$ tensor

$D(\Lambda)\,1\,D^{-1}(\Lambda) = 1$ scalar

$\sigma^{ij\,\dagger} = \sigma^{ij}$ SO(3,1)

$\sigma^{0i\,\dagger} = -\sigma^{i0}$ SO(3) – compact 回転 有限

non-compact

↑ unitary ってなんだ！

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$\sigma^{\mu\nu\,+} \neq \sigma^{\mu\nu}$ $\qquad \psi' = D\psi \qquad \psi^+\psi' \neq \psi\psi'$

$\sigma^{\mu\nu\,+}\beta = \beta\sigma^{\mu\nu}$ $\qquad \psi' = D\psi$

$D(\Lambda)^+\beta = \beta D(\Lambda)^{-1} \qquad \bar\psi'\psi = \psi'^+\beta\psi'$

$= \psi^+ D^+\beta D\psi = \bar\psi\psi$ quasi-unitary

Dirac 方程式の covariance

$(i\gamma^\mu\partial_\mu - m)\psi(x) = 0 \qquad \partial_\mu = \frac{\partial}{\partial x^\mu} = \frac{\partial x'^\nu}{\partial x^\mu}\frac{\partial}{\partial x'^\nu} = \Lambda^\nu{}_\mu\,\partial'_\nu$

$\psi'(x') = D(\Lambda)\psi(x)$

$(i\underbrace{D\gamma^\mu D^{-1}}_{\Lambda_\rho{}^\mu\gamma^\rho}\partial_\mu - m)\underbrace{D\psi}_{\psi'} = 0$

$\underbrace{i\gamma^\rho\Lambda_\rho{}^\mu(\Lambda^{-1})_\mu{}^\nu}_{\delta_\rho^\nu} \qquad (i\gamma\partial' - m)\psi' = 0$

Fourier 展開 $\psi(x) = \sum_{s=\pm}\int d^3p\; a(p,s)\,U_{ps}(x) + b^+(ps)\,V_{ps}(x)$

$\sim e^{-ipx} \qquad \sim e^{ipx}$

$\psi^+(x) = \Sigma\int\; a^+\; U^*\; b\; V^*$

$\bar\psi(x) = \Sigma\int\; a^+\; \bar U\; b\; \bar V$

$(i\partial\!\!\!/ - m)U = 0 \qquad (p\!\!\!/ - m)U = 0 \qquad \bar U(p\!\!\!/ - m) = 0$

$V \qquad (-p\!\!\!/ - m)V = 0 \qquad \bar V(-p\!\!\!/ - m) = 0$

Dirac eq の解

$\begin{cases}(p\!\!\!/ - m)\,u(p,s) = 0 & \bar u(p\!\!\!/ - m) = 0\\ (p\!\!\!/ + m)\,v(p,s) = 0 & \bar v(p\!\!\!/ - m) = 0\end{cases}$

$u(p,s) = \sqrt{\frac{E+m}{2m}}\begin{pmatrix}\chi\\ \frac{\vec\sigma\vec p}{E+m}\chi\end{pmatrix}$ $\quad \chi = \begin{pmatrix}1\\0\end{pmatrix}\uparrow \quad \chi = \begin{pmatrix}0\\1\end{pmatrix}\downarrow$

$v(p,s) = \sqrt{\quad}\begin{pmatrix}\frac{\vec\sigma\vec p}{E+m}\chi\\ \chi\end{pmatrix}$ $\quad \chi = \begin{pmatrix}1\\0\end{pmatrix}\downarrow \quad \chi = \begin{pmatrix}0\\1\end{pmatrix}\uparrow$

$\bar u(p,s)\,u(p,s') = \delta_{ss'}$

$\bar v(p,s)\,v(p,s') = -\delta_{ss'}$

$\sum_s u(ps)\bar u(p,s) = \frac{p\!\!\!/ + m}{2m} \qquad \sum_s\left[u(p,s)\bar u(p,s) - v(p,s)\bar v(p,s)\right] = 1$

$\sum_s v(p,s)\bar v(p,s) = \frac{p\!\!\!/ - m}{2m}$ 完全系

canonical quantisation $\quad \{\psi(x,t),\, i\psi^+(x',t)\} = i\delta^3(x-x')$

$\{a(p,s)\,a^+(p's')\} = \delta^3(p-p')\,\delta_{ss'}$

$\{b \quad b^+\} = \,''$

$U_{ps}(x) = \frac{1}{(2\pi)^{3/2}}\sqrt{\frac{m}{E_p}}\,u(p,s)\,e^{-ipx}$

$V_{ps}(x) = \,''\qquad''\qquad e^{+ipx}$

$H = \int d^3x\,\psi^+\underbrace{(-i\alpha\nabla + \beta m)}_{i\frac{\partial}{\partial t}}\psi = \int d^3x\,\psi^+(i\tfrac{\partial}{\partial t})\psi = \int d^3x\,\frac{1}{(2\pi)^3}\int d^3p\,d^3p'\sqrt{\frac{m}{E_p}\frac{m}{E_{p'}}}\,E_{p'}\,(a^+(\quad)e^{ipx} + b(\quad)e^{-ipx})$

$\times(a$

$\times\left(a^+(ps)\,\bar u(ps)\gamma_0 e^{ipx} + b(ps)\,\bar v(p,s)\gamma_0 e^{-ipx}\right)$

$\times\left(a(ps')\,u(p's')\,e^{-ipx} + b^+(p's')\,v(p',s')\,e^{ipx}\right)$

$= \sum_{s=\pm}\int d^3p\;E_p\left[a^+(p,s)\,a(p,s) - b(p,s)\,b^+(p,s)\right]$

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$\vec{P} = \sum_{\pm s} \int d^3p\ \vec{p}\ [a^+a - bb^+]$
$+ b^+b - \{b, b^+\}$

Normal ordering　creation op が annihilation op の必ず左にくるように

$:i\psi^+\psi:$　☺ すべてのエネルギー vacuum に relative

$M^{\mu\rho\sigma} = (-i)\,i\bar{\psi}\gamma^\mu \frac{1}{2}\sigma^{\rho\sigma}\psi - (T^{\mu\rho}x^\sigma - T^{\mu\sigma}x^\rho)$

$M^{0\rho\sigma} = \psi^+ \frac{\sigma^{\rho\sigma}}{2}\psi - (\underbrace{T^{0\rho}}_{p^\rho}x^\sigma - \underbrace{T^{0\sigma}}_{p^\sigma}x^\rho)$

$J^{\rho\sigma} = \int d^3x\, M^{0\rho\sigma} = \int d^3x \left[(x^\rho p^\sigma - x^\sigma p^\sigma) + \psi^+ \frac{\sigma^{\rho\sigma}}{2} \psi \right]$
orbital　　spin　spin 1/2

ex) $J_3 = J^{12}$

$J_3\, a^+(p,\pm)|0\rangle = \pm\frac{1}{2} a^+(p,\pm)|0\rangle$ を $p=(m,\vec{0})$ rest frame について示せ

D May 19

$\{\psi_\alpha(x,t), \psi_\beta^+(x',t')\}$
$= \int \frac{d^3p}{(2\pi)^3}\sqrt{\frac{m^2}{E_p E_{p'}}} \{ (a(p,s)u_\alpha(p,s)e^{-ipx} + b^+(p,s)v_\alpha(p,s)e^{ipx}),$
$\{a, a^+\} = \delta_{ss'}\delta^3_{(p-p')}$　$(a^+(p',s')(\bar{u}(p',s')\gamma_0)_\beta e^{ip'x'} + b(p',s')(\bar{v}(p',s')\gamma_0)_\beta e^{-ipx}) \}$

$= \sum_s \int \frac{d^3p}{(2\pi)^3}\frac{m}{E_p} \left[\underbrace{u_\alpha(p,s)(\bar{u}(p,s)\gamma_0)_\beta}_{\left(\frac{\not{p}+m}{2m}\gamma_0\right)_{\alpha\beta}} e^{-ip(x-x')} + \underbrace{v_\alpha(p,s)(\bar{v}(p,s)\gamma_0)_\beta}_{\left(\frac{\not{p}-m}{2m}\right)_{\alpha\beta}} e^{ip(x-x')} \right]$

$= \int \frac{d^3p}{(2\pi)^3}\frac{1}{2E_p}(i\not{\partial}+m)\underbrace{\gamma_0}\,[e^{-ip(x-x')} - e^{ip(x-x')}]$

$= (i\not{\partial}+m)\gamma_0\, i\Delta(x-x')$

$\{\psi_\alpha(x)\bar{\psi}_\beta(x')\} = (i\not{\partial}+m)\,i\Delta(x-x') = -iS(x-x')$　　$S(x-x') = -(i\not{\partial}+m)_{\alpha\beta}\Delta(x-x')_{\alpha\beta}$

$\langle 0|[\psi_\alpha(x), \bar{\psi}_\beta(x')]|0\rangle = (i\not{\partial}+m)\Delta_1(x-x')$
$= -S_1(x-x')_{\alpha\beta}$　　$S_1(x-x') = -(i\not{\partial}+m)\Delta_1(x-x')$

Spin と統計
comutator で量子化　$[\psi(x), \bar{\psi}(x')] = (i\not{\partial}+m)\Delta_1(x-x')$

$\bar{\psi}\psi$
$[\Theta(x), \Theta(y)] = \sum \bar{\psi}\psi\,[\underbrace{\bar{\psi}\psi}_{\Delta_1}] \neq 0 \quad (x-y)^2 < 0$　micro causality を破る

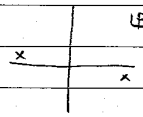
$\langle 0|T(\psi_\alpha(x), \bar{\psi}_\beta(y))|0\rangle = iS_F(x-y)_{\alpha\beta}$

$= \theta(x^0-y^0)\langle 0|\psi_\alpha(x)\bar{\psi}_\beta(y)|0\rangle \pm \theta(y^0-x^0)\langle 0|\bar{\psi}_\beta(y)\psi_\alpha(x)|0\rangle$

$= \int \frac{1}{(2\pi)^3}\frac{m}{E_p} d^3p\ \theta(x^0-y^0)\left(\frac{\not{p}+m}{2m}\right)_{\alpha\beta} e^{-ip(x-y)} \pm \theta(y^0-x^0)\left(\frac{\not{p}-m}{m}\right)_{\alpha\beta} e^{ip(x-y)}$

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$= (i\not{\partial}+m)_{\alpha\beta}\int \frac{1}{(2\pi)^3}\frac{1}{2E_p}\left[\theta(x^0-y^0)\,e^{-ip(x-y)} \mp \theta(y^0-x^0)\,e^{ip(x-y)}\right]$



$= i\int \frac{d^4p}{(2\pi)} \frac{(\not{p}+m)_{\alpha\beta}}{p^2-m^2+i\epsilon} e^{-ip(x-y)}$

$S_F(x-y) = \int \frac{d^4p}{(2\pi)^4} \frac{\not{p}+m}{p^2-m^2+i\epsilon} e^{-ip(x-y)}$

Noether current

$\begin{cases} \psi \to e^{i\lambda}\psi \\ \bar\psi \to i\bar\psi e^{-i\lambda} \end{cases}$ $\quad \partial_\mu(\bar\psi\gamma^\mu\psi)=0$ $\quad J^\mu = \bar\psi\gamma^\mu\psi$ $\quad Q = \int d^3x\, J^0 = \int d^3x\, \psi^\dagger\psi = \sum_s \int d^3p \left[a^\dagger(ps)a(ps) + b(ps)b^\dagger(ps)\right]$

e electronic current $\quad :: \quad a^\dagger a - b^\dagger b$

lepton current

charge conjugation

$\psi(x) = \int \frac{d^3p}{(2\pi)^{3/2}}\sqrt{\frac{m}{E_p}}\left(a(ps)u(ps)e^{-ipx} + b^\dagger(ps)v(ps)e^{+ipx}\right)$

$\psi^c = C\psi C^{-1}$ $\quad \underbrace{Ca\,C^{-1}}_{\eta^* b} u\, e^{-ipx} \quad \underbrace{Cb^\dagger C^{-1}}_{\eta^* a^\dagger} v\, e^{+ipx}$ $\quad \psi^c = C\psi e^{-1} = C\bar\psi^T$

$\bar\psi(x) = \int \cdots \left(a^\dagger(ps)\bar u(ps)e^{ipx} + b(ps)\bar v(ps)e^{-ipx}\right)$ $\quad \eta^* u = C\bar v^T$ $\quad \eta^* v = C\bar u^T$

$= C\bar\psi^T$ $\quad a^\dagger(ps)C\bar u^T(ps)e^{ipx} + b(ps)C\bar v^T(ps)e^{-ipx}$

$\begin{cases} Ca\,e^{-1} = \eta^* b \\ Ca^\dagger e^{-1} = \eta b^\dagger \\ Cb^\dagger e^{-1} = \eta^* a^\dagger \\ Cb\,e^{-1} = \eta a \end{cases}$

C を決める $\quad (i\not{\partial}-m)\psi = 0$

$Ce^{-1}(i\not{\partial}-m)C\bar\psi^T = 0$

$C^{-1}(iC^{-1}\gamma^\mu C\partial_\mu - m)\bar\psi^T = 0$ $\quad C^{-1}\gamma^\mu C = -(\gamma^\mu)^T$

$\bar\psi(-i\gamma^\mu\overleftarrow{\partial}_\mu - m) = 0$ $\quad C(\gamma^\mu)^T C^{-1} = -\gamma^\mu$

$(-i\gamma^T\partial - m)\bar\psi^T = 0$

$C^T = -C$ を示す $\quad C^{-1}\gamma^{\mu T}(C^T)^{-1} = -\gamma^\mu$

$C^T C^{-1}\gamma^\mu C(C^T)^{-1} = \gamma^\mu$ $\quad$ 全ての Dirac 行列と交換

$C^T C^{-1}\gamma^\mu (C^T C^{-1})^T = \gamma^\mu$ $\quad C^T C^{-1} \propto 1$

$\forall\gamma^\mu \quad [C^T C^{-1}, \gamma^\mu] = 0$

Dirac-Pauli

$(\gamma^0)^T = \gamma^0 \quad (\gamma^2)^T = \gamma^0$ $\quad \gamma^\mu C = -C\gamma^{\mu T}$

$(\gamma^1)^T = -\gamma^1 \quad (\gamma^3)^T = -\gamma^3$ $\quad \gamma^0\gamma^2$ と反可換 $\quad C = i\gamma^2\gamma^0 = \begin{pmatrix} 0 & & & -1 \\ & & 1 & \\ & -1 & & \\ 1 & & & 0 \end{pmatrix}$

$\gamma^1\gamma^3$ 可換

$C = -C^T = -C^{-1} = -C^\dagger$

(表示による)

$C^T C^{-1} = -1$ (ようない)

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$C\psi_\alpha C^{-1} = C_{\alpha\beta}\bar{\psi}_\beta$

$\underbrace{CC}_{1}\psi_\alpha\underbrace{C^{-1}C^{-1}} = C_{\alpha\beta}C\bar{\psi}_\beta C^{-1}$ $\quad C\bar{\psi}_\beta C^{-1}$ $\quad C^2 = -1$

$C\bar{\psi}_\beta C^{-1} = (C^{-1})_{\beta\alpha}\psi_\alpha$ $\quad C\bar{\psi}_\alpha C^{-1} = -\psi_\beta C^{-1}_{\beta\alpha}$

$= -(C^{-1})^T_{\beta\alpha}\psi_\alpha$

$= -C^{-1}_{\alpha\beta}\psi_\alpha$

$= -\psi_\alpha C^{-1}_{\alpha\beta}$

$Cu_1 = -\bar{v}_1^T$ $\quad u_1 = \begin{pmatrix} \chi \\ \frac{\sigma p}{E+m}\chi \end{pmatrix}$

$u_1 = +C\bar{v}_1^T$
$u_2 = -C\bar{v}_2^T$
$v_1 = C\bar{u}_1^T$
$v_2 = -C\bar{u}_2^T$

ここで v_2 に $-$ sign をつけて再定義 $\rightarrow \eta = \eta^* = 1$ ととるとよい

$\Psi_{e^+e^-} = \int d^3p\, d^3p'\; a^\dagger(p,s)\, b^\dagger(p's')\, \Psi(ps\,p's')$

$C\Psi_{e^+e^-}C^{-1} = b^\dagger(ps)\, a^\dagger(p's')\,\Psi(\cdots)$
$\quad - a^\dagger(ps)\, b^\dagger(p's')\,\Psi(p's'\,ps)$

もし $\Psi(ps\,p's') = \Psi(p's'\,ps)$

$C\Psi_{e^+e^-}C^{-1} = -\Psi_{e^+e^-}$ intrinsic C parity は odd

	C	decay
$^3S_1$ spin sym, orbit sym	$-$	3γ
$^{2S+1}L_J$		
$^1S_0$ spin antisym, orbit sym	$+$	2γ

$C = (-1)^{L+S}$

π^0 $C = +$

ρ^0 $C = -$

$j^\mu = \bar{\psi}\gamma^\mu\psi$

$:j^\mu: = [\bar{\psi}, \gamma^\mu\psi]$

$C[\bar{\psi}_\alpha(\gamma^\mu)_{\alpha\beta}\psi_\beta]C^{-1} = (\gamma^\mu)_{\alpha\beta}[C\bar{\psi}_\alpha C^{-1}, C\psi_\beta C^{-1}] = (C^{-1}\gamma^\mu C)_{\rho\sigma}[\bar{\psi}_\sigma\psi_\rho] = (-)[\bar{\psi}_\sigma(\gamma^\mu)_{\sigma\rho}\psi_\rho]$

$[-\psi_\rho(C^{-1})_{\rho\alpha}\cdot C_{\alpha\sigma}\bar{\psi}_\sigma]$ $\quad \overline{(\gamma^{\mu T})_{\rho\sigma}}$

ex)

$C^{-1}[\bar{\psi}, \theta\psi]C = [\bar{\psi}, \theta'\psi]$

P	1	γ^μ	$\sigma^{\mu\nu}$	γ^5	$\gamma^5\gamma^\mu$
C	+	$-$	$-$	+	+

$\begin{pmatrix} +1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}$ $\quad \det\Lambda = -1$ $\quad$ proper $\rightarrow$ improper

Parity $\in$ Lorentz 群 $\quad P\gamma^\mu P^{-1} = \Lambda_\nu{}^\mu(P)\gamma^\nu$ $\quad P = \gamma^0 = \beta$

$P\psi(x)P^{-1} = P\psi(\bar{x})$ $\quad \Lambda x = \bar{x} = (t, -\vec{x})$

$P\psi^\dagger(x)P^{-1} = \psi^\dagger(\bar{x})P^\dagger$

$P\bar{\psi}(x)P^{-1} = \bar{\psi}(\bar{x})P$

$P\,a(ps)\,P^{-1} = c\,(-p,s)$

$P\,b^\dagger(ps)P^{-1} = -b^\dagger(-p,s)$

$P\,a^\dagger(ps)P^{-1} = a^\dagger(-p,s)$

$P\,b(ps)P^{-1} = -b(-p,s)$

$\psi\psi^\dagger$ の relative parity は $\ominus$

$\psi(x) = a(ps)\,u(ps)\,e^{-ipx} + b^\dagger(ps)\,v(ps)\,e^{ipx}$

PP^{-1} $\quad Pa(ps)P^{-1}u(ps)e^{-ipx} + Pb^\dagger P^{-1}v(ps)e^{ipx}$

$P\psi(\bar{x}) = a(ps)\underset{\beta}{P}u(ps)e^{-ip\bar{x}} + b^\dagger(ps)\underset{\beta}{P}v(ps)e^{ipx}$

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Time reversal (∈ Lorentz 群)

$\Lambda^0{}_0 > 0 \rightarrow \Lambda^0{}_0 < 0$ $\quad J i P^\mu J^{-1} = i \Lambda_\nu{}^\mu P^\nu \quad \Lambda_\nu{}^\mu = (-1, 1, 1, 1)$

$\mu = 0 \quad J H J^{-1} = -H \quad H \geq 0$

$J i P^\mu J^{-1} = -i J P^\mu J^{-1} \rightarrow J H J^{-1} = H$

J : anti linear, anti unitary

$$\begin{cases} J(\xi\Phi + \eta\Psi) = \xi^* J\Phi + \eta^* J\Psi \\ (J\Phi, J\Psi) = (\Phi, \Psi)^* \end{cases}$$

$i\frac{\partial \varphi}{\partial t} = [\varphi, H]$

$i\frac{\partial \varphi}{\partial(-t)} = -[\varphi, H]$

$H \rightarrow -H$ or $i \rightarrow -i, \varphi \rightarrow \varphi^* \quad \varphi^*(t) = \varphi(-t)$

$J\psi J^{-1} = T\psi(-t, \vec{x})$ $\quad (i\gamma^\mu \partial_\mu - m)\psi = 0$

$C C^{-1}$ $\quad (i\gamma^\mu \partial_\mu - m) C \bar{\psi}^T$ $\quad \gamma_0\gamma_5 C(\psi^\dagger \gamma^{0T})$

γ^5 $\quad (-i\gamma^\mu \partial_\mu - m)\gamma_5 C \bar{\psi}^T$

γ^0 $\quad (-i\gamma^0\partial_0 + i\gamma^i \partial_i - m)\underbrace{\gamma_0\gamma_5 C \bar{\psi}^T}_{\psi(\vec{x}, -t)} = 0$ $\quad T = -i\gamma_5 C = i\gamma^1\gamma^3 = T^\dagger = T^{-1}$

$$\begin{cases} -i\gamma_5 C u_1(\vec{p}) = -i u_2^*(-\vec{p}) & -i\gamma_5 C v_1(\vec{p}) = +i v_2^*(-\vec{p}) \\ -i\gamma_5 C u_2(\vec{p}) = +i u_1^*(-\vec{p}) & -i\gamma_5 C v_2(\vec{p}) = -i v_1^*(-\vec{p}) \end{cases}$$

$$\begin{cases} T u(p,s) = \xi\, u(-p,-s) \\ T v(p,s) = \xi^* v(-p,-s) \end{cases} \qquad \begin{cases} J a(ps) J^{-1} = \xi\, a(-p,-s) \\ J b(ps) J^{-1} = \xi^C b(-p,-s) \end{cases}$$

- **Vector field** (cf photon field)

(1) $(\Box + m^2) A^\mu = 0 \rightarrow$ spin $0, 1$

(2) $\partial_\mu A^\mu = 0$ 既約条件 (Lorentz 条件)

↳ Scalar field の除外

$\mathcal{L} = \frac{1}{2}\left(-\frac{\partial A^\mu}{\partial x^\nu}\frac{\partial A_\mu}{\partial x_\nu} + m^2 A^\mu A_\mu\right)$

$[\mathcal{L}]_{A^\mu} = m^2 A_\mu + \partial_\nu(\partial^\nu A_\mu) = 0 \qquad (\Box + m^2) A_\mu^\mu = 0 \rightarrow \partial_\mu A^\mu = 0$ が出ない… negative energy

$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} m^2 A^\mu A_\mu = -\frac{1}{2}(\partial_\mu A_\nu - \partial_\nu A_\mu)\frac{\partial A^\nu}{\partial x_\mu} + \frac{1}{2} m^2 A^\mu A_\mu$

$(F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$

$[\mathcal{L}]_{A^\nu} = m^2 A_\nu + \partial^\mu F_{\mu\nu} = \partial^\mu(\partial_\mu A_\nu - \partial_\nu A_\mu) + m^2 A_\nu = \Box A_\mu - \partial_\nu \partial^\mu A_\mu + m^2 A_\nu = 0$

$\partial^\mu F_{\mu\nu} = -m^2 A_\nu$

$\partial^\nu \partial^\mu F_{\mu\nu} = -m^2 \partial^\nu A_\nu$ $\quad (\underbrace{\partial^\nu \partial^\mu F_{\mu\nu}}_{0} = 0)$

$\begin{cases} \partial_\mu A^\mu = 0 & \text{if } m^2 \neq 0 \\ (\Box + m^2) A^\mu = 0 \end{cases}$

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A^ν の conjugate $\pi_\nu = \frac{\partial \mathcal{L}}{\partial \dot{A}^\nu} = -F_{\nu 0} = -F^{\nu 0}$

$\pi_0 = 0$ 第0成分の comited relation は決定不可能

$A_0 = -\frac{1}{m^2}\partial_\mu F^{\mu 0} = +\frac{1}{m^2}\nabla\vec{\pi}$ manifest Lorentz coveriance は壊れている

$\pi_k = -\left(\frac{\partial A_k}{\partial x^0} - \frac{\partial A_0}{\partial x^k}\right) = \partial_k A_0 + \dot{A}^k_k \quad \rightarrow \quad \dot{A}^k_k = \pi_k - \partial_k A_0$

$\mathcal{H} = \pi_\nu A^\nu - \mathcal{L} = \pi_k(\pi_k - \partial_k A_0) + \frac{1}{4}F_{ij}F^{ij} - \frac{1}{2}\pi_k^2 - \frac{m^2}{2}(A^0)^2 + \frac{m^2}{2}(A^k)^2$

$= \frac{1}{2}\pi_k^2 + \frac{1}{4}(F_{ij})^2 - \frac{1}{2}\frac{1}{m^2}(\nabla\vec{\pi})^2 + \frac{m^2}{2}(A^k)^2 - \partial_k(\pi_k A_0) + (\partial_k \pi_k)A_0$

$= \frac{1}{2}\vec{\pi}^2 + \frac{1}{2}(\nabla\times A)^2 + \frac{1}{2}m^2\vec{A}^2 + \frac{1}{2}\frac{1}{m^2}(\nabla\vec{\pi})^2 +$ 表面　　$\mathcal{H}$ は $A^1 \cdots A^3, \pi^1 \cdots \pi^3$ でかけてる

positive definite しかし manifest Lorentz cov

canonical quantisation $[A_j(x,t)\ \pi_k(x',t)] = i\delta_{jk}\delta^3(x-x')$

$[\dot{A}_j(x,t), A_k(x,k)] = [\pi_j - \frac{1}{m^2}\partial_j(\nabla\pi), A_k] = -i\left(\delta_{jk} - \frac{\partial_j\partial_k}{m^2}\right)\delta^3(x-x')$

$[A_j(x), A_k(y)] = \left(\delta_{jk} - \frac{\partial_j\partial_k}{m^2}\right) i\Delta(x-y)$

↓

$[A_\mu(x), A_\nu(y)] = \left(-\eta_{\mu\nu} - \frac{\partial_\mu\partial_\nu}{m^2}\right) i\Delta(x-y)$

$\mu,\nu = 0$ でも成り立つ

exercise)

① $[A_\mu A_\nu] = -i\Delta(x-y)$ を $\mu=\nu=0$ に拡張できることを示せ

② $(\Box + m^2)A_0 = 0$

③ $\partial_\mu A^\mu = 0$

formal に Lorentz coveriance が保存する

Fourier 展開

$A^\mu = \int \frac{d^3p}{(2\pi)^3}\frac{1}{2p^0}\left(a(p\sigma)\varepsilon(p,\sigma)e^{-ipx} + a^+(p,\sigma)\varepsilon^*(p,\sigma)e^{ipx}\right)$　　$D(\Lambda)^+ A^\nu(x) = A'^\mu(x')$

$\Lambda^\mu{}_\nu A^\nu(x) = A'^\mu(x')$

rest frame

$\varepsilon^\mu(0,1) = \begin{pmatrix}0\\1\\0\\0\end{pmatrix}$　$\varepsilon^\mu(0,2) = \begin{pmatrix}0\\0\\1\\0\end{pmatrix}$　$\varepsilon^\mu(0,3) = \begin{pmatrix}0\\0\\0\\1\end{pmatrix}$　$k^\mu = \begin{pmatrix}m\\0\\0\\0\end{pmatrix}$

$\varepsilon^\mu(0,\sigma)\ k^\mu$　　$k_\mu\varepsilon^\mu = 0$ Lorentz 条件

$\varepsilon^\mu(p,\sigma)$

$\Lambda = \begin{pmatrix}\cosh\varphi & 0 & 0 & \sinh\varphi\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ \sinh\varphi & 0 & 0 & \cosh\varphi\end{pmatrix}$　$\cosh\varphi = E/m$　$\sinh\varphi = P/m$

$\varepsilon^\mu(p,3) = \begin{pmatrix}P/m\\0\\0\\E/m\end{pmatrix}$　$k^\mu = \begin{pmatrix}E\\0\\0\\P\end{pmatrix}$　$\varepsilon^\mu k_\mu = P/m\cdot E - pE/m = 0$

$\varepsilon^\mu(p,\sigma) = \begin{pmatrix}0\\1\\0\\0\end{pmatrix}$

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$\left(\frac{p}{m}\right)^2 = -1 + \frac{E^2}{m^2}$ $\mu = \nu = 3$ $\left(\frac{E}{m}\right)^2 = 1 + \frac{p^2}{m^2}$ $\mu = \nu = 1, 2$ 1

on shell off shell

$\sum_\sigma \varepsilon^\mu(p,\sigma)\,\varepsilon^{*\nu}(p,\sigma) = -\eta^{\mu\nu} + \frac{p^\mu p^\nu}{m^2} \qquad -\eta^{\mu\nu} - \frac{\partial^\mu \partial^\nu}{m^2}$

ex)

$J^{\rho\sigma} = \int M^{0\rho\sigma} d^3x$ で Λ^μ が spin = 1 であることを explicit に示せ

▷ May 26

Feynmann propagator

$-(\Box + m^2)\Delta_F = 1 \qquad \Delta_F = \frac{-1}{\Box + m^2} \to \frac{1}{(2\pi)^4}\int d^4p \frac{e^{-ipx}}{p^2 - m^2 + i\varepsilon}$

$(i\not\partial - m)\Delta_F = 1 \qquad S_F = \frac{i\not\partial + m}{-\Box + m^2}$

$[\mathcal{L}]_{\varphi^a} = c^{ab}\varphi_b \qquad c^{ab}\Delta_{F\,bc} = \delta^a_c \qquad \Delta^{\nu\lambda} = a\eta^{\nu\lambda} + b\partial^\nu\partial^\lambda$

$(\Box + m^2)\,a\,\delta^\lambda_\mu + b(\Box + m^2)\partial_\mu\partial^\lambda$

$[\mathcal{L}]_{A_\nu} = \underbrace{[(\Box + m^2)\eta_{\mu\nu} - \partial_\mu\partial_\nu]}_{c_{\mu\nu}} A^\nu = 0 \qquad -a\partial_\mu\partial^\lambda - b\Box\partial_\mu\partial^\lambda = \delta^\lambda_\mu$

$c_{\mu\nu}\,c_{\mu\nu}\Delta^{\nu\lambda} = \delta^\lambda_\mu \qquad a = \frac{1}{\Box + m^2} \qquad b = \frac{a}{m^2}$

$\Delta_F^{\nu\lambda} = \frac{\eta^{\nu\lambda} + \frac{1}{m^2}\partial^\nu\partial^\lambda}{\Box + m^2}$

電磁場 gauge 不変性

A_μ 量 $|A_\mu|^2$ = photon 数 古典 $\partial_\mu\Lambda$ $-ik_\mu\Lambda$ (Λ: c-number, decouple)

$A_\mu + \partial_\mu\lambda$ photon 自由度 2 量子 Λ field

A^μ $\mu = 0, \cdots, 3$ ← 4つ ↳ ghost field

Lorentz A^μ mix

$\partial_\mu A^\mu = 0$ が出ない

解決

1. Lorentz 共変性を破る。$\mu = 1, 2,$ → S 行列の Lorentz 不変性
2. unphysical な photon → negative prob, energy
3. massive $m \to 0$ 3成分 long mode decouple

massless Lorentz 不変性 ⇔ current 保存

massive renormalisibility

$F_{\mu\nu}$ ν / μ

0	E^1	E^2	E^3
$-E^1$	0	$-H^3$	H^2
$-E^2$	H^3	0	$-H^1$
$-E^3$	$-H^2$	H^1	0

古典論

$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \qquad A^\mu = (\vec{\Phi}, \vec{A}^\mu)$

$\vec{E} = -\nabla\vec{\Phi} - \dot{\vec{A}} \qquad E_1 = F_{01}$

$\vec{H} = \nabla \times \vec{A} \qquad H_3 = F_{21}$

$\mathcal{L} = \frac{1}{2}(\vec{E}^2 - \vec{H}^2)$ Heviside unit

$\frac{1}{8\pi}$ Gaussian unit

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Field eq $-\frac{\partial \mathcal{L}}{\partial[\partial_\mu A_\nu]} = +\partial_\mu F^{\mu\nu} = 0$ Maxwell

$\partial_\mu(\partial^\mu A^\nu - \partial^\nu A^\mu) = 0$ $\vec{H} = \nabla \times \vec{A}$ $\nabla \cdot \vec{H} = 0$

$\Box A^\nu - \partial_\mu \partial_\nu A^\mu = 0$ $\vec{E} = -\nabla \phi - \dot{A}$ $\nabla \times \vec{E} = -\frac{\partial H}{\partial t}$

$A^\mu \to A^\mu - \partial^\mu \Lambda$ として 残りの2本 $\Box A^\mu = j^\mu$ $\partial_\mu A^\mu = 0$

$\partial_\mu \partial_\nu A^\mu = 0$ Lorentz gauge Lorentz gauge 自由度 $\Box \Lambda = 0$

$\Box A^\nu = 0$

$\pi^\mu = \frac{\partial \mathcal{L}}{\partial \dot{A}_\mu} = -F^{0\mu} = E^\mu$

$\pi^0 = 0$ massive の時 A^0 は従属変数、A^i は他の項の従属変数にならない $[A^0, \pi^0] = ?$

A^0 が消去出来る時 → Coulomb gauge

1 Coulomb gauge による定式化

$\nabla \vec{E} = -\Delta \dot{A}^0 - \nabla \dot{A} = 0$

A^0 は c-number

$\nabla \dot{A}$ も 〃 (zero mode を除く)

$A_\mu \to A_\mu - \partial_\mu \Lambda$

$A_\mu' = A_\mu - \partial_\mu \int_0^t A^0(t', x)\, dt'$

$A_0' = 0$

$\nabla \vec{A}'' = \nabla \vec{A}' - \Delta \Lambda = 0$ となる様に Λ

$\Lambda = \frac{1}{4\pi} \int \frac{\nabla A}{|x' - x|} d^3x$

$\vec{A}(x) = \int \frac{1}{(2\pi)^{3/2}} \frac{1}{\sqrt{2k^0}} [\varepsilon(k\lambda) a(k\lambda) e^{-ikx} + \varepsilon^*(k\lambda) a^\dagger(k\lambda) e^{+ikx}]$

polar vector $\vec{\varepsilon}(k\lambda) \cdot \vec{k} = 0$ $\lambda = 1, 2$ $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ $\begin{cases} \nabla \cdot \vec{A} = 0 \\ A^0 = 0 \end{cases}$ transverse のみ 2成分

$\varepsilon(k\lambda) \varepsilon^*(k\lambda') = \delta_{\lambda\lambda'}$

completeness $\sum_\lambda \varepsilon^i(k\lambda) \varepsilon^j(k\lambda) = \delta^{ij} - \frac{k^i k^j}{k^2}$

canonical quantization $[a(k\lambda), a^\dagger(k'\lambda')] = \delta_{\lambda\lambda'} \delta^3(k - k')$

$H = \frac{1}{2} \int d^3x (E^2 + H^2) = \frac{1}{2} \int d^3k\, k^0 \{a(k\lambda), a^\dagger(k\lambda)\}$

Feynman propagator $iD_{F\mu\nu}(xy) = \langle 0 | T(A_\mu(x) A_\nu(y)) | 0 \rangle = \int \frac{d^4k}{(2\pi)^4} e^{-ikx} \sum \varepsilon_i(k\lambda) \varepsilon_j^*(k\lambda) \frac{1}{k^2 + i\epsilon}$

四次元化 $\eta^\nu = (1\,0\,0\,0)$ $k^\mu = (k\,0\,0\,k)$ $\epsilon^\mu (1\,0\,0\,1)$ $k^\mu \epsilon_\mu = 0$

$\hat{\epsilon}^\mu_{k^\mu} = \frac{k^\mu - (k\eta)\eta^\mu}{\sqrt{(k\eta) - k^2}}$ $(0\,0\,0\,k)$ をつくりたい

$\sum_{\lambda=0}^{3} \varepsilon^\mu(k\lambda) \varepsilon^{*\nu}(k\lambda) = -\eta^{\mu\nu} + \eta^\mu \eta^\nu - \hat{k}^\mu \hat{k}^\nu = -\eta^{\mu\nu} - \frac{k^2 \eta^\mu \eta^\nu - (k\eta)(k^\mu \eta^\nu + k^\nu \eta^\mu) + k^\mu k^\nu}{(k\eta)^2 - k^2}$

$\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ $\begin{pmatrix} -1 \\ +1 \\ +1 \\ +1 \end{pmatrix}$

$$D_{F\mu\nu}(x-y) = \int \frac{d^4x}{(2\pi)^4} \frac{e^{-ikx}}{k^2+i\epsilon}$$

$\partial_\mu j^\mu = 0$ $\partial_\mu D_F^{\mu\nu} = 0$ $D^{\mu\nu} \propto k^\mu$ に比例する部分は current 保存があれば落ちるはず

$$\int \frac{d^4k}{(2\pi)^4} e^{-ik(x-y)} \frac{k^2}{k^2+i\epsilon} \frac{1}{k^2} = -\frac{1}{4\pi} \frac{\delta(t-t')}{|x-x'|}$$ Coulomb と同型

↓ Coulomb と cancel

$$D_{\mu\nu F} = \int \frac{d^4k}{(2\pi)^4} e^{-ikx} \frac{-\eta_{\mu\nu}}{k^2+i\epsilon}$$

2) Fermi の方法

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{\alpha}{2} (\partial_\mu A^\mu)^2 \qquad \alpha = 1$$

cf. $\partial_\mu A^\mu = 0$

$$[\mathcal{L}]_{A_\nu} = \partial_\mu F^{\mu\nu} + \frac{1}{\alpha} \partial_\nu (\partial_\mu A^\mu) = \left(\Box \eta^{\mu\nu} - \left(1 - \frac{1}{\alpha}\right) \partial^\mu \partial^\nu\right) A_\mu = 0$$

$\pi_i = \frac{\partial \mathcal{L}}{\partial \dot{A}^i} = \dot{A}^i + \partial_i A^0 = -E^i$ 電場

$\pi_0 = \frac{\partial \mathcal{L}}{\partial \dot{A}^0} = -\frac{1}{\alpha} (\partial_\mu A^\mu)$

$[A^\mu, \pi_\nu] = i \delta^\mu{}_\nu \delta^3(x-x')$

exercise) $[A^\mu \pi_\nu] =$

$\alpha = 1$ Hamiltonian を計算

$\dot{\pi}_0 = i[H, \pi_0]$

$\dot{\pi}_i = i[H, \pi_i]$

$\dot{A}^0 = -\nabla A_0 - \pi_0$

$\dot{A}^i =$

$\ddot{A}^0 = i[H, \dot{A}^0]$

$\Box A^\mu = 0$ を示せ

$$\mathcal{L} = -\frac{1}{2} (\partial_\mu A_\nu - \partial_\nu A_\mu) \partial^\mu A^\nu - \frac{1}{2} (\partial_\mu A^\mu)(\partial_\nu A^\nu)$$

$$= -\frac{1}{2} (\partial_\mu A_\nu)(\partial^\mu A^\nu) + \frac{1}{2} \partial_\nu [A_\mu (\partial^\mu A^\nu) - A^\nu (\partial_\mu A^\mu)]$$

表面積分

- $\partial_\mu A^\mu = 0$ が出る
- negative energy

$\pi_\mu = \frac{\partial \mathcal{L}}{\partial \dot{A}^\mu} = -\partial_0 A_\mu$ $H = \pi_\mu \dot{A}^\mu - \mathcal{L} = -(\partial_0 A^0)^2 + (\partial_0 A^i)^2 - \frac{1}{2} (\partial_\mu A_\nu)(\partial^\mu A^\nu)$

$= \frac{1}{2} [\dot{A}_\mu \dot{A}^\mu - (\nabla A_\mu)(\nabla A^\mu)]$

$$A^\mu = \int \frac{d^3k}{(4\pi)^{3/2}} \frac{1}{\sqrt{2k^2}} \left(a(k\lambda) \varepsilon(k\lambda) e^{-ikx} + a^\dagger(k\lambda) \varepsilon^*(k\lambda) e^{ikx}\right)$$

$\varepsilon^\mu(k,1)$ $\varepsilon^\mu(k,2)$ $\vec{\varepsilon} \cdot \vec{k} = 0$

$\varepsilon^\mu(k,2) = \vec{k}/|k|$

$\varepsilon^\mu(k,0) = (1, \vec{0})$

orthogonality $\varepsilon^{*\mu}(k\lambda) \varepsilon_\mu(k\sigma) = -\eta_{\lambda\sigma}$

Completeness $\sum_{\lambda=0}^{3} \varepsilon^\mu(k\lambda) \varepsilon^\nu(k\lambda) = -\eta^{\mu\nu}$

$[A^\mu \pi_\nu] = i \delta^\mu{}_\nu \delta^3(x-x')$

$[a(k\lambda) \; a^\dagger(k'\lambda')] = \delta_{\lambda\lambda'} \delta^3(k-k')$

$$H = \frac{1}{2} \int d^3x \, k^0 \left(\sum_{\lambda=1}^{3} \{a(k\lambda) a^\dagger(a\lambda)\} - \{a(k0) a^\dagger(k0)\}\right)$$

$\vec{P} = \int \vec{p} \cdots$

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Lorentz condition　physical state を制限　$\partial_\mu A^\mu(x)|\Psi\rangle = 0$　supplementary condition

$A^\mu = \cdots \sum_{\lambda=0}^{3} \varepsilon^\mu(k\lambda)$

$\partial_\mu A^\mu = -ik_\mu \varepsilon^\mu(k3)\, a(k3)\, e^{-ikx} + ik_\mu \varepsilon^\mu(k0)\, a(k0)\, e^{-ikx}$ $\Big|\Psi\rangle = 0$

$[a(k3) + a^\dagger(k0)]|\Psi\rangle = 0$

$[a^\dagger(k2) + a(k0)]|\Psi\rangle = 0$

第3成分のみ　$|\Psi\rangle = |\Psi_\perp \times \Psi\rangle$　($\perp$: (1,2), Ψ: (3,0))

$|\Psi\rangle = \sum \alpha(n_3\, n_0)|n_3\, n_0\rangle$

$\alpha(n_3\, n_0)\left(\sqrt{n_3+1}\,|n_3+1, n_0\rangle + \sqrt{n_0}\,|n_3\ n_0-1\rangle\right) = 0$　(n_3, n_0)

$\alpha(n_3-1\ n_0)\sqrt{n_3} + \sqrt{n_0+1}\,\alpha(n_3\ n_0+1) = 0$

$$\begin{cases} \alpha(n_3\, n_0) = -\sqrt{\dfrac{n_0+1}{n_3+1}}\,\alpha(n_3+1,\, n_0+1) \\ \alpha(n_3\, n_0) = -\sqrt{\dfrac{n_3+1}{n_0+1}}\,\alpha(n_3+1,\, n_0+1) \end{cases}$$

$n_0 = n_3$ のみ値をもつ

$\alpha(n_3\, n_0) = (-1)^{n_0}\delta_{n_3 n_0}$

$|\Psi\rangle = |0.0\rangle - |11\rangle + |21\rangle - |33\rangle + \cdots$

norm $= \infty$

構造は完全に決まる

E は positive definite

$D_{\mu\nu} \propto -\eta_{\mu\nu}$

必要条件

$\forall x,\ \partial_\mu A^\mu(x)|0\rangle = 0$

$[\partial_\mu A^\mu(x)\ \partial_\nu A^\nu(y)]|0\rangle = 0$

$= \partial_\mu\partial_\nu [A^\mu A^\nu]$

$= -\eta^{\mu\nu} D(x-y)$

難点　$\langle\Psi|[\partial_\mu A^\mu(x)\ A^\nu(y)]|\Psi\rangle = 0$

$\partial_\mu D^{\mu\nu}(x-y) = \partial^\nu D(x-y) \neq 0$

$|\Psi\rangle$ が normalizable でなかった

(3) massive な formalism で $m \to 0$

$-\eta^{\mu\nu} + \dfrac{\partial^\mu\partial^\nu}{m^2}$

Stueckelberg の Lagrangian　$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \overbrace{\frac{k^2}{2}A_\mu A^\mu}^{\text{mass}} + A^\mu\partial_\mu B$　補助的な場 (kinetic term のない場)

gauge inv ×

$A_\mu \to A_\mu - \partial_\mu\Lambda \quad B \to B + k^2\Lambda$

$-k^2 A_\mu \partial^\mu\Lambda + \frac{k^2}{2}(\partial_\mu\Lambda)(\partial^\mu\Lambda) + A^\mu k^2 \partial_\mu\Lambda$　gauge 不変性回復

$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{k^2}{2}A_\mu A^\mu + A^\mu\partial_\mu B + \frac{\alpha}{2}B^2$

① $[\mathcal{L}]_{A^\mu} = \partial^\nu F_{\mu\nu} + \partial_\mu B + k^2 A_\mu$

② $[\mathcal{L}]_B = \alpha B - \partial_\mu A^\mu$ → $B = \frac{1}{\alpha}(\partial_\mu A^\mu)$　$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{k^2}{2}A_\mu A^\mu - \frac{1}{2\alpha}(\partial_\mu A^\mu)^2$

∂^μ① $\Box B + k^2\partial^\mu A_\mu = 0$

+② $(\Box + k^2\alpha)B = 0$

$[\mathcal{L}]_{A^\mu} = C_{\mu\nu}A^\nu$　$C_{\mu\nu} = (\Box + k^2)\eta_{\mu\nu} + (1/\alpha - 1)\partial_\mu\gamma_\nu$

$C_{\mu\nu}\Delta^{\nu\lambda} = \delta_\mu^\lambda$

$\Delta_F^{\mu\nu} = \dfrac{-\eta^{\mu\nu} + (1-\alpha)\frac{\partial^\mu\partial^\nu}{\Box + \alpha k^2}}{-\Box - k^2}$　$k \to 0$ smooth

$= \displaystyle\int \frac{d^4k}{(2\pi)^4} \frac{-\eta^{\mu\nu} + (1-\alpha)\frac{k^\mu k^\nu}{k^2 - \alpha k^2}}{k^2 - \kappa^2 + i\epsilon}$　smooth な $k \to 0$

IR behabive が regularise されている

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$\alpha = 1$ Feynman gauge $-\eta^{\mu\nu}$ α くりこみを受ける

$\alpha = 0$ Landau gauge $-\eta^{\mu\nu} + \frac{k^\mu k^\nu}{k^2}$ $\frac{1}{k^4}$ dipole ghost

Interacting Field

Heisenberg 描像 state $t = -\infty \sim \infty$ すべての情報
(picture) 特定の時刻で specify

$t \to -\infty$ で自由粒子の集合 $\begin{cases} |0\rangle_{in} \\ |p\rangle_{in} \\ |p\cdot q\rangle_{in} \\ \vdots \end{cases}$ $t \to +\infty$ で散乱後の状態の情報

$t \to +\infty$ で自由 $\begin{cases} |0\rangle_{out} \\ |p\rangle_{out} \\ |p\cdot q\rangle_{out} \end{cases}$

in, out 同じ Hilbert 空間の異なった表式

各々の orth. normal set $|\alpha\rangle_{out} = S^\dagger |\beta\rangle_{in}$ と書く ${}_{out}\langle\alpha| = {}_{in}\langle\beta| S$

${}_{out}\langle\beta|\alpha\rangle_{in} = {}_{in}\langle\beta| S |\alpha\rangle_{in} = S_{\alpha\beta}$ Lorentz transform $U(\Lambda) S U(\Lambda)^{-1} = S$

${}_{out}\langle\beta|\alpha\rangle_{out} = \delta_{\alpha\beta}$

$= {}_{in}\langle\beta| S S^\dagger |\alpha\rangle_{in}$ $SS^\dagger = 1$

$|p\rangle_{in} = \underbrace{a^\dagger_{in}(p)}_{\text{定義}} |0\rangle$ $\phi_{in}(x) = \underbrace{\int \cdots (a_{in}(p) e^{-ipx} + a^\dagger_{in}(p) e^{ipx})}_{\text{定義}}$

$S a_{out} S^\dagger |p\cdots\rangle_{in} = S a_{out} |p\cdots\rangle_{out} = S\sqrt{n}\,|\cdots\rangle_{out} = a_{in} |p\cdots\rangle_{in}$ $a_{in} = S a_{out} S^{-1}$

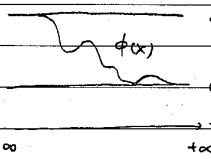
$\phi_{in} = S \phi_{out} S^{-1}$

$(\Box + m^2)\phi_{in}(x) = 0$

$(\Box + m^2)\phi_{out}(x) = 0$

0コ $|0\rangle_{in} = |0\rangle_{out} = |0\rangle$ } 安定

1コ $|p\rangle_{in} = |p\rangle_{out} = |p\rangle$ }



Interpoleting field $(\Box + m^2)\phi = j(x)$

$\phi(x) = U^\dagger(x_0)\phi_{in}(x) U(x_0)$

なる U がなるとする

$U(-\infty) = 1$, $U(+\infty) = S$

$\lim_{t\to+\infty} \phi(x) = \phi(x)_{out}$

$\lim_{t\to+\infty} \phi(x) = \phi(x)_{in}$

$\frac{\partial}{\partial t} UU^{-1} = 0$ $\dot{U}U^{-1} + U\frac{\partial}{\partial t}U^{-1} = 0$ $\frac{\partial}{\partial t}(U^{-1}) = -U^{-1}\dot{U}U^{-1}$

$\frac{\partial \varphi_{in}}{\partial t} = i[H_{in}(\varphi_{in}), \varphi_{in}]$ $\dot{\varphi}_{in} = \frac{\partial}{\partial t}(U\phi U^{-1}) = \dot{U}U^{-1}U\phi U^{-1} + U i[H(\varphi), \varphi]U^{-1} - \underbrace{U\phi U^{-1}}_{\phi_{in}} UU^{-1}$

$\frac{\partial \varphi}{\partial t} = i[H(\varphi), \varphi]$

$\dot{\varphi}_{in} = [\dot{U}U^{-1}, \phi_{in}] + i[H, \phi_{in}] = i[H_{in}(\varphi_{in}), \phi_{in}]$

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$\dot{U}U^{-1} + iH = iH_{in} + \text{const}$ $\quad U[H(\varphi), \varphi]U^{-1}$

$\dot{U} = -i(H - H_{in})U$ $\quad H(U\varphi U^{-1})$ $\underbrace{\varphi_{in}}$ $\underbrace{U\varphi U^{-1}}_{\varphi_{in}}$

$i\frac{\partial}{\partial t}U(x) = H'(\varphi_{in})U(x)$

$H' = H - H_{in}$

$U(t,t') = U(t)U^{-1}(t')$ initial coordinate を t' に set

$U(t,t) = 1$

$U(t,t')U(t't'') = U(tt'')$

$U(t,t') = U^{-1}(t't)$

$S = U(+\infty, -\infty)$ $\quad U(t,t) = 1 - i\int_{t'}^{t} dt_1 H'(t_1)$

$= 1 - i\int_{t'}^{t} dt_1 H'(t_1) + (-i)^2 \int_{t'}^{t} dt_1 \int_{t'}^{t_1} dt_2 H'(t_1)H(t_2)$

$+ \cdots + (-i)^k \int_{t'}^{t} dt_1 \int \cdots \int_{t'}^{t_{n-1}} dt_n \, H(t_1)\cdots H(t_{n-1})$

$$U(t,t') = \sum_{n=0}^{\infty} (-i)^n \frac{1}{n!} \int_{t'}^{t} dt_1 \cdots \int_{t'}^{t} dt_n \, P(H(t_1)\cdots H(t_n))$$

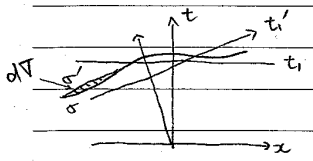
Dyson series

Feynmann graph ▷ Jun 2

1. S 行列の Lorentz 不変性 — integrability $[H_I(x), H_I(y)] = 0$, $(x_0-y_0)^2 - (\vec{x}-\vec{y})^2 < 0$
2. $H_I(x)$ の例
   - $H_I = eA_\mu^{in}(x)\bar{\psi}^{in}(x)\gamma^\mu\psi^{in}(x)$ QED
   - $eA_0(x)\psi^*(x)\psi(x) = eA_0(x)\rho(x)$
   - $H_I = g\bar{\psi}\psi\phi$ } Yukawa interaction ($J^P = 0^+$ for φ)
   - $ig\bar{\psi}\gamma^5\psi\phi$ } 〃 ($J^P = 0^-$ 〃)
3. Wick's theorem (T 積, P 積 → Normal 積)
4. Feynmann rules
   - 4.1. 外線
   - 4.2. 内線
   - 4.3. vertex
   - 4.4. その他
5. 微分 断面積
6. 例 { Compton 散乱 ($e + \gamma \to e' + \gamma'$) Klein-Nishina の公式
   $e^+e^- \to \mu^+\mu^-$

積分 $\int d\Gamma$

$\frac{\partial U}{\partial t} = -i\,\underbrace{H_I(t)U}$



$\frac{\delta U}{\delta\sigma(x)} = \frac{U(\sigma') - U(\sigma)}{d\Gamma} = -i\,\underbrace{\mathcal{H}_I(x)}_{\text{局所的}}U(\sigma)$ $\quad \frac{\delta^2 U}{\delta\sigma(x)\delta\sigma(y)} = \frac{\delta^2 U}{\delta\sigma(y)\delta\sigma(x)}$ integrability

$\sigma(x)$: 同時刻

super multiple time formalism

$(-i)^2 H_I(y)H_I(x)U = (-i)^2 H_I(x)H_I(y)U$

① $[H_I(y), H_I(x)] = 0$ for $(y-x)^2 < 0$

② $H_I(x)$: Lorentz scalar

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$\mathcal{L}^f = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi$ $\left\{\begin{array}{l}\psi \to e^{i\theta}\psi \\ \bar{\psi} \to \bar{\psi}e^{-i\theta}\end{array}\right.$ global gauge 変換

$\frac{1}{2}(E^2-H^2)$

$D_\mu = (\partial_\mu + ieA_\mu)$ 共変微分 covariant derivative

$\mathcal{L}^{total} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\gamma^\mu D_\mu - m)\psi$

$\hookrightarrow \mathcal{L}_I = -e\bar{\psi}\gamma^\mu\psi A_\mu$

$\mathcal{H} = \Sigma\dot{\varphi}\frac{\partial\mathcal{L}}{\partial\dot{\varphi}} - \mathcal{L} = \mathcal{H}^{free} - \mathcal{L}_I$

$\mathcal{H}_I$

3. Wick's theorem

$$S = \sum_{n=0}^{\infty}\frac{(-i)^n}{n!}\int d^4x_1 \cdots \int d^4x_n\, \mathcal{P}(\mathcal{H}_I(\varphi_{in}(x_1)) \cdots \mathcal{H}_I(\varphi_{in}(x_n)))$$

$S_{fi} = \langle \gamma', e', out \mid \gamma, e, in\rangle$

$= {}_{out}\langle 0 \mid a_{out}(e')\, a^-_{out}(\gamma')\; a^+_{in}(\gamma)\, a^+_{in}(e) \mid 0\rangle_{in}$

$a_{out} = S^+ a_{in} S$ $\quad {}_{in}\langle 0 \mid$

$= {}_{in}\langle 0 \mid S^+ (a_{out}(e')\, a_{out}(\gamma'))\, S\, a^+_{in}(\gamma)\, a^+_{in}(e) \mid 0\rangle_{in}$

$= \underline{\langle 0 \mid a_{in} \cdots a_{in} S a^+_{in} a^+_{in} \mid 0\rangle}$ ③

$\langle 0 \mid S \mid 0\rangle$ → 後述 (Feynman 図で、真空の図は落とせ)

$= \langle 0 \mid a_{in}(e')\, a_{in}(\gamma')\, \frac{(-i)^n}{n!}\int d^4x_1 \cdots \int d^4x_n\, T(eA_\mu(x_1)\bar{\psi}(x_1)\gamma^\mu\psi(x_1)$

(右辺へ

$\cdots eA_\mu(x_n)\bar{\psi}(x_n)\gamma^\mu\psi(x_n))$

$A^a + A^c$ $\quad e^{-i\varepsilon t}$ $\quad e^{i\varepsilon t}$ $\quad \times a^+_{in}(\gamma)\, a^+_{in}(e) \mid 0\rangle$

$V = (2\pi)^3$ $\quad a_{in} A^{in}_\mu(x_1) = [a, A_\mu(x_1)]$

$= f^*_\mu = \frac{1}{(2\pi)^{3/2}}\frac{1}{\sqrt{2\omega}}\epsilon_\mu e^{ik'x_1}$

$\psi_{in}\, a^+_{in} \equiv \{\psi_{in}(x), a^+_{in}\}$

$= \psi(p) = \frac{1}{(2\pi)^{3/2}}\sqrt{\frac{m}{E}}\, u e^{-ipx}$ $\quad A_\mu(x_2)\, a^+_{in} = [A_{in}, a^+_{in}] = f_\mu(x_2) = \frac{1}{(2\pi)^{3/2}}\frac{1}{\sqrt{2\omega}}\epsilon_\mu e^{-ikx_2}$

$a_{in}\bar{\psi} \equiv \{a_{in}, \bar{\psi}\} = \bar{\psi}(p') = \frac{1}{(2\pi)^{3/2}}\sqrt{\frac{m}{E'}}\,\bar{u}\, e^{ipx}$

$T\psi^\alpha(x_1)\bar{\psi}^\beta(x_2) = T(\psi^{\alpha A} + \psi^{\alpha C})(\bar{\psi}^{\beta A} + \bar{\psi}^{\beta C}) = \theta(t_1-t_2)(\psi^A+\psi^C)(\bar{\psi}^A+\bar{\psi}^C)$

$- \theta(t_2-t_1)(\bar{\psi}^A+\bar{\psi}^C)(\psi^A+\psi^C)$

$= \theta(t_1-t_2)(\psi^A\bar{\psi}^A + \psi^C\bar{\psi}^C + \psi^C\bar{\psi}^A + \underline{\psi^A\bar{\psi}^C}$

$-\bar{\psi}^C\psi^A + \{\psi^A, \bar{\psi}^C\}$

$+\theta(t_2-t_1)(\cdots\cdots \underline{-\bar{\psi}^A\psi^C}$

$\psi^C\bar{\psi}^A - \{\bar{\psi}^A, \psi^C\}$

$= :\psi^\alpha\bar{\psi}^\beta: + \underbrace{\theta(t_1-t_2)\{\psi^A, \bar{\psi}^C\} - \theta(t_2-t_1)\{\bar{\psi}^A, \psi^C\}}_{\langle 0 \mid T\psi^\alpha(x_1)\psi^\beta(x_2) \mid 0\rangle}$ $\quad \langle 0 \mid T\psi\bar{\psi} \mid 0\rangle$ = C-number

a^+ は左, a は右

$\langle 0 \mid : \cdots : \mid 0\rangle = 0$

$T\varphi_1(x_1)\varphi_2(x_2) = :\varphi_1(x_1)\varphi_2(x_2): + :\overbrace{\varphi_1\ \varphi_2}: \quad \overbrace{\varphi_1\varphi_2} = \langle 0 \mid T\varphi_1\varphi_2 \mid 0\rangle$

$T\varphi_1\varphi_2\varphi_3 = \theta(t_1t_2t_3)\varphi_1\varphi_2\varphi_3 = :\varphi_1\varphi_2\varphi_3: + :\overbrace{\varphi_1\varphi_2}\varphi_3: + :\varphi_1\overbrace{\varphi_2\varphi_3}: + :\overbrace{\varphi_1}\varphi_2\overbrace{\varphi_3}:$

$+\theta(t_1t_3t_2)\varphi_1\varphi_3\varphi_2$

$\vdots$

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$T\varphi_1\varphi_2\cdots\varphi_n = :\varphi_1\cdots\varphi_n: + \sum_{\text{all pair}} :\varphi_1\cdots\overbrace{\varphi_i\cdots\varphi_j}\cdots\varphi_n: + \sum_{\text{all-2pair}} :\varphi_1\cdots\overbrace{\varphi_i\cdots\varphi_j}\cdots\varphi_n: + \cdots$

$\langle 0|\, a_{in}^{f_1}\cdots a_{in}^{f_n}\, S\, a_{in}^{+}\, a_{in}^{+}\, |0\rangle$

= ① 外線 $\longleftarrow_{p}$ $\sqrt{\frac{m}{E}}\frac{1}{(2\pi)^{3/2}}u\,e^{-ipx}$ ② 内線 $x_1 \longleftarrow x_2$ $iS(x_1-x_2)$

$\longleftarrow_{p}$ $\sqrt{\frac{m}{E}}\frac{1}{(2\pi)^{3/2}}\bar{u}\,e^{ipx}$ $\nu\sim\mu$ $x_1\ x_2$ $i\Delta_{\nu\mu}(x_1-x_2)$

$\frac{1}{\sqrt{2E}}\frac{1}{(2\pi)^{3/2}}\epsilon_\mu e^{-ipx}$ x_1 x_2 $i\Delta(x_1-x_2)=\int\frac{i}{p^2-m^2}e^{-ip(x_1-x_2)}\frac{d^4p}{(2\pi)^{3/2}}$

$\frac{1}{\sqrt{2E}}\frac{1}{(2\pi)^{3/2}}\epsilon_\mu e^{ipx}$ ③ vertex μ $-ie\gamma^\mu$

$\frac{1}{\sqrt{2E}}\frac{1}{(2\pi)^{3/2}}e^{-ipx}$ ④ loop $\times(-1)$ $T\bar{\psi}\psi\bar{\psi}\psi\bar{\psi}\psi$ $-\psi\bar{\psi}$

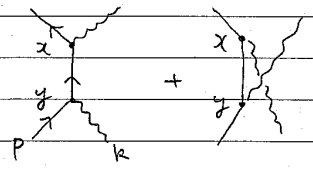
e^+ $\sqrt{\frac{m}{E}}\frac{1}{(2\pi)^{3/2}}\bar{v}\,e^{-ipx}$ z, x, y

e^+ $\sqrt{\frac{m}{E}}\frac{1}{(2\pi)^{3/2}}v\,e^{ipx}$

$\langle 0|\, aa\,\frac{(-i)^2}{(2!)^2}\int d^4x\int d^4y\,(eA_\mu^{(x)}\bar{\psi}\gamma^\mu\psi\quad eA_\nu(y)\bar{\psi}(y)\gamma^\nu\psi(y))\,a^+a^+|0\rangle$

$=\frac{(2!)^2}{(2!)^2}(-i)^2\quad \bar{\psi}_e(p',x)\left\{f_\mu^+(k_2)\gamma^\mu iS(x-y)\gamma^\nu f_\nu(y)\psi_e(p,y)+f_\mu(k,x)\gamma^\mu iS(x-y)\gamma^\nu f_\nu(k',y)\right\}\psi_e(p,x)$

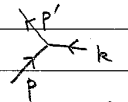
$\downarrow\ \frac{1}{(2\pi)^{3/2}}\frac{1}{\sqrt{2\omega}}\epsilon_\mu$



$A=\left(\frac{1}{(2\pi)^{3/2}}\right)^4\sqrt{\frac{m}{E'}}\sqrt{\frac{m}{E}}\frac{1}{\sqrt{2\omega'}}\frac{1}{\sqrt{2\omega}}\bar{u}'\left(\epsilon'_\mu\gamma^\mu\frac{i}{\not p+\not k-m}\epsilon_\nu\gamma^\nu+\epsilon_\mu\gamma^\mu\frac{i}{\not p-\not k-m}\epsilon'_\nu\gamma^\nu\right)u$

$+\epsilon_\mu\gamma^\mu\frac{i}{\not p'-\not k-m}\epsilon'_\nu\gamma^\nu)\,u$

積分 $\int d^4x_1\cdots\int d^4x_n$

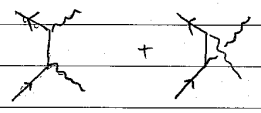
vertex  $(2\pi)^4\delta^{(4)}(\Sigma\pm p)$

propergator の $\int\frac{d^4p}{(2\pi)^4}$ $S=(2\pi)^4\delta^4(\Sigma p_f-\Sigma p_i)\left\{\frac{1}{(2\pi)^{3/2}}\right\}^{n_f+n_i}\prod_{i,f}\left(\frac{1}{\sqrt{2\omega}}\text{ or }\sqrt{\frac{m}{E}}\right)A$

$A=(\text{vertex rule})\times\left(\text{propergator}\ \frac{i}{\not p-m}\ \frac{-i\left(g_{\mu\nu}-(1-\alpha)\frac{p_\mu p_\nu}{p^2}\right)}{p^2}\right)$

$\times$(外線 $u,\bar{u},v,\bar{v},\epsilon^\mu$)

Compton 散乱



$$A = -ie^2 \bar{u}\left(\not\epsilon' \frac{1}{\not p + \not k - m} \not\epsilon + \not\epsilon \frac{1}{\not p - \not k' - m} \not\epsilon' \right) u$$

$$\left(\not p = p_\mu \gamma^\mu \right)$$

$$\frac{\not\epsilon \not\epsilon' \not k'}{-2pk'} = \frac{\not\epsilon \not k' \not\epsilon'}{2pk'}$$

$$\frac{1}{\not p + \not k - m} = \frac{\not p + \not k - m}{(p+k)^2 - m^2} = \frac{\not p + \not k + m}{2pk} = \frac{\not p + \not k + m}{2m\omega}$$

$$\frac{1}{\not p - \not k' - m} = \frac{\not p - \not k' + m}{2m\omega'}$$

ゲージ：クーロンゲージ

$\epsilon = (0, \vec{\epsilon})$ $\epsilon^0 = 0$

$p = (m; \vec{0})$ $p\epsilon = 0$

$k\epsilon = 0$, $k'\epsilon' = 0$ $p\epsilon' = 0$

$$\not\epsilon' \frac{\not p + \not k + m}{2pk} \not\epsilon = \frac{\not\epsilon' \left| \not\epsilon (-\not p - \not k + m) + 2\epsilon k + 2\epsilon \not p \right.}{2pk} = -\frac{\not\epsilon' \not\epsilon \not k}{2pk} = \frac{\not\epsilon' \not k \not\epsilon}{2pk}$$

$$A = \bar{u}\left(\frac{\not\epsilon' \not k \not\epsilon}{2pk} + \frac{\not\epsilon \not k' \not\epsilon'}{2pk'} \right) u$$

$k k = k^2 = 0$

$$T = (2m)^2 \sum_{\text{all-spin}} |A|^2 = T_r (\not p' + m)\left(\frac{\not\epsilon' \not k \not\epsilon}{2pk} + \frac{\not\epsilon \not k' \not\epsilon'}{2pk'} \right)(\not p + m)\left(\frac{\not\epsilon \not k \not\epsilon'}{2pk} + \frac{\not\epsilon' \not k' \not\epsilon'}{2pk'} \right)$$

$$= 2e^4 \left(\frac{pk'}{pk} + \frac{pk}{pk'} + 4(\varepsilon\varepsilon')^2 - 2 \right)$$

$$d\sigma = \frac{\sum_{f} |S|^2}{f\left(\frac{v_{rel}}{V}\right) \cdot T} \qquad S = \langle 0 | \underbrace{(2\pi)^4 \delta^4}_{V \cdot T} | 0 \rangle$$

時間

$$= \sum_{f} |\langle 0 | |^2 (2\pi)^4 \delta^{(4)} \frac{V^2}{v_{rel}} = \sum_{spin} \int d^3p' \int d^3k' \, |\langle 0 | |^2 (2\pi)^4 \delta^{(4)} (2\pi)^6$$

$$v_{rel} = \sqrt{|\vec{v}_1 - \vec{v}_2|^2 - \left| \frac{\vec{v}_1 \times \vec{v}_2}{c^2} \right|^2} = \frac{\sqrt{(P_1 P_2)^2 - m_1^2 m_2^2}}{E_1 E_2}$$

$$\langle (\varepsilon\varepsilon')^2 \rangle = \frac{1 + \cos^2\theta}{2}$$

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2} \left(\frac{\omega'}{\omega} \right)^2 \frac{1}{m^4} \left(\frac{1}{4} \sum_{\substack{\text{all-spin} \\ \text{all-polar}}} T \right) = \frac{\alpha^2}{4m^2} \left(\frac{\omega'}{\omega} \right)^2 \left(\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - 2 + 4(\varepsilon\varepsilon')^2 \right)$$

$\omega, \omega' \to 0$ Thomson 散乱 $\omega = \omega'$ $\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4m^2}(1 + \cos^2\theta) \times 2$ $\sigma \to \frac{8\pi}{3} \frac{\alpha^2}{m^2}$

▷ Jun 9

復習

1. 与えられた in-out state n 個の vertex を含む topological に異なる図形を書く
2. 内線 $i\Delta_F(p) = i \frac{1}{p^2 - m^2 + i\epsilon}$

   $i S_F(p) = i \frac{\not p + m}{p^2 - m^2 + i\epsilon}$

   $i D_F(p)_{\mu\nu} = i \frac{-\eta_{\mu\nu} + (1-\alpha)\frac{k_\mu k_\nu}{k^2}}{p^2 + i\epsilon}$
3. vertex は各 field を除いたもの $-ie\gamma^\mu$
4. fermion の spinor は矢印の逆向きに並べる $\bar{u} \leftarrow$
5. fermion loop には $-$ sign

coop Filing Leaf 6mm×41行

6. in-out の wave fun $\boxed{\frac{1}{(2\pi)^{3/2}}\frac{1}{\sqrt{2p}}}$, $\boxed{\frac{1}{(2\pi)^{3/2}}\sqrt{\frac{m}{E}}}u(p)\cdots$

7. $\boxed{(2\pi)^4\delta^4(P_f-P_i)}$ ← □の部分は遷移確率 w の中に繰込んでしまってよい

Transition rate W_{fi}

$\xrightarrow{c}\ \xleftarrow{c}$

$S_F = \delta_{fi} - (2\pi)i\,\delta(E_f-E_i)\,T_{fi}$ $\quad \sigma_{fi} = \frac{V}{v_{rel}}W_{fi}$ $\quad v_{rel} = v_1 + v_2 = 2c$

$W_{fi} = 2\pi\rho_f\,\delta(E_f-E_i)\,|T_{fi}|^2$ $\quad \Gamma_{fi} = W_{fi}$

covariant

$S_{fi} = \delta_{fi} - (2\pi)^4 i\,\delta^4(P_F-P_i)\,t_{fi}$ $\quad |t_{fi}|^2 = \frac{1}{(2\pi)^3 2p_1^0}\frac{1}{(2\pi)^3 2p_2^0}|M|^2\frac{1}{(2\pi)^3 2p_1'^0}\cdots\frac{1}{(2\pi)^3 2p_n'^0}$

散乱 $P_1 + P_2 \to p_1' + \cdots + p_n'$ $\quad \frac{1}{(2\pi)^3}\frac{m}{E}$

$\sigma_{fi} = (2\pi)^4\delta^4(P_f-P_i)\frac{(2\pi)^6}{v_{rel}}\int d^3p_1\,d^3p_2\cdots d^3p_n\,|t_{fi}|^2$ $\quad$ 崩壊 $P = (2\pi)^4\delta^4(P_f-P_i)(2\pi)^3\int\cdots$

Compton 散乱



$\omega \to 0$ 極限 (NR limit)

$\frac{d\sigma_{compt}}{d\Omega} \longrightarrow \frac{d\sigma_{Th}}{d\Omega} = \frac{\alpha^2}{m^2}\frac{1+\cos^2\theta}{2} = r_e^2\frac{1+\cos^2\theta}{2}$ $\quad r_e = \frac{\alpha}{m}$ $\quad \alpha = \frac{e^2}{4\pi}$

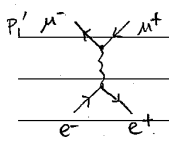
$\sigma_{Th} = \int d\Omega\frac{d\sigma}{d\Omega} = 2\pi\int_{-1}^{1}dz\,r_e^2\frac{1+\cos^2\theta}{2} = \frac{8\pi}{3}r_e^2 = 0.665\times10^{-24}\ cm^2$

* 高次補正項が現れない　e の定義に使える (Dyson 流)

* 大きな cross section $\sigma(\gamma + H \to \gamma + H)$

$\sigma(\gamma + e \to \gamma + e)$

$e^+e^- \to \mu^+\mu^-$ $\quad q = P_1 + P_2 = P_1' + P_2'$



$M = \bar{u}(P_1')(-ie\gamma^\mu)v(P_2')\,iD_{F\mu\nu}(q)\,\bar{v}(P_2)(-ie\gamma^\nu)u(P_1)$

$D_{F\mu\nu} = \frac{-\eta_{\mu\nu} + (1-\alpha)\frac{q_\mu q_\nu}{q^2}}{q^2+i\epsilon}$

$\bar{u}(P_1')\gamma^\mu v(P_2')(P_1'+P_2') = 0$ $\quad$ 考えなくてよい

$\not{P}_1$ $\quad$ conserved current

$\bar{u}\not{P}_1 = \bar{u}m$ $\quad \not{P}_2'v(P_2') = -mv(P_2)$

$[\bar{u}(p)\gamma^\mu v(p')]^+ = u'^+\gamma^{\mu+}\gamma^0 u = u'\gamma^\mu u(p')$ より $\quad |M|^2 = \left|-i\frac{e^2}{q^2}\right|^2\bar{u}(P_1')\gamma^\mu v(P_2')\,\bar{v}(P_2')\gamma^\nu u(P_1')$

$\bar{v}(P_2)\gamma_\mu u(P_1)\,\bar{u}(P_1)\gamma_\nu v(P_2)$

Trace formula

$\frac{1}{4}\sum_{spin}|M|^2 = \frac{1}{4}\left(\frac{e^2}{q^2}\right)^2 Tr\left[\frac{\not{P}_1'+m_\mu}{2m_\mu}\gamma^\mu\frac{\not{P}_2'-m_\mu}{2m_\mu}\gamma^\nu\right]Tr\left[\frac{\not{P}_2-m_e}{2m_e}\gamma_\mu\frac{\not{P}_1+m_e}{2m_e}\gamma_\nu\right]$

$Tr\,\not{a}\not{b} = 4ab$

$Tr\,\not{a}\not{b}\not{c}\not{d} = 4\{(ab)(cd)+(ad)(bc)-(ac)(bd)\}$

$= \frac{1}{4}\left(\frac{e^2}{q^2}\right)^2\left(\frac{1}{4m_\mu^2}\right)\left(\frac{1}{4m_e^2}\right)\times$

$\{Tr(\not{P}_1'\gamma^\mu\not{P}_2'\gamma^\nu - m^2\gamma^\mu\gamma^\nu)$

$\{Tr(\not{P}_1\gamma_\mu\not{P}_2\gamma_\nu - m^2\gamma_\mu\gamma_\nu)$

$S = (P_1+P_2)^2 = 2(m_e^2 + P_1P_2) = 2(m_\mu^2 + P_1'P_2') = q^2$

CMS 系 $\vec{P}_1 + \vec{P}_2 = 0$ $\quad P_1'P_2 = \frac{s}{4}(1-\beta_{p'}\beta_p\cos\theta)$

$P_1^0 = P_2^0 = \frac{\sqrt{s}}{2}$

$P_1'P_1 = P_1'^0P_1^0 - |\vec{P}'||\vec{P}|\cos\theta = P_1'^0P_1^0(1-\beta_{p'}\beta_p\cos\theta)$

$= \frac{s}{4}(1-\beta_{p'}\beta_p\cos\theta)$

$4(P_1'^\mu P_2'^\nu + P_1'^\nu P_2'^\mu - [(P_1'P_2') + m^2]\eta^{\mu\nu})$

$4(P_1^\mu P_2^\nu + P_1^\nu P_2^\mu - [(P_1P_2) + m^2]\eta_{\mu\nu})$

$m_e \ll m_\mu$ $\quad \beta_p = 1$, $\beta_{p'} = \beta$ とおく

$16\{2(P_1'P_1)(P_2'P_2) + 2(P_1'P_2)(P_2'P_1) - 2(P_1'P_2')[P_1P_2 + m_\mu^2]$

$+ 4(P_1'P_2' + m_\mu^2)(P_1P_2 + m_e^2) - 2(P_1P_2)[P_1'P_2' + m_e^2]$

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$$\frac{1}{4}|M|^2 = \frac{1}{4}\left(\frac{e}{s}\right)^2 \left\{2\left(\frac{s}{4}\right)^2(1-\beta\cos\theta)^2 + 2\left(\frac{s}{4}\right)^2(1+\beta\cos\theta)^2 - 2\left(\frac{s}{2}-m_e^2\right)\frac{s}{2} - 2\left(\frac{s}{2}-m_\mu^2\right)\frac{s}{2} + 4\frac{s}{2}\frac{s}{2}\right\}$$

$$= \frac{1}{32}\left(\frac{e^2}{s}\right)^2 s^2 \left\{2(1+\beta^2\cos^2\theta) + 8m_\mu^2\frac{1}{s}\right\} = \frac{e^4}{16}\left(1+\beta^2\cos^2\theta + 4\frac{m_\mu^2}{s}\right) \qquad \beta = \sqrt{1-\frac{4m_\mu^2}{s}}$$

$$d\sigma = \frac{1}{v_{rel}}\frac{m_e}{p_1^0}\frac{m_e}{p_2^0}\int \frac{d^3p_1' m_\mu}{(2\pi)^3 p_1'^0}\frac{d^3p_2' m_\mu}{(2\pi)^3 p_2'^0}(2\pi)^4\delta^4(p_1'+p_2'-p_1-p_2)|M|^2$$

$v_{rel} = 2$

$$\int \frac{d^3p_1'}{2p_1'^0}\frac{d^3p_2'}{2p_2'^0}\delta^4(p_1'+p_2'-p_1-p_2) = \int \frac{d^3p_1}{2p_1^0 2p_2^0}\delta(p_1^0+p_2^0-p_1'^0-p_2'^0) = \int \frac{p_1^2 dp_1 d\Omega}{2p_1^0 2p_2^0}\frac{1}{\left(\frac{|p_1|}{p_1^0}+\frac{|p_2|}{p_2^0}\right)} = \frac{|\vec{p}_1|}{4(p_1^0+p_2^0)}\int d\Omega$$

$\leftarrow p_1^2 dp_1 d\Omega$

$|p_1| = |p_2|$

$$\frac{d\sigma}{d\Omega} = \frac{1}{2}\frac{m_e}{p_1^0}\frac{m_e}{p_2^0}\frac{1}{(2\pi)^2}m_\mu^2\left(\frac{|\vec{p}|}{p_1'^0+p_2'^0}\right)\frac{1}{16}\frac{1}{m_e^2}\frac{1}{m_\mu^2}e^4\left(1+\beta^2\cos^2\theta+4\frac{m_\mu^2}{s}\right)$$

$\beta/2$

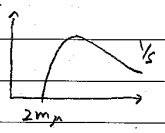
$$= \frac{e^4}{4\pi^2}\frac{1}{16}\frac{\beta}{s}\left(1+\beta^2\cos^2\theta+(1-\beta^2)\right) = \frac{\alpha^2}{4}\frac{1}{s}\beta\left(\underbrace{1+\cos^2\theta}_{\beta\approx 1} + \underbrace{(1-\beta^2)\sin^2\theta}_{\approx 0}\right)$$

spin ½ 粒子の production
⇒ quark の存在の証明になった！
（ハドロンのシャワー）

$$\sigma_{\mu\nu} = \int \frac{d\sigma}{d\Omega}d\Omega \qquad \cos\theta = z$$

$$= 2\pi\frac{\alpha^2}{4s}\beta\int_{-1}^{1}dz = \frac{4\pi}{3}\frac{\alpha^2}{s}\beta\left(1+\frac{1-\beta^2}{2}\right) \quad \leftarrow [cm^2]$$

$$= \frac{86.9}{s}\ nb \quad (s \text{ in GeV})$$

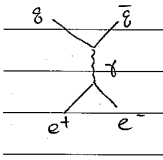


$\mathcal{L}$ luminosity $[cm^{-2}\cdot s^{-1}]$ $\sigma\mathcal{L}$ [event/s]

$\sim 10^{31}\ cm^{-2}s^{-1}$

$s = (3\ GeV)^2$ 1.5 GeV をぶつける $\sigma = \frac{87}{9}\times 10^{-24}\ cm^2$ $\sigma\mathcal{L}\cdot 86400\ sec/day = 8400\ events/day$

1TeV 〃 $= 0.1\ events/day$



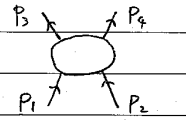
u d s c

$\left(\frac{2}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2 = \frac{10}{9}$

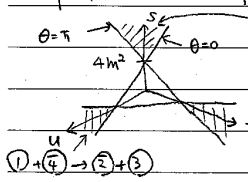
$u_1\ u_2\ u_3 \rightarrow \frac{10}{3}$

$$R = \frac{\sigma(e^+e^- \rightarrow \text{all hadron})}{\sigma(e^+e^- \rightarrow \mu^+\mu^-)} = \frac{10}{3}$$ 実験結果！ → color の自由度

$\Sigma q\bar{q}$

Mandelstam variable		CMS	lab
	$s = (P_1+P_2)^2$	$= 4E$	$= m_1^2 + m_2^2 + E_2^{lab} m_1$
	$t = (P_1-P_3)^2$	$= m_1^2+m_2^2 - 2E_1E_3(1-\beta_1\beta_3\cos\theta)$	
	$u = (P_2-P_4)^2$	$=$	

equal mass の時 $e^+e^- \rightarrow \mu^+\mu^-$



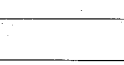
ex) $\gamma + e^- \rightarrow \gamma + e^-$ の時 physical region を図示せよ

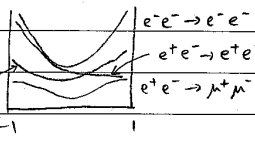
exercise

1) $e^- + e^- \rightarrow e^- + e^-$ $\frac{d\sigma}{d\Omega}$ を計算して $\frac{d\sigma}{d\Omega}(e^+e^- \rightarrow \mu^+\mu^-)$ と共に図示せよ

統計に注意 (Møller 散乱)

2) $e^+e^- \rightarrow e^+e^-$ (Bhabler 散乱)





3) $e^+e^- \rightarrow \gamma\gamma$

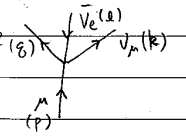
特に ER, NR の振舞い

4) $e^+e^- \rightarrow \phi^+\phi^-$ ϕ charged spin 0 粒子

$\frac{d\sigma}{d\Omega}$? threshold 2ヶの振舞い 何故か？

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例題 $\mu^- \to e^- + \bar{\nu}_e + \nu_\mu$ $\quad [\bar{u}'\gamma^\mu(1-\gamma_5)u]^\dagger = [\bar{u}\gamma^\mu(1-\gamma_5)u']$

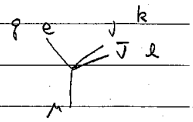


$$\mathcal{L}_{eff} = \frac{G_F}{\sqrt{2}} \bar{\psi}_{\nu_\mu} \gamma^\mu (1-\gamma_5) \psi_\mu \bar{\psi}_e \gamma_\mu (1-\gamma_5) \psi_{\nu_e}$$

$$M = i \frac{G_F}{\sqrt{2}} \bar{u}_{\nu_\mu}(k) \gamma^\mu (1-\gamma_5) u_\mu(p) \bar{u}_e(q) \gamma_\mu (1-\gamma_5) v_{\nu_e}(l)$$

$$|M|^2 = \frac{G_F^2}{2} \mathrm{Tr}\left[k\gamma^\mu(1-\gamma_5)(\not{p}+m)\gamma^\nu(1-\gamma_5)\right] \rightsquigarrow \mathrm{Tr}\left[2\not{k}\gamma^\mu\not{p}\gamma^\nu(1-\gamma_5)\right]$$

$$\mathrm{Tr}\left[\not{q}\gamma_\mu(1-\gamma_5)\not{k}\gamma_\nu(1-\gamma_5)\right] \rightsquigarrow \mathrm{Tr}\left[2\not{q}\gamma_\mu\not{k}\gamma_\nu(1-\gamma_5)\right]$$



$\mathrm{Tr}(\gamma_5 \not{a}\not{b}\not{c}\not{d})$

$= 4i\,\varepsilon_{\alpha\beta\gamma\delta} a^\alpha b^\beta c^\gamma d^\delta$

$\varepsilon^{\alpha\beta\gamma\delta}\varepsilon_{\alpha\beta\gamma\delta} = -4!$

$\varepsilon^{\alpha\beta\gamma\mu}\varepsilon_{\alpha\beta\gamma\nu} = -3!\,\delta^\mu_\nu$

$\varepsilon^{\alpha\beta\mu\nu}\varepsilon_{\alpha\beta\sigma\rho} = -2! \begin{vmatrix} \delta^\mu_\sigma & \delta^\nu_\sigma \\ \delta^\mu_\rho & \delta^\nu_\rho \end{vmatrix}$

$\varepsilon^{\alpha\mu\nu\lambda}\varepsilon_{\alpha\sigma\rho\tau} = -1! \begin{vmatrix} \cdots \\ \vdots \end{vmatrix}$

$= -0! \begin{vmatrix} \cdots \\ \vdots \end{vmatrix}$

$$\frac{1}{2}\sum |M|^2 = 64\, G_F^2 (kq)(pl)$$

$$\Gamma(\mu \to e\nu\bar{\nu}) = \frac{1}{2m_\mu} \int \frac{d^3q}{(2\pi)^3 2q^0} \frac{d^3k}{(2\pi)^3 2k^0} \frac{d^3l}{(2\pi)^3 2l^0} (2\pi)^4 \delta^4(p \to q-k-l) \left(\frac{1}{2}\sum |M|^2\right)$$

$$\int d^4r\, \delta^4(r-k-l) = 1$$

$$\left(\int \frac{d^3l}{2l^0} \frac{d^3k}{2k^0} \delta^4(k+l-r)\, k^\alpha l^\beta = \frac{\pi}{24}\left[\eta^{\alpha\beta} r^2 - 2r^\alpha r^\beta\right] \right)$$

$r = p - q$

$r^2 = m_\mu^2 - 2m_\mu E_e$

$rp = m_\mu^2 - m_\mu E_e$

$rq = m_\mu E$

$$= \frac{1}{2m_\mu} \frac{1}{(2\pi)^5} \int \frac{d^3q}{2q^0} \delta^4(p+q-r) \frac{\pi}{24} 64\, G_F^2 \left[(qp) r^2 + 2(rp)(rq)\right]$$

$\int d^4r$; $\quad m_\mu^2 E_e \left[3 - 4\frac{E_e}{m_\mu}\right]$

$$= \frac{1}{2m_\mu} \frac{1}{(2\pi)^5} \frac{4\pi}{2} \int E_e dE_e \frac{\pi}{24} 64 G_F^2 m_\mu^2 E_e \left[3 - 4\frac{E_e}{m_\mu}\right]$$

$$\frac{d\Gamma}{dE_e} = \frac{G_F^2 m_\mu^2 E_e^2}{12\pi^3}\left(3 - 4\frac{E}{m_\mu}\right)$$

$m_\mu = 105.658389(37)\,\mathrm{MeV}$

$\tau_\mu = 2.19703(4)\,\mu s$

$\Gamma = \tau^{-1}$

$$\Gamma = \int_0^{\frac{m_\mu}{2}} \frac{d\Gamma}{dE_e} dE_e = \frac{G_F^2 m_\mu^5}{192\pi^3}$$

正確にやると $\Gamma = \Gamma_0 \times P\left(\frac{m_e}{m_\mu}\right)\left(1 + \frac{\alpha}{2\pi}\left(\pi^2 - \frac{25}{4}\right)\right)$ ← 0.4%

$\rightarrow G_F = 1.16637(2) \times 10^{-5}\ \mathrm{GeV}^{-2}$

ex 5 $\mu \to e\nu\bar{\nu}$ の ν_μ と $\bar{\nu}_e$ の spectrum を求めよ

ex 6 $m_e \neq 0$ ならどうなるか？ 初等解析関数

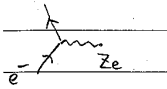
ex 7 $(\pi \to e\bar{\nu}_e)$ $\quad \mathcal{A} = \bar{\psi}_e \gamma^\mu (A - B\gamma_5)\psi_\nu \partial_\mu \phi$ $\quad \frac{\Gamma(\pi \to e\bar{\nu}_e)}{\Gamma(\pi \to \mu\bar{\nu}_\mu)}$ の比を実験と比べよ

を計算せよ ; A, B: free parameter

ex 8 $\sigma(\nu_\mu + e^- \to \mu^- + \nu_e)$ $\quad$ flux $\phi(\nu_e) = 10^{10}/s$ $\quad E(\nu_\mu) \simeq 30\,\mathrm{MeV}$ とする

G_F cm^2 で表せ $\quad$ まともな exp に必要な transport の総量 ; 100 count/day

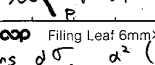
ex 9 Coulomb 散乱 (外場による散乱)



$A^0 = -\frac{Ze}{r}$ $\quad \vec{A} = 0$

ex 10 Rosenbluth formula

$q = p' - p$



$$\bar{u}(p')\Gamma_\mu(p'p)u(p) = \bar{u}(p')\left[\gamma_\mu F_1(q^2) + i\sigma_{\mu\nu}\frac{q^\nu}{2M} F_2(q^2)\right] u(p)$$

coop Filing Leaf 6mm×41行

Ans $\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4E^2} \frac{\left(F_1^2 - \frac{q^2}{4M^2}F_2^2\right)\cos^2\frac{\theta}{2} - \frac{q^2}{2M^2}(F_1+F_2)^2 \sin^2\frac{\theta}{2}}{\left(1 + \frac{2E}{M}\sin^2\frac{\theta}{2}\right)\sin^4\frac{\theta}{2}}$ $\quad$ E: electron energy

$e + p \to e + p$

ex 11 Hydrogen atom

2p → 1s の考察 NR. Coulomb gauge

$\vec{F} = \vec{I} + \vec{S}$

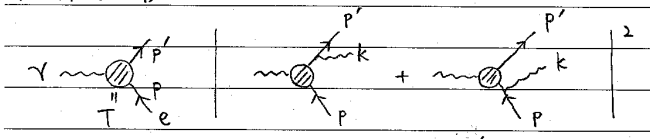
F = 1 → F = 0 の考察

1S — F=1 / F=0 準位 Splitting を求める

ex 12 回転振動 考察 (n=2) → (n=1) + γ

▷ Jun 16

赤外発散



$$M = (-ie)\bar{u}(p')\left[\gamma^\mu \frac{\not p' + \not k' + m}{(p'+k')^2 - m^2}\not\varepsilon + \not\varepsilon \frac{\not p - \not k + m}{(p-k)^2 - m^2}\gamma^\mu\right]u(p)\varepsilon_\mu$$

$k \to 0$

$$\bar{u}(p')\gamma_\mu \frac{\not p' + \not k' + m}{(p'+k')^2 - m^2} = \bar{u}(p')\frac{p_\mu}{p'k} \qquad = e\bar{u}(p)\not\varepsilon u(p)\left(\frac{p'_\mu}{p'k} - \frac{p_\mu}{pk}\right)\varepsilon^\mu$$

$$|M|^2 = e^2 |\bar{u}(p)\not\varepsilon u(p)|^2 \left(\frac{p'_\mu}{p'k} - \frac{p_\mu}{pk}\right)\left(\frac{p'_\nu}{p'k} - \frac{p_\nu}{pk}\right)\cdot\varepsilon^\mu\varepsilon^\nu$$

$$[\sigma_B]_{IR} = \sigma_0 e^2 \int \frac{d^3k}{(2\pi)^3 2k}\left[-\frac{m^2}{(p'k)^2} - \frac{m^2}{(pk)^2} + 2\frac{(pp')}{(p'k)(pk)}\right](-\eta^{\mu\nu})$$

$$\int d^3k \frac{1}{2k}\frac{m^2}{(p'k)^2} = \int \frac{k^2 dk}{2k} 2\pi \int_{-1}^{1} dz \frac{1}{(1-\beta' z)^2}\frac{m^2}{(E'k)^2} \qquad \beta' = \frac{p'}{E'}$$

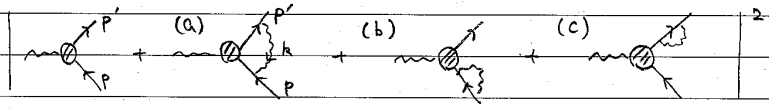
$$= \int \frac{dk}{k} 2\pi \underbrace{\frac{1}{2}\int_{-1}^{1} dz \frac{1}{(1-\beta' z)}\left(\frac{m}{E'}\right)^2}_{1} = 2\pi \int \frac{dk}{k}$$

$$\int d^3k \frac{1}{2k}\frac{(pp')}{(pk)(p'k)} = \int \frac{dk}{k}\frac{1}{2}\int d\Omega \frac{1 - \vec{\beta}\vec{\beta}}{(1-\vec{\beta}\hat{k})(1-\vec{\beta}'\hat{k})}$$

$$g(\vec{\beta},\vec{\beta}') = -1 + \frac{1}{4\pi}\int d\Omega \ \text{〃}$$

$$[\sigma_B]_{IR} = \sigma_0 \frac{\alpha}{2\pi}[\ln k]_{k_{min}}^{k_{max}} g(\vec{\beta},\vec{\beta}')$$

virtual γ emittion



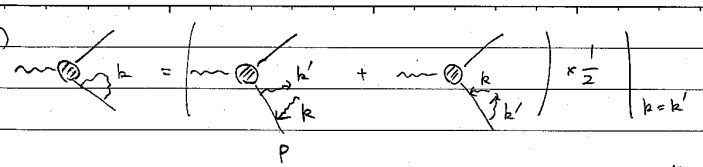
$$= |\bar{u}\not\varepsilon u|^2(1 + (Rc) + (Rc)^*) \qquad Rc = (a) + (b) + (c)$$

$$(a)\ M^{(a)} = (-ie)^2 \int \frac{d^4k}{(2\pi)^4}\bar{u}(p')\not\varepsilon \frac{\not p' + \not k' + m}{(p'+k')^2 - m^2}\not\varepsilon \frac{\not p + \not k + m}{(p+k)^2 - m^2}\gamma^\nu u(p) \frac{-\eta^{\mu\nu}}{k^2 + i\epsilon}$$

$$= -ie^2 \int \frac{d^4k}{(2\pi)^4}\frac{p'^\mu}{p'k}\frac{p^\nu}{pk}\eta_{\mu\nu}\,\bar{u}(p)\not\varepsilon u(p)\frac{1}{k^2 + i\epsilon}$$

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(c)



$$\frac{1}{2}\left(\frac{P_\mu}{pk}\frac{P_\nu}{p(k-k')}+\frac{P_\nu}{(-pk')}\frac{P_\mu}{(-pk')p(k-k')}\right)=\frac{1}{2}\frac{P_\mu P_\nu}{p(k-k')}\left[\frac{1}{pk}-\frac{1}{pk'}\right]=-\frac{1}{2}\frac{P_\mu P_\nu}{(pk)^2}$$

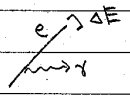
$$M^{(b)}+M^{(c)}=(-ie)^2\int\frac{d^4k}{(2\pi)^4}i^2\left[-\frac{m^2}{2(pk)^2}-\frac{m^2}{2(p'k)}\right]\bar{u}(p')\Gamma u(p)\cdot i\frac{1}{k^2+i\epsilon}\qquad \frac{1}{k^2+i\epsilon}=\frac{P}{k^2}-i\pi\delta(k^2)$$

$$|M|^2=\left\{1+e^2\int\frac{d^3k}{(2\pi)^3}\frac{1}{2k}\left[\frac{m^2}{pk}+\frac{m^2}{p'k}-\frac{pp'}{(pk)(p'k)}\right]\right\}|\bar{u}\Gamma u|^2$$

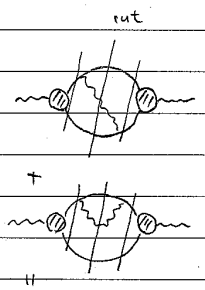
$$\sigma_{virtual}\Big|_{IR}=\sigma_0 e^2\int\frac{d^3k}{(2\pi)^3}\frac{1}{2k}\left[\frac{m^2}{p^2}+\cdots\right]$$

$$\sigma_B\Big|_{IR}=\sigma_0 e^2\int\qquad\left[-\frac{m^2}{pk}\cdots\right]$$

$$[\sigma_{virtual}+\sigma_B]_{IR}=0$$

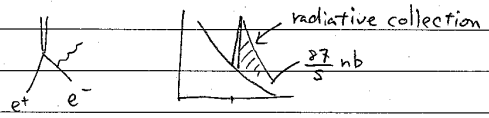


$$\sum_{cut}$$



一般的なcutting ruleの特殊な例

$e^+e^-\longrightarrow$ resonance state $\longrightarrow \mu^+\mu^-$



$J/\psi(3100)\quad 1^{--}$

$$\sigma_0(E)=A\frac{\Gamma/2}{(E-m_R)^2+(\Gamma/2)^2}\approx A\pi\delta(E-m_R)\qquad \Gamma_{e^+e^-}$$

$$\sigma(E)=\int_0^E d\varepsilon\,\sigma_0(E-\varepsilon)\frac{dI(\varepsilon)}{d\varepsilon}\qquad I(\varepsilon)=\frac{2\alpha}{\pi}g(\vec{\beta},\vec{\beta}')\int_0^\varepsilon\frac{d\varepsilon'}{\varepsilon'}\qquad E_\gamma<\varepsilon$$

$$\sigma(E)=\pi A\left.\frac{dI}{d\varepsilon}\right|_{\varepsilon=E-m_R}\qquad g(\beta,\beta')=-1+\frac{1}{4\pi}\int d\Omega_k\frac{1-\vec{\beta}\vec{\beta}'}{(1-\vec{\beta}\hat{k})(1-\vec{\beta}'\hat{k})}$$

$$\beta=(0,0,v)\quad\beta'=(0,0,-v)\quad=-1+\frac{1}{2}\int_{-1}^{1}dz\frac{1+v^2}{1-v^2z^2}=-1+\frac{1}{2v}\ln\left|\frac{1+v}{1-v}\right|>0$$

multiphoton emittion $\quad I(\varepsilon)=\int_0^\varepsilon\frac{2\alpha}{\pi}\,q\left(\frac{\varepsilon'}{E}\right)^{-\frac{2\alpha}{\pi}g}\frac{d\varepsilon'}{\varepsilon'}$

CPT定理

Lorentz不変なlocal field theoryはCPT不変

$$CPT\,\phi(x)\,T^{-1}P^{-1}C=\underbrace{\xi^*\xi^*\eta^*}_{-1}\phi^\dagger(-x)$$

$$P\,A_\mu(x)\,P^{-1}=-\eta^*P^\nu{}_\mu A^\lambda(t,-x)$$

$$C\,A_\mu(x)\,C^{-1}=\xi^*A_\mu^+$$

$$J\,A_\mu\,J^{-1}=\xi^*P^\nu{}_\mu A^\lambda(-t,x)$$

$$\varepsilon^\mu(p,\sigma)=\Lambda^\mu{}_\nu\,\varepsilon^\nu(0,\sigma)$$

$$\varepsilon^\mu(-p,\sigma)=\Lambda^\mu{}_\nu(-p)\,\varepsilon^\nu(0,\sigma)$$

$$P^\mu{}_\lambda\Lambda^\lambda{}_\sigma(p)\underbrace{P^\sigma{}_\nu\,\varepsilon^\nu(0,\sigma)}_{-\varepsilon^\nu(0,\sigma)}=-P^\mu{}_\lambda\,\varepsilon^\lambda(p,\sigma)$$

Dirac field

$P\psi(x)P^{-1} = \gamma_0 \psi(t,-x)$ $\quad PCJ\psi_\alpha(x) J^{-1}C^{-1}P^{-1} = i(\gamma_5)_{\alpha\beta}\bar{\psi}^*_\beta(-x)$

$C\psi_\alpha(x)C^{-1} = C_{\alpha\beta}\bar{\psi}_\beta$ $\quad PCJ\bar{\psi}_\alpha(x)J^{-1}C^{-1}P^{-1} = -i\psi_\beta(\gamma_5\gamma_0)_{\beta\alpha}$

$J\psi(x)J^{-1} = -i\gamma_5 C\psi(-t,x)$

$PCJ\ \bar{\psi}^A_\alpha \Gamma_{\alpha\beta}\psi^B_\beta T^{-1}C^{-1}P^{-1} = -\psi^A_\alpha \Gamma'_{\beta\alpha}\bar{\psi}^B_\beta$ $\quad PCJ :\bar{\psi}^A_\alpha T_{\alpha\beta}\psi^B_\beta: J^{-1}C^{-1}P^{-1} = \bar{\psi}^B P'\psi^A$

$P' = (\gamma^5\gamma^0 P^* \gamma^0\gamma^5)^T$

$\Gamma = 1 \quad i\gamma_5 \quad \gamma_\mu \quad \gamma_5\gamma_\mu \quad \sigma_{\mu\nu}$

$+ \quad + \quad - \quad - \quad +$

Lorentz inv $\mathcal{L}$ 定義 connect

PCT 定義について -sign

$\partial_\mu \to \theta\partial_\mu$

$A_\mu \to -A_\mu$

$CPT\,\mathcal{L}\,CPT = \mathcal{L}(-x)$

$CPT\,\varphi\,CPT = +\varphi \qquad \mathcal{H} = \dot{\varphi}\pi - \mathcal{L}$

$\dot{\varphi} \quad -$

$\pi \quad -$

$CPT\,\psi(x)(CPT)^{-1} = H(-x)$

$\left.\begin{array}{l} \langle e^-|H|e^-\rangle = m_{e^-} \\ \langle e^+|H|e^+\rangle = m_{e^+} \end{array}\right\} \quad m_{e^+} = m_{e^-}$

証) $\langle a|H|b\rangle = \langle b^{CPT}|\underset{\parallel}{H}{}^{CPH}|a^{CPT}\rangle \qquad \vec{P}=0$

H

$\langle\alpha|e^{-iHt}|\beta\rangle \overset{\text{at rest}}{=} e^{-imt-Pt}$ $\quad m = m^{CPT}$ $\quad m_i = m_{\bar{i}}$ $\quad$ total width (lifetime)

$\langle\beta^{CPT}|e^{-iHt}|\alpha^{CPT}\rangle = e^{-im^{CPT}t - P^{CPT}t}$ $\quad P = P^{CPT}$ $\quad \Gamma_i = \Gamma_{\bar{i}}$ に対して particle width

$\tau_{\pi^+} = \tau_{\pi^-}$ とは限りません.

粒子と反粒子の particle width が等しくなるためには

一般には CP conservation が必要

Reduction formula (Lehmann Symantik Zimmenwann)

$|p_1 \cdots p_n\rangle_{in} = a^\dagger_{in}(p_1)\cdots a^\dagger_{in}(p_n)|0\rangle$

$\varphi_{in}(x) = \int d^3p\,(f_p a_p + f_p^* a_p^\dagger)$

$a^\dagger_{p\,in} = -i\int d^3x\, f_p(x) \overleftrightarrow{\partial_0} \varphi_{in}(x)$

$\lim_{t\to\infty}\langle\alpha|\varphi(t)|\beta\rangle = \langle\alpha|\varphi_{in}(t)|\beta\rangle \qquad t\to-\infty \quad \varphi(t)\to\varphi_{in}(t)$

reduction formula

$S(i\to f) = \underset{out}{\langle f|}\, i\,p\,\rangle_{in}$

$= {}_{out}\langle f|a^\dagger_{in}(p)|i\rangle_{in} = \underset{out}{\langle} f|\overbrace{a^\dagger_{out} + a^\dagger_{in}(p) - a^\dagger_{out}(p)}^{0}|i\rangle$

$= (-i)_{out}\langle f|\int d^3x\, f(x)\overleftrightarrow{\partial_0}(\varphi_{in}(x) - \varphi_{out}(x))|i\rangle_{in}$

$= i\left(\lim_{t\to+\infty} - \lim_{t\to-\infty}\right)\int d^3x\ {}_{out}\langle f|f(x)\overleftrightarrow{\partial_0}\varphi(x)|i\rangle_{in}$

$= i\int_{-\infty}^{\infty} dx^0 \int d^3\vec{x}\ \partial_0\ {}_{out}\langle f|f(x)\overleftrightarrow{\partial_0}\varphi(x)|i\rangle_{in}$

$f\ddot{\varphi}(x) - \ddot{f}\varphi(x)$

$(\partial^2 - m^2)f$

$= i\int d^4x\, f_p(x)(\Box + m^2)\,{}_{out}\langle f|\varphi(x)|i\rangle_{in}$

coop Filing Leaf 6mm×41行

$f = f' + p'$ $\quad {}_{out}\langle f| = {}_{out}\langle f'\, p'|$

${}_{out}\langle f'\, p'|\varphi(x)|i\rangle_{in} = {}_{out}\langle f'|\, a_{out}\varphi(x)|i\rangle_{in}$ $\quad$ if $p' \neq i$

$= {}_{out}\langle f'|\, a_{out}\varphi(x) + \varphi(x)a_{in}^{(p')} - \varphi(x)a_{in}|i\rangle_{in}$

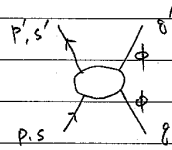
$= {}_{out}\langle f'|\, a_{out}\varphi(x) - \varphi(x)a_{in}|i\rangle_{in}$

$= +i\int d^3x\, f_{p'}^{*} \overleftrightarrow{\partial_0}\, {}_{out}\langle f'|\varphi_{out}(x)\varphi(x) - \varphi(x)\varphi_{in}(x)|i\rangle_{in}$

$= (+i)\left(\lim_{t\to+\infty} - \lim_{t\to-\infty}\right)\int d^3x\, f_{p'}^{*} \overleftrightarrow{\partial_0} \langle f'|T(\varphi(x)\varphi(x))|i\rangle$

$= +i\int_{-\infty}^{\infty} dt\, \partial_0 \int d^3x\, f_{p'}^{*} \overleftrightarrow{\partial_0} \langle f'|T(\varphi\varphi)|i\rangle = +i\int d^4x\, f_{p'}^{*}(\Box+m^2)\langle 0|T\varphi(x)\varphi(x)|0\rangle$

${}_{out}\langle p_1' p_2' \cdots p_{n'}'|p_1 \cdots p_n\rangle_{in} = \prod_{p'} \int i f_{p'}^{*}(x')(\Box+m^2) \prod_{p} i\int f_p(x)(\underbrace{\Box+m^2}_{\text{逆ずり}})\langle 0|T(\varphi\cdots\varphi)|0\rangle$ $\quad n+n'$



$S_{fi} = {}_{out}\langle p's'\, q'|p\, s\, q\rangle_{in} = \delta_{fi} - \int d^4x\, d^4x'\, d^4y\, d^4y' \sqrt{\frac{1}{(2\pi)^3 2q^0}}\sqrt{\frac{1}{(2\pi)^3 2q^0}}\sqrt{\frac{m}{(2\pi)^3 p'^0}}\sqrt{\frac{m}{(2\pi)^3 p^0}}$

$\overrightarrow{(\Box_{x'}+m^2)}\bar{u}(p's')\overrightarrow{(i\partial_{y'}-m)}\langle 0|T(\bar{\psi}(y')\psi(y)\varphi(x')\varphi(x)|0\rangle$

$\overleftarrow{(-i\partial_{y'}-m)}u(ps)\overleftarrow{(\Box_x+m^2)}\, e^{i(p'x'+q'x'-py-qx)}$

unitarity

largest time equation

$F(x_1 \cdots x_n)$ diagram 

underlined diagram $F(x_1 \cdots \underline{x_i} \cdots \underline{x_j} \cdots x_n)$

① underline $\pm$: $i\Delta_F(x_i - x_k)$ $\quad i\Delta_F(x) = \frac{i}{(4\pi)^4}\int \frac{e^{-ipx}}{p^2 - m^2 + i\varepsilon} d^4p$

② i underlined, k not " : $\Delta^+(x_i - x_k)$ 

$\Delta^+(x) = \int \frac{d^4x}{(4\pi)^3}\theta(\pm p^0)\delta(p^2 - m^2)e^{-ipx}$

③ i not, k underlined : $\Delta^-(x_i - x_k)$ 

$i\Delta_F(x) = \theta(x^0)\Delta^+(x) + \theta(-x^0)\Delta^-(x)$

④ i, k 共に underlined : $[i\Delta_F(x_i - x_k)]^*$

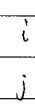
$(\Delta^+(x))^* = \Delta^-(x)$

⑤ underlined vertex

$i \to (-i)$

 +  $= 0$ 

x_i^0 が largest time とする

$F(x_1 \cdots x_i \cdots x_n) = -F(x_1 \cdots \underline{x_i} \cdots x_n)$  $i\Delta_F(x_i - x_j) + (-)\Delta^+(x_i - x_j) = 0$

$\sum_{\text{underlined}} F(x_1 \cdots x_n) = 0$ // $\quad \Delta + \Delta + \Delta + \Delta$

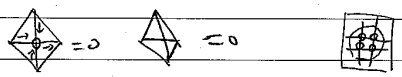
$F(x_1 \cdots x_n) + F(\underline{x_1} \cdots \underline{x_n}) = -\sum_{\text{underline}}' F(x_1 \cdots x_n)$

ψ, ϕ

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S行列 外線が付いている {x} について積分されている → $\Delta_F(p)$ =

momentum conservation



circled の連結領域が外線とつながっている

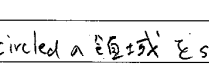
$S = 1 + iT$

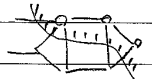
$S^+S = 1$

$i(T - T^+) = -T^+T$

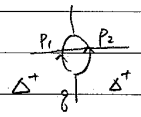
$i(\langle a|T|b\rangle - \langle a|T^+|b\rangle) = -\sum_c \langle a|T^+|c\rangle\langle c|T|b\rangle$

$2\mathrm{Re}\,F$ (Tで考えるIm) ; antiunp



in  circled の領域を shadow に入れる

$F + F^* = -\sum_{\text{antiunp}} F$



$$\int i\,i\,\frac{1}{p_1^2-m^2}\,i\,\frac{1}{p_2^2-m^2}\,i\,(2\pi)^4\delta^4(p_1+p_2-g)$$

$$\int \frac{1}{(2\pi)^3}\delta(p_1^2-m^2)\,\theta(p_1^0)\,\frac{1}{(2\pi)^3}\delta(p_2^2-m^2)\,\theta(p_2^0)\,(2\pi)^4\delta(p_1+p_2-g)\,d^4p_1\,d^4p_2 = \int \frac{1}{(2\pi)^2}\frac{d^3p_1}{2p_1^0}\frac{d^3p_2}{2p_2^0}\,\delta^4(p_1+p_2-g)$$

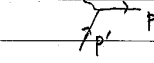
= ∩ の phase space on shell

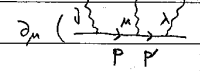
ex) $\frac{d\sigma}{d\Omega} = |f(\theta)|^2$

$\sigma_{\text{tot}} = \frac{4\pi}{|p|}\mathrm{Im}\,f(0)$

cophcal theorem

Ward identity





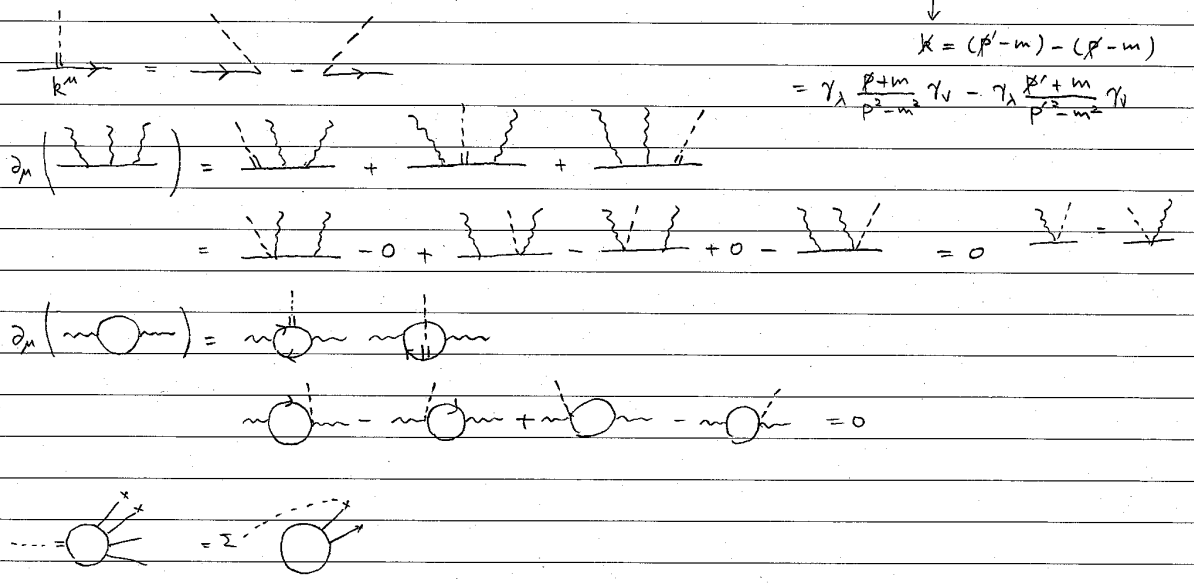
$p' = p + k$

▷ Jun 23

$\varepsilon_\mu \to k_\mu$ とした時 diagram = 0 $\left(\gamma_\lambda \frac{\not p' + m}{p'^2-m^2}\gamma_\mu \frac{\not p + m}{p^2-m^2}\gamma_\nu\right)$

$\not k = (\not p' - m) - (\not p - m)$

$= \gamma_\lambda \frac{\not p + m}{p^2-m^2}\gamma_\nu - \gamma_\lambda \frac{\not p' + m}{p'^2-m^2}\gamma_\nu$



Path integral による量子化

canonical formalism $Q_a(t) = e^{iHt} Q e^{-iHt}$

$Q_a(t)|q,t\rangle = q_a|q,t\rangle$ $\quad \langle q,t|p,t\rangle = \frac{1}{\sqrt{2\pi}} e^{ip_a q_a}$

$|q,t\rangle = e^{iHt}|q\rangle$

$H(Q(t),P(t)) = e^{iHt} H e^{-iHt} = H(Q,P)$

$\langle q' t'|q,t\rangle = \int dq_0 \cdots dq_N \langle q' t|q_N t_N\rangle \langle q_N t_N|q_{N-1} t_{N-1}\rangle \vdots \langle q_1 t_1|q,t\rangle$

N+1 個の積

$= \int \prod_{a,k} (dq_a)_N \frac{\prod_a dp_a}{2\pi} \exp\left[-\sum H(q_k,p_k)\Delta t + \sum p_{ak}(q_{ak}-q_{ak-1})\right]$

$q_{ak} = q_a(t_k)$

$p = \dot{q}\Delta t$

$= \int \prod_a dq_a(t) \frac{\prod_a dp_a}{2\pi} e^{i\left\{-\int_t^t dt\, H(q,p) - \sum_a p_a(t) q_a(t)\right\}}$ $k \to 0$ 連続となる極限. unique に決まるのは q'' が q に等しいとき

↳ E-L eq

$\langle q' t'|q,t\rangle = \int dq_1 \cdots dq_N \langle q' t|q_N t_N\rangle \langle q_N t_N|q_{N-1} t_{N-1}\rangle \vdots \langle q_1 t_1|q,t\rangle$

$= \int \prod_{a,k} (dq_a)_N \frac{\prod_a dp_a}{2\pi} \exp\left[-\sum H(q_k,p_k)\Delta t + \sum p_{ak}(q_{ak}-q_{ak-1})\right]$

$= \int \prod_a dq_a(t) \frac{\prod_a dp_a}{2\pi} e^{i\left\{-\int_t^t dt\, H(q,p) - \sum_a p_a(t) q_a(t)\right\}}$

operator

$\langle q' t+\Delta t|\Theta(P(t)\,Q(t))|q,t\rangle = \int$

$\langle q' t'|\Theta_A(P(t_a)Q(t_a))\,\Theta_B(P(t_b),Q(t_b))\cdots|q,t\rangle$

$= \int_{BC} \prod dq(t) \int \frac{\prod dp(\tau)}{2\pi} \Theta_A(P(t_a)Q(t_a))\,\Theta_B(\quad)\, e^{\int i\{\dot{q}p - H(q,p)\}dt}$

Gaussian integral

complex positive definite by real part

$\int_{-\infty}^{\infty} d\xi\, e^{-a\xi^2} = \sqrt{\frac{2\pi}{a}}$ $\quad \xi,\eta$ n dim space の vector $\int_{-\infty}^{\infty} d^n\xi\, e^{-(\xi, A\xi)} = \frac{1}{(\det A/2\pi)^{1/2}}$

$\int [d\xi]\, e^{-\frac{1}{2}(\xi A\xi)} = \frac{1}{(\det A/2\pi)^{1/2}}$

2次形式

$K(\xi) = \frac{1}{2}(\xi A\xi) + B\xi + C$ $\quad \frac{\partial K(\xi)}{\partial \xi} = 0$ $\quad \xi = -A^{-1}B$

$= \frac{1}{2}(\xi - \bar{\xi}, A(\xi - \bar{\xi})) + K(\bar{\xi})$

$K(\bar{\xi}) = -\frac{1}{2}(B, A^{-1}B) + C$

$\int [d\xi]\, e^{-K(\xi)} = e^{-K(\bar{\xi})} \frac{1}{(\det A/2\pi)^{1/2}}$

$q_a \to \phi(x)$

$p_a \to \pi(x)$

$\langle \phi(x't') t|\, \overline{\pi}\{\Theta(\pi,\phi)\cdots\}\, |\phi(x,t)\, t\rangle = \int [d\phi(x)] \left[\frac{d\pi(x)}{2\pi}\right] \Theta \cdots e^{i\int_t^{t'} dt \int d^3x (\phi(x)\pi(x) - H(\phi,\pi))}$

H が π の二次形式で表す

$H = \frac{1}{2} A(\phi)_{xy} \pi(x)\pi(y) + B(\phi(x))\pi(x) + C(\phi(x))$

$$\int\left[\frac{d\pi(x)}{2\pi}\right] e^{i\int_t^{t'}dt\int d^3x\,[\dot\phi(x)\pi(x)-\mathcal{H}]} = \int\left[\frac{d\pi(x)}{2\pi}\right] e^{i\,I[\phi,\pi]} = \det(2\pi i A)^{-1/2}\, e^{i\,I[\phi,\pi]}$$

$$\frac{\partial}{\partial\pi(x)}\left[\dot\phi(x)\pi(x)-\mathcal{H}(\phi,\pi)\right]=0 \qquad \dot\phi(x)=\left.\frac{\partial H}{\partial\pi(x)}\right|_{\pi=\bar\pi}$$

$$\dot\phi(x)\bar\pi(x)-\mathcal{H}(\phi(x),\bar\pi(x)) = \mathcal{L}(\phi(x),\dot\phi(x))$$

$$\int\left[\frac{d\pi(x)}{2\pi}\right] e^{i\,I(\phi,\pi)} = (\det 2\pi i A)^{1/2}\, e^{+i\int_t^{t'}dt\int d^3x\,\mathcal{L}(\phi,\dot\phi)}$$

operator の期待値 $\quad\Theta(\pi(x),\varphi(x))$ $\quad\pi$はたいがい$\dot\varphi$

$$\langle\phi(t')t'|\,T(\Theta_1(\phi)\Theta_2(\phi)\cdots)\,|\phi,t\rangle = \int[d\phi]\,(\det 2\pi i A)^{1/2}\,\Theta_1(\phi)\Theta_2(\phi)\cdots e^{+i\int_t^{t'}dt\int d^3x\,\mathcal{L}(\phi)}$$

Aが field を含む時は無視出来ない

$$\mathcal{L}=\int d^3x\,(\partial_\mu\vec\phi)(\partial_\mu\vec\phi) \qquad \vec\phi^{\,2}=1$$

$$\vec\phi=(\phi_1\cdots\phi_N)$$

Non-linear σ model $\qquad \phi_N=\sqrt{1-(\phi_1^2+\phi_2^2+\cdots\phi_{N-1}^2)}$

$$\int d^3x\sum_{i,j=1}^{N-1} g_{ij}(\phi)\,\partial_\mu\phi_i(x)\,\partial_\mu\phi_j(x)$$

$$g_{ij}=\delta_{ij}-\frac{\phi_i\phi_j}{1-\sum_{i=1}^{N-1}\phi_i^2}$$

O(N) Heisenberg $\quad\sum_i\vec S_i\vec S_{i+1} \qquad \vec S_{i+1}^{\,2}=\vec S_i^{\,2}=1$

$$=\sum(S_{i+1}-S_i)^2-2$$

$$=\sum(\nabla\vec S_i)^2-2$$

$$e^{i\mathcal{L}(\phi)/\hbar} \longleftrightarrow e^{-H/kT}$$

S 行列

$$\langle\beta_{out}|\,T\{\Theta\cdots\}\,|\alpha_{in}\rangle$$

QM $\longleftrightarrow$ static

$\hbar \longleftrightarrow k_B$

$$=\int\prod_t d\phi\,\langle\beta_{out}|\phi(t')\rangle\langle\phi(t')|\,T\{\Theta\cdots\}\,|\phi(t')\rangle\langle\phi(t')|\alpha_{in}\rangle$$

($|\phi(t')\rangle$: $+\infty$ 側, $|\phi(t')\rangle$: $-\infty$ 側)

α, β が vacuum 考える

$$a_{in}(\vec p)\,|0_{in}\rangle=0 \qquad a_{in}(\vec p)=\frac{1}{(2\pi)^{3/2}}\int\frac{d^3x}{\sqrt{2p^0}}\,e^{+ipx}\overleftrightarrow{\partial_0}\varphi(x) \qquad \pi=-i\frac{\delta}{\delta\varphi}$$

$$\langle\phi(-\infty)|\,a_{in}(\vec p)\,|0_{in}\rangle \qquad \dot\varphi-ip^0\varphi$$

$$\int d^3x\,\frac{1}{\sqrt{2E}}\,e^{-i\vec p\vec x}\left(\frac{\delta}{\delta\phi(x)}+E_p\phi(x)\right)\langle\phi(-\infty)|0_{in}\rangle=0 \qquad \downarrow\pi \quad \downarrow E_p$$

$$\left(\frac{d}{dx}+ax\right)\phi=0 \qquad \langle\phi(-\infty)|0_{in}\rangle = N\exp\left(-\frac{1}{\hbar}\int d^3x\,d^3y\,\phi(x)F(x-y)\phi(y)\right)$$

$$-\int d^3x\int d^3y\,F(x-y)\,\phi(y)e^{-i\vec p\vec x}+E_p\int e^{-i\vec p\vec x}\phi(x)=0$$

$$\int d^3y\left[\int d^3x\,F(x-y)e^{-i\vec p\vec x}-E_p e^{-i\vec p\vec y}\right]\phi(y)=0$$

$$F(x-y)=\frac{1}{(2\pi)^3}\int e^{ip(x-y)}d^3p\,E_p$$

$\langle 0_{out} | \phi(+\infty)\ t=+\infty \rangle \langle \phi(-\infty),\ t=-\infty | 0_{in} \rangle$ $\qquad f(+\infty) - f(-\infty) = \lim_{\epsilon \to 0} \epsilon \int dt\, e^{-\epsilon|t|} f(t)$

$= |N|^2 \exp\left(-\frac{1}{2} \int d^3x \int d^3y\, F(x-y) \underbrace{[\phi(x\,\infty)\,\phi(y\,\infty) + \phi(x\,-\infty)\,\phi(y\,-\infty)]}_{\epsilon \int_{-\infty}^{\infty} d\tau\, \phi(\tau)\, e^{-\epsilon|t|}} \right.$

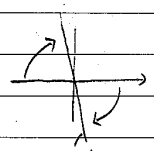
$\langle 0_{out} | T\{ O \cdots \} | 0_{in} \rangle$

$= \int [d\phi(x)] (\det 2\pi i A)^{-\frac{1}{2}}\, O \cdots$

$\exp i \left[\int d^4x\, \mathcal{L}(\phi) + \frac{i}{2} \epsilon \int d^3x\, d^3y\, d\tau\, F(x-y)\, e^{-\epsilon|t|}\, \phi(x,\tau)\, \phi(y,\tau) \right]$

$\epsilon \to 0$

$+i\epsilon$

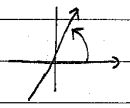


Euclid 化 $x^0 = t \to -ix^4$ $\quad x_E = (x^4, \vec{x})$

$\mathcal{L} = \frac{1}{2} \{ (\partial_0 \phi)^2 - (\nabla\phi)^2 - m^2\phi^2 \}$ $\quad e^{ikx}$ $\quad k_0 \to ik_4$

$-(\partial_0\phi)^2$

$\int e^{+\mathcal{L}}$ conv $\qquad$ Wick rotation



Green's functions $\qquad$ Schwinger's source function

$\mathcal{L}(x) \to \mathcal{L}(x) + J(x)\phi(x)$

$Z[J] = \langle 0_{out} | 0_{in} \rangle = e^{iW[J]} \qquad W[J] = \sum \frac{1}{n!} \int d^4x_1 \cdots d^4x_n\, G^n(x_1 \cdots x_n)\, J(x_1) \cdots J(x_n)$

$= \int [d\phi]\, e^{+i\int d^4x (\mathcal{L} + J\phi)}$

$-i \log Z[J]$

$i \frac{\delta W[J]}{\delta J(x_1)} = \frac{i \int [d\phi]\, \phi(x_1)\, e^{i\int d^4x [\mathcal{L} + J\phi]}}{\int [d\phi] e^{i\int d^4x (\cdots)}} = \frac{i \langle 0_{out} | \phi(x_1) | 0_{in} \rangle}{\langle 0_{out} | 0_{in} \rangle} = \langle \phi(x_1) \rangle = 0$

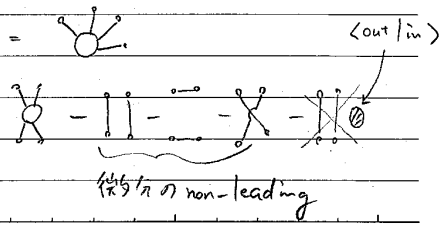
$i \frac{\delta^2 W[J]}{\delta J(x_1) \delta J(x_2)} = \frac{(i)^2 \int [d\phi]\, \phi(x_1)\phi(x_2) e^{i\int \cdots}}{\int [d\phi] \cdots} - \underline{(i) \int [d\phi]\, \phi(x_1) e^{\cdots}}\ \underline{(i) \int [d\phi]\, \phi(x_2) e^{\cdots}}$

$= (i)^2 \left\{ \frac{\langle 0_{out} | \phi(x_1)\phi(x_2) | 0_{in} \rangle}{\langle 0_{out} | 0_{in} \rangle} - \frac{\langle 0_{out} | \phi(x_1) | 0_{in} \rangle \langle 0_{out} | \phi(x_2) | 0_{in} \rangle}{\langle 0_{out} | 0_{in} \rangle^2} \right\}$

$= (i)^2 \langle \phi(x_1)\phi(x_2) \rangle$

$i \frac{\delta^n W[J]}{\delta J(x_1) \cdots \delta J(x_n)} \Big|_{J=0} = i^n \langle 0_{out} | T(\phi(x_1) \cdots \phi(x_n)) | 0_{in} \rangle_{conn}$

$W =$



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One particle irreducible (1PI) な Green func

$\varphi_c(x) = \frac{\delta W[J]}{\delta J(x)}$ Jについて解く. Legendre 変換 → φ_c

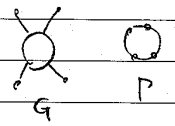
$-\Gamma[\varphi_c] = \int d^4x\, \varphi_c(x) J(x) - W[J]$

$-\frac{\delta \Gamma[\varphi_c]}{\delta \varphi_c(x)} = J(x) + \int d^4x \underbrace{\varphi_c \frac{\delta J}{\delta \varphi} - \frac{\delta W}{\delta J}\frac{\delta J}{\delta \varphi}}_{0}$

$\Gamma[\varphi_c] = \sum \frac{1}{n!} \int d^4x_1 \cdots d^4x_n\, \Gamma^n(x_1 \cdots x_n)\, \varphi_c(x_1) \cdots \varphi_c(x_n)$

Γ^n ~~amputated~~ の PI) Green func

amputated



$\varphi_c(x) = \frac{\delta W}{\delta J(x)}$

$\frac{\delta}{\delta \varphi_c(y)} \delta^4(x-y) = \frac{\delta}{\delta \varphi_c(y)} \frac{\delta W}{\delta J(x)} = \int \frac{\delta J(z)}{\delta \varphi_c(y)} \frac{\delta^2 W}{\delta J(z) \delta J(x)} d^4z = \int \frac{\delta^2 \Gamma}{\delta \varphi_c(y) \delta \varphi_c(z)} \frac{\delta^2 W}{\delta J(z) \delta J(x)} d^4z$

$= \int d^4z\, \Gamma^{(2)}(y,z)\, G^{(2)}(z,x)$

$\frac{\delta}{\delta \varphi_c(w)} \times$

$0 = \int \frac{\delta^3 \Gamma}{\delta \varphi_c(w) \delta \varphi_c(y) \delta \varphi_c(z)} \frac{\delta^2 W}{\delta J(z) \delta J(x)} + \frac{\delta J(u)}{\delta \varphi_c(w)} \frac{\delta^2 W}{\delta J(w) \delta J(z) \delta J(x)}$

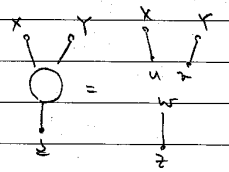
$\frac{\delta \Gamma}{\delta \varphi_c(y) \delta \varphi_c(z)} - \frac{\delta \Gamma}{\delta \varphi_c(u) \delta \varphi_c(w)}$

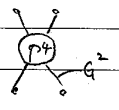
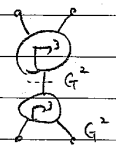
$\int dz\, \Gamma^3(w,y,z)\, G^2(z,x) = \int dw dz\, \Gamma^2(y,z)\, \Gamma^2(w,u)\, G^3(u,z,x)$

$\times \int \Gamma^3(x,v)\, dx$

$\Gamma^{(3)}(w,y,v) = \int \Gamma^{(2)}(w,z)\, \Gamma^2(y,z)\, \Gamma^2(v,x)\, G(u,z,x)\, du dz dx$

$G^3(x,y,z) = \int du dv dw\, G^2(xy)\, G^2(xu)\, G^2(zw)\, \Gamma^3(u,v,w)$



$G^4 =$  $+ \sum$ 

$G^n(x_1 \cdots x_n) = \int \frac{d^4p_1}{(2\pi)^4} \cdots \frac{d^4p_n}{(2\pi)^4} e^{i(p_1x_1 + \cdots p_nx_n)} \bar{G}(p_1 \cdots p_n)$

$\bar{G}(p_1 \cdots p_n) = \int d^4x_1 \cdots d^4x_n\, e^{-i(p_1x_1 + \cdots p_nx_n)} G(x_1 \cdots x_n)$

$(2\pi)^4 \delta^4(p_1 + \cdots p_n)\, G(p_1 \cdots p_n)$ Γ を用いた

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$\Gamma(\phi)$ を action integral とおく $= \int (\text{eff } \mathcal{L}_1) d^4x$

$$\int [d\phi] e^{i[\Gamma(\phi) + \int J\varphi]/\hbar}$$

$\hbar \to 0$ $\left.\frac{\partial \Gamma(\varphi)}{\partial \varphi}\right|_{\varphi=\varphi_c} = -J$ tree が $\hbar \to 0$ で あらわれる

$e^{iW[J]}$ or $\hbar \to 0$ loop 展開 $\Gamma(\varphi)$ effective pot

▷ Jun 30

$$\int [d\xi] \underbrace{e^{-\left[\frac{1}{2}(\xi A \xi) + (b\xi) + c\right]}}_{g(\xi)} = g\left(-\frac{\partial}{\partial b}\right) \int [d\xi] e^{\cdots}$$

ex $\int [d\xi] \xi_{\alpha_1} \cdots \xi_{\alpha_{2r}} e^{-\frac{1}{2}(\xi A \xi)} = \frac{1}{(\det A/2\pi)^{1/2}} \sum_{\text{neiboring of } \alpha_1 \cdots \alpha_{2r}} \prod_{\alpha'\alpha''} (A)^{-1}_{\alpha'\alpha''}$ ($\xi = A^{-1/2}\xi'$)

2 point function (free field)

$\mathcal{L} = \frac{1}{2}(\partial_\mu \varphi)^2 - \frac{1}{2} m^2 \varphi^2$ $\quad e^{-\frac{1}{2}(\varphi A \varphi)}$

状態和 $Z = \int [d\varphi(x)] e^{i\int d^4x \left\{\frac{1}{2}(\partial_\mu \varphi)^2 - \frac{1}{2} m^2 \varphi^2\right\}}$

$$\frac{\langle 0 \text{ out} | T[\varphi(x), \varphi(y)] | 0 \text{ in}\rangle}{\langle 0 \text{ out} | 0 \text{ in}\rangle} = \underbrace{\frac{i}{-\partial^2 - m^2 + i\epsilon}}_{\left(\frac{1}{A}\right)_{xy}} = i \frac{1}{(2\pi)^4} \int d^4p \frac{e^{-ip(x-y)}}{p^2 - m^2 + i\epsilon} = i\Delta_f(x-y)$$

gauge field

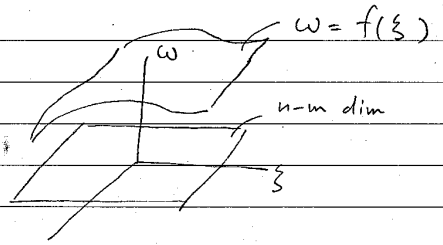
$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} = A_\mu \underbrace{A}_{(\partial^2 \eta^{\mu\nu} - \partial^\mu \partial^\nu)} A_\nu$ $\qquad \mathcal{L} \to \mathcal{L} - \frac{1}{2\alpha}(\partial_\mu \Lambda^\mu)^2$

gauge の問題を正しく扱う

m 個

$[\varphi] = [\xi] + [\omega]$

physical gauge



$$Z = \int [d\xi] e^{iI} = \int [d\varphi] e^{iI} \delta^m[\omega] = \int [d\varphi] e^{iI} \delta^m[\omega - f(\xi)]$$

gauge function $F(\varphi) = 0 \to \omega - f(\xi) = 0$

$$\delta(F(\varphi)) \det\left|\frac{\partial F}{\partial \omega}\right| = \delta(\omega - f(\xi))$$

$$Z = \int [d\varphi] \prod_m \delta(F^m(\varphi)) \det\left|\frac{\partial F_m}{\partial \omega}\right| e^{iI}$$

em field $m = 1$

$$F = \partial_\mu A^\mu - f = 0$$

$\delta A_\mu = -\partial_\mu \theta$

$\delta F = -\partial^2 \theta$

$\frac{\delta F}{\delta \theta} = -\partial^2$

$Z = \int [dA_\mu]\, e^{iI}\, \delta(\partial_\mu A^\mu - f)\, \det(-\partial^2)$

$\det$ を $\int df\, G(f)$ から

$G(f) = \exp\left(-\frac{i}{2\alpha}\int d^4x\, f\right)$ とおくことにする

$= N \int [dA_\mu]\, e^{i\left(I - \frac{1}{2\alpha}\int \partial_\mu A^\mu d^4x\right)}$

fermion field

scalar — c-number

Dirac — anhicommuting c-number (反可換)

Grassmann algebra

$\int e^{\bar\eta A \eta + \bar\eta \xi + \bar\xi \eta}\, d\bar\eta\, d\eta$ ($\xi, \bar\xi$: source field)

$= e^{-\bar\xi A^{-1} \xi} \int e^{(\bar\eta + \bar\xi A^{-1})A(\eta + A^{-1}\xi)}\, d\bar\eta\, d\eta$

$\int d\eta\, d\bar\eta\, e^{+\bar\eta A \eta} = \int d\bar\eta\, d\eta\, (1 + \bar\eta A \eta)$

$= -A$

自由度 $2n$ の場合

$\eta_1 \cdots \eta_n,\ \bar\eta_1 \cdots \bar\eta_n$

$\int \prod_i (d\bar\eta_i\, d\eta_i)\, f(\bar\eta, \cdots \eta, \cdots) = f_{1\cdots n \bar n \cdots \bar 1}$

f を展開 $f_{1\cdots n\bar n\cdots\bar 1}$ は $\eta_1 \cdots \eta_n \bar\eta_n \cdots \bar\eta_1$ の係数

自由度 1 のとき $\eta, \bar\eta$

$\eta\bar\eta + \bar\eta\eta = 0,\ \eta^2 = 0,\ \bar\eta^2 = 0$

function $f(\eta, \bar\eta) = f_{00} + f_{10}\bar\eta + f_{01}\eta + f_{11}\eta\bar\eta$

derivative $\partial f = \frac{\partial f}{\partial \eta} = f_{01} + f_{11}\bar\eta,\ \partial^2 f = 0$

$\bar\partial f = \frac{\partial f}{\partial \bar\eta} = f_{10} - f_{11}\eta,\ \bar\partial^2 f = 0$

integral $\int d\eta\, \eta = 1,\ \int d\bar\eta\, \bar\eta = 1,\ \int d\eta\, 1 = \int d\bar\eta\, 1 = 0$

$\int d\eta\, f(\eta\bar\eta) = f_{01} + f_{11}\bar\eta = \partial f$

$\int d\bar\eta\, f(\eta\bar\eta) = \bar\partial f$

$\int d\bar\eta\, d\eta\, f(\eta\bar\eta) = \bar\partial \partial f$

exercise

$\int \prod (d\bar\eta_i)(d\eta_i)\, e^{\sum \bar\eta_i A_{ij} \eta_j} = (-)^n \det A$

$\int \prod d\bar\eta_i\, d\eta_i\, e^{\sum \bar\eta_i A_{ij} \eta_j + \sum(\bar\eta_i \xi_i + \bar\xi_i \eta_i)} = e^{-\bar\xi_i (A^{-1})_{ij} \xi_j}\, (-1)^n \det A$

$Z_J = \int [d\psi][d\bar\psi]\, e^{-i\int d^4x \left[\bar\psi(x)(i\partial\!\!\!/ - m)\psi(x) + \bar\psi(x)\xi(x) + \bar\xi\psi\right]}$

$\frac{\langle 0\ \text{out} | T(\psi(x)\bar\psi(y)) | 0\ \text{in}\rangle}{\langle 0\ \text{out} | 0\ \text{in}\rangle} = \frac{i}{i\partial\!\!\!/ - m}$

$G^2(p,-p) = i\langle\varphi\varphi\rangle = -\Delta_F(p) = \dfrac{-1}{p^2-m^2}$

$P^2(p-p) = \dfrac{1}{\Delta_F(p)} = p^2-m^2$

$\varphi P^2(p-p)\varphi = \varphi(-\partial^2-m^2)\varphi$

摂動論

$$Z[J] = \int [d\varphi(x)]\, e^{i\int d^4x \left[\frac{1}{2}(\partial_\mu\varphi)^2 - \frac{1}{2}m^2\varphi^2 + J(x)\varphi(x)\right]} \qquad (\varphi A \varphi)$$

$$= e^{-\frac{1}{2}\int d^4x\, d^4y\, J(x)\, i\Delta(x-y)\, J(y)} \qquad (A^{-1},\ -i\frac{\delta}{\delta J(x)})$$

$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}'$ $\quad Z[J] = \langle 0\ out | 0\ in\rangle_J = \int [d\varphi]\, e^{i\int d^4x \mathcal{L}'(\varphi)}\, e^{-\frac{1}{2}\cdots}$

$$= \exp\left(i\int d^4x\, \mathcal{L}\left(-i\frac{\delta}{\delta J(x)}\right)\right) \exp\left\{-\frac{1}{2}\int d^4x\, d^4y\, J(x)\, i\Delta(x-y)\, J(y)\right\}$$

- $f\left(-i\frac{\partial}{\partial x}\right) g(x) = g\left(-i\frac{\partial}{\partial y}\right) f(y)\, e^{i(x,y)}\Big|_{y=0}$

証明) Fourier transform

$f(x) = e^{i(a\cdot x)}$ $\qquad e^{a\frac{\partial}{\partial x}} F(x) = \left(1 + a\frac{\partial}{\partial x} + \frac{a^2}{2!}\frac{\partial^2}{\partial x^2} + \cdots\right) F(x) = F(x+a)$

$g(x) = e^{i(b\cdot x)}$

$LHS = e^{\left(a\frac{\partial}{\partial x}\right)} e^{i(bx)} = e^{i(bx+a)}$

$RHS = e^{\left(b\frac{\partial}{\partial y}\right)} e^{i(ax)+i(xy)} = e^{b\frac{\partial}{\partial y}} e^{i(a+x,y)} = e^{i(a+x\ \ b+y)}$

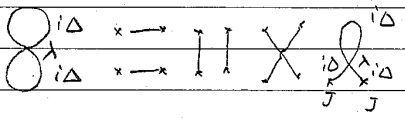
$x \to J,\ y \to \varphi$

$$Z[J] = e^{\frac{1}{2}\int d^4x\, d^4y\, i\Delta(x-y)\frac{\delta}{\delta\varphi(x)}\frac{\delta}{\delta\varphi(y)}}\, e^{i\int d^4z\,[\mathcal{L}'[\varphi(z)] + J(z)\varphi(z)]}\Big|_{\varphi=0}$$

$$Z[J] = \left[1 + \frac{1}{2}\int i\Delta \frac{\delta}{\delta\varphi}\frac{\delta}{\delta\varphi} + \frac{1}{2}\left(\frac{1}{2}\int i\Delta\frac{\delta}{\delta\varphi}\frac{\delta}{\delta\varphi}\right)(\cdots)\right] \qquad \mathcal{L}' = -\frac{1}{4!}\lambda\varphi^4$$

$$\left[1 + i\left(-\frac{\lambda}{4!}\varphi^4 + J\varphi\right) + \frac{1}{2!}i^2\left(-\frac{\lambda}{4!}\varphi^4 + J\varphi\right) + \cdots\right]$$

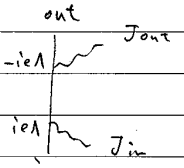
$J(x) \xrightarrow{i\Delta} J(x)$



S行列

$J \to J_{in}(x) = \lim_{p^2\to m^2} f_p(x)(\Box^2+m^2)$

$\to J_{out}(x) = \lim_{p^2\to m^2} f_p^{\lambda}(x)(\Box+m^2)$



exersise

Compton 散乱の Feynmann graph を書け (factorial 注意)

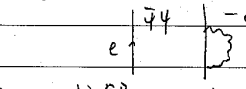
$$\frac{1}{6!}(-ieA + JA + \bar{\xi}\psi + s\bar{\psi})^6$$

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stability of vaccumm $|0\ out\rangle = |0\ in\rangle$ $S_{00}=1$

" 1 particle state $|p\ out\rangle = |p\ in\rangle$ $S_{pp}=1$

our lagrangian $\mathcal{L} = \mathcal{L}_0 + \mathcal{L}'$



$|p\ out\rangle \neq |p\ in\rangle$

lagrangian の分解は unique でない $\mathcal{L} = \mathcal{L}^0 + \mathcal{L}'$ $\quad -\delta m\bar\psi\psi + \cdots$

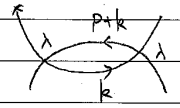
$= \mathcal{L}^{*0} + \mathcal{L}^{*\prime}$ $\quad \mathcal{L}^{*\prime} = \mathcal{L}' + \delta m\bar\psi\psi$ $\quad S_{pp}=1$

$\mathcal{L}^{*0} = \mathcal{L}' - \delta m\bar\psi\psi$

$-m_0\bar\psi\psi$ $\quad m_0+\delta m = m$

example $\lambda\varphi^4$ theory

$$\mathcal{L} = \frac{1}{2}(\partial_\mu\varphi)^2 - \frac{1}{2}m^2\varphi^2 - \frac{\lambda}{4!}\varphi^4$$



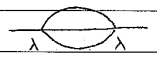
$= \frac{1}{2}$

$$\frac{1}{2}\int \lambda^2 \frac{d^4k}{(k^2-m^2)((p+k)^2-m^2)} = \frac{i}{32\pi^2}\ln\Lambda^2 \int^\Lambda d^4k + O(\text{finite})$$

$\frac{1}{2!}\ \frac{1}{4!}\ \frac{1}{4!}$ $\quad 8_{C3}\times 4\times 3\times 2$

counter term $\Delta\mathcal{L} = \left(-\frac{\lambda^2}{32\pi^2}\ln\Lambda^2\right)\frac{\lambda}{4!}\varphi^4 = -C\frac{\lambda}{4!}\varphi^4$

$\mathcal{L}' + \Delta\mathcal{L}$:



$\longrightarrow -\frac{1}{2}\left[A(\partial_\mu\varphi)^2 - B\varphi^2\right]$

$i\Delta\mathcal{L}$

$\Delta\mathcal{L} = -\frac{\lambda}{4!}\varphi^4 + \frac{1}{2}A(\partial_\mu\varphi)^2 - \frac{1}{2}B\varphi^2$

$\mathcal{L}' \to \mathcal{L}' + \Delta\mathcal{L}$ $\quad Z_3$

$\mathcal{L} \to \mathcal{L} + \Delta\mathcal{L} = \frac{1}{2}(1+A)(\partial_\mu\varphi)^2$

$-\frac{1}{2}(m^2+B)\varphi^2$

$Z_3\delta m^2$ $\quad -(1+C)\frac{\lambda}{4!}\varphi^4$

Z_1

$$\mathcal{L} + \Delta\mathcal{L} = \frac{1}{2}(\partial\varphi_0)^2 - \frac{1}{2}m_0^2\varphi_0^2 - \frac{\lambda_0}{4!}\varphi_0^4$$

$m_0\ \lambda_0$ は cutoff dependent

$$\begin{cases} \varphi_0 = \sqrt{Z_3}\,\varphi \\ \quad = \frac{1}{2}Z_3\left[(\partial_\mu\varphi)^2 + (m^2+\delta m^2)\varphi^2 + \frac{Z_1}{Z_3^2}\lambda\varphi^4\right] \\ m^2+\delta m^2 = m_0^2 \\ \frac{Z_1}{Z_3^2}\lambda = \lambda_0 \end{cases}$$

Z_1 Z_3 renormalization constant

何か ∞ を消す条件では unique でない

Dyson → $p^2=m^2$ で "free" → その finite term が決まる

Bogalubov → $p=0$ "

⋮ renormalization condition

Superficial degree of divergence

D = (# mon in numerator) − (# mon in denumerator)

$\mathcal{L}' = \sum \mathcal{L}_i'(\underbrace{\partial\cdots}_{d_i}\ \underbrace{\varphi\cdots}_{b_i}\ \underbrace{\psi\cdot\bar\psi\cdots}_{f_i})$

B 外線が Boson の数 $\quad \sum n_i b_i = B + 2IB$

IB 内 " $\quad \sum n_i f_i = F + 2IF$

n_i vertex i の数 $\quad$ F 外線 fermion

IF 内 "

$D = \sum n_i d_i + (4-2)\underbrace{I_B}_{B - \sum n_i b_i} + (4-1) I_F - 4(\sum n_i - 1)$

$\delta^4(I_p)$

$D = \sum n_i d_i - B - \frac{3}{2}F + 4 - \sum n_i b_i + \frac{3}{2}\sum n_i f_i - 4\sum n_i = -B - \frac{3}{2}F + 4 + \sum n_i \delta_i$

$d_i + b_i + \frac{3}{2} f_i - 4$

$\delta_i = d_i + b_i + \frac{3}{2} f_i - 4$

$\delta = 0$ D は外線数無にまる　発散は field res by counter term に吸収できる

$\delta > 0$ D は vertex 数にまる　無限の counter term くりこみ不可

Formal argument

1PI Green function $\Gamma(p, p', \cdots)$　$\Gamma = a + b\not{p} + b'\not{p}' + \cdots + cp^2 + c'\not{p}\not{p}' + c''p'^2 + \cdots$

1) 摂動力の多次で 1PI を計算

$D < 0$

2) $D \geq 0$ Taylor 展開の D 次項以下が打消すように counter term

3) correcteal lagrangian で次の order を計算

Regularization

1. momentum cut-off $\int^{\Lambda} dk$

gauge inv にわず

2. Pauli-Villars の方法

gauge inv みたす

regulator field を導入

$\frac{1}{k^2 - m^2 + i\epsilon} \to \sum_i c_i \frac{1}{k^2 - M_i^2 + i\epsilon}$　$c_1 = 1$　$M_1 = m$

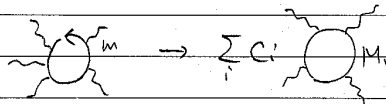
$\sum_i c_i \frac{1}{k^2 - M_i^2 + i\epsilon} = \sum_i c_i \frac{1}{k^2} \frac{1}{1 - \frac{M_i^2}{k^2}} = \sum c_i \frac{1}{k^2}\left(1 + \frac{M_i^2}{k^2} + \left(\frac{M_i}{k}\right)^4 \cdots\right)$

$c_1 = 1$　$M_1 = m$　$\sum c_i = 0$　leading cancl

$c_2 = -1 - \varepsilon$　$M_2 = m/\sqrt{\varepsilon} = M$　$\sum c_i M_i^2 = 0$　next leading cancel

$c_3 = \varepsilon$　$M_3 = m/\varepsilon$　計算後に aux field の mass $\to \infty$

$M \to \infty$



conversion integral $\int \frac{d^4k}{(k^2 - c + i\epsilon)} = \frac{\pi^2}{2i\epsilon}$

divergent integral $\int \frac{(k^2)^{m-2}}{(k^2 - c + i\epsilon)^n} d^4k = \frac{i\pi^2}{(-c)^{n-m}} \frac{\Gamma(m)\Gamma(n-m)}{\Gamma(n)}$

$\frac{1}{(k^2 - \Delta_i^2)} \to \sum c_i \int d^4k \frac{1}{(k^2 - \Delta_i^2 + i\epsilon)^2} = -i\pi^2 \sum c_i \ln \Delta_i^2$

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3) n次元法 ▷ Jul 7

$g_{\mu\nu} = (-1, +1, +1, +1)$ $\quad 4 \to n$

$$I(m) = \int \frac{d^4k}{(2\pi)^4} \frac{1}{(k^2-\Delta^2+i\varepsilon)^m} \to \int \frac{d^nk}{(2\pi)^n} \frac{1}{(k^2-\Delta^2+i\varepsilon)^m} \sim \Gamma(m-\tfrac{n}{2})$$

$2m = n$ log

Dirac algebra

$\eta_{\mu\nu}\eta^{\mu\nu} = n$

$\gamma_\mu \gamma^\mu = n \cdot 1$

$\{\gamma_\mu, \gamma_\nu\} = 2\eta_{\mu\nu}$

$\gamma_\mu \gamma^\alpha \gamma^\mu = (2-n)\gamma^\alpha$

$\gamma_\mu \gamma^\alpha \gamma^\beta \gamma^\mu = 4\eta^{\alpha\beta} + (n-4)\gamma^\alpha\gamma^\beta$

$\gamma_\mu \gamma^\alpha \gamma^\beta \gamma^\gamma \gamma^\mu = -2\gamma^\gamma\gamma^\beta\gamma^\alpha + (n-4)\gamma^\alpha\gamma^\beta\gamma^\gamma$

trace formula 4Dの時と同じ

Sym integer $k^\mu k^\nu \to \frac{1}{4}\eta^{\mu\nu}k^2 \to \frac{1}{n}\eta^{\mu\nu}k^2$

$$I(r,m) = \int \frac{d^nk}{(2\pi)^n} \frac{(k^2)^r}{(k^2-\Delta^2+i\varepsilon)^m} = i\frac{(-)^{r-m}}{(4\pi)^{n/2}} (\Delta^2)^{r-m+\frac{n}{2}} \frac{\Gamma(r+\frac{n}{2})\Gamma(m-r-\frac{n}{2})}{\Gamma(\frac{n}{2})\Gamma(m)}$$

$\dim[\mathcal{L}] = n$ $\quad [\phi] = \frac{n-2}{2}$ $\quad [\psi] = \frac{n-1}{2}$ $\quad [e] = n-(n-1)-\frac{n-2}{2} = \frac{4-n}{2}$

$\int d^nx\,\mathcal{L}$ $\qquad (\partial_\mu\phi)^2$ $\qquad e\bar{\psi}A\psi$

$k^0 = ik^4$ $\quad \vec{k} = \vec{k}$ $\quad i\int \frac{d^nk_E}{(2\pi)^n} \frac{(k_E^2)^r}{(k_E^2+\Delta^2)^m} (-)^{m-r}$ $\qquad \int \frac{d^nx}{(2\pi)^n} \frac{x^{2r}}{(x^2+1)^m} (-1)^{m-r} (\Delta^2)^{r-m+\frac{n}{2}}$

$k_E = \Delta \cdot x$

$$\int d^nx = \int x^{n-1}dx \int_0^{2\pi} d\phi \int_0^{\pi} d\theta_3 \sin\theta_3 \int_0^{\pi} d\theta_4 \sin\theta_4 \cdots \int_0^{\pi} d\theta_n \sin\theta_n \qquad \int_0^{\pi} \sin^n\theta\, d\theta = \pi^{1/2} \frac{\Gamma(\frac{n+1}{2})}{\Gamma(\frac{n}{2}+1)}$$

$$= \int z^{n-1}dz \frac{2\pi^{n/2}}{\Gamma(n/2)}$$

$\Gamma(1) = 1$ $\quad \Gamma(1/2) = \sqrt{\frac{1}{\pi}}$? $\quad \Gamma(n+1) = n\Gamma(n)$ $\quad \Gamma(n) = \frac{2^{n-1}}{\sqrt{n}} \Gamma(\frac{n}{2})\Gamma(\frac{n+1}{2})$

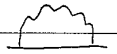
$\Gamma'(1) = -\gamma = -0.5772$

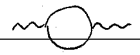
4) lattice regularization $\qquad$ eg. regularization → 何かをこわす

$M \to \infty$ にした時 recover し易いもの

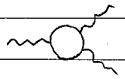
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QED divergent なグラフ


D=+1 log
counter term は 2ヶ


D=+2 conserved current
−2 D=0 log


D=0 log


D=+1
=0 Furry の定理

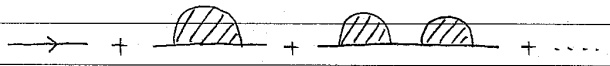

D=0 conv
−4

くりこみの定式化

$\mathcal{L} = \bar{\psi}_0 (i\not{\partial} - m_0)\psi_0 + int$

0 : bare = unrenormalized

electron self energy



$iS_F'(p)\ \ iS_F^0(p) + iS_F^0(p)(-i\Sigma(p))S_F^0(p) + \cdots$ → $\frac{1}{\not{p}-m}\frac{1}{1+B}$

$iS_F'(p) = iS_F(p)\frac{1}{1-\Sigma(p)S_F(p)} = \frac{i}{S_F^{-1}(p)-\Sigma(p)}$

$S_F^{-1}(p) - \Sigma(p) = \not{p} - m_0 - A - (-\not{p}+m_0)B + \tilde{\Sigma}(p)$

$= (1+B)\left[(\not{p}-m_0) - \frac{A}{1+B}\right]$

$m_0 + \delta m$

$\Sigma(p) = A + (-\not{p}+m_0)B + \tilde{\Sigma}(p)$

$\tilde{\Sigma}(p) = O(-\not{p}+m)^2$

renormalised pert'n theory

$\psi_0 = Z_2^{1/2}\psi$

$m_0 = m - \delta m$

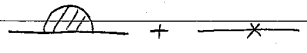
$\mathcal{L} = \bar{\psi}(i\not{\partial} - m)\psi + int$

$-i\Sigma = -i\left(A + B(-\not{p}+m) + \tilde{\Sigma}(p)\right)$

$\mathcal{L}_c = a\bar{\psi}i\not{\partial}\psi + b\bar{\psi}\psi$

log 発散

$-i\Sigma = -i(A + B(-\not{p}+m) + \tilde{\Sigma}(p) + i(a\not{p}+b)$



$S_F'(p) = \frac{1}{S_F^{-1}(p) - \Sigma(p)}$

$S_F'(p)\big|_{\not{p}=m} = \frac{1}{\not{p}+m}$ くりこみ条件

$\Sigma(m) = 0,\ \frac{\partial}{\partial \not{p}}\Sigma(p)\Big|_{\not{p}=m} = 0$

$\mathcal{L}_c = -B\bar{\psi}(i\not{\partial}-m)\psi + A\bar{\psi}\psi = -(1-Z_2)\bar{\psi}(i\not{\partial}-m)\psi + Z_2\delta m\bar{\psi}\psi$

とかく

$A - (am+b) = 0$

$B + a = 0$

$\begin{cases} a = -B \\ b = A + Bm \end{cases}$

$S_F'(p) = \frac{1}{S_F^{-1}(p) - \tilde{\Sigma}(p)}$

$\mathcal{L}_0 = \bar{\psi}(i\partial\!\!\!/ - m)\psi$ (phys)

$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}' = Z_2\bar{\psi}(i\partial\!\!\!/ - m)\psi + Z_2\delta m\,\bar{\psi}\psi = \bar{\psi}_0(i\partial\!\!\!/ - m)\psi_0 + \delta m\,\bar{\psi}_0\psi_0 = \bar{\psi}_0(i\partial\!\!\!/ - m_0)\psi_0$

cut-off dependent

photon self-energy

$\mathcal{L} = -\frac{1}{4}F^0_{\mu\nu}F^{0\mu\nu} + \text{int} - \frac{1}{2\alpha_0}(\partial_\mu A^\mu)^2$

$iD'_{F\mu\nu} = iD_{F\mu\nu} + iD_{F\mu\lambda}\overbrace{i\pi^{\lambda\rho}}^{1PI} iD_{F\rho\nu} + \cdots$



$D^0_{F\mu\nu} = \frac{-\eta_{\mu\nu} + \frac{g_\mu g_\nu}{g^2}}{g^2} - \alpha_0\frac{1}{g^2}\frac{g_\mu g_\nu}{g^2}$

$\pi^{(g)}_{\mu\nu} = (-\eta_{\mu\nu}g^2 + g_\mu g_\nu)e_0^2\,\pi(g^2)$



∂_μ (…) $= 0$

vacuum polarization tensor
strictly gauge invariant
photon は massless

$D'_{F\mu\nu}(g) = \frac{-\eta_{\mu\nu} + \frac{g_\mu g_\nu}{g^2}}{g^2[1 + e_0^2\pi(g^2)]} - \alpha_0\frac{1}{g^2}\frac{g_\mu g_\nu}{g^2}$

perturbation → renormalized pert'n theory

$A^0_\mu = Z_3^{1/2}A_\mu$

$e_0^2\pi(g^2) = C + \underbrace{Dg^2 + \cdots}_{e_0^2\tilde{\pi}(g)}$

$\frac{1}{\log}$

$\mathcal{L}_0 = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\alpha}(\partial_\mu A^\mu)^2$

$\mathcal{L}_c = c\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$

$e^2\Pi_{\mu\nu}(g) = (-\eta_{\mu\nu}g^2 - g_\mu g_\nu)(\overbrace{C - c + \tilde{\pi}(g^2)}^{\pi(g^2)})$

ren cond. $g^2 \to 0$ $D_F' \sim \frac{1}{g^2}$ $\tilde{\pi}(g^2) \sim O(g^2)$

$c = C = 1 - Z_3$

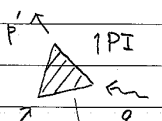
$\mathcal{L}_c = (1 - Z_3)\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$

$\mathcal{L}_0 =$ $\alpha_0 = Z_3\alpha$

$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_c = -\frac{Z_3}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\alpha}(\partial_\mu A^\mu)^2 = -\frac{1}{4}F^0_{\mu\nu}F^{0\mu\nu} - \frac{1}{2\alpha_0}(\partial_\mu A^{0\mu})^2$

Vertex part

$\langle 0|\bar{\psi}\psi A|0\rangle = iS_F\,iS_F\,iD_F\,\overset{\wedge}{-e}\,iP$



$= -ie\Gamma_\mu(p'p)$

$\Gamma_\mu(p'p) = \gamma_\mu + \Lambda_\mu(p'p)$

$\Lambda_\mu = D\gamma_\mu + \tilde{\Lambda}_\mu(p'p)$

log 発散 / 有限

$\bar{u}(p)\bar{\Lambda}_\mu(p,p)u(p) = 0$

$\langle\rangle$ = 対条件

$\bar{u}(p)\Gamma_\mu(pp)u(p) = \bar{u}(p)\gamma_\mu u(p)$

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$\mathcal{L}_c = D e \bar{\psi} A \psi$

$-ie\Gamma_\mu = -ie(\gamma_\mu + D\gamma_\mu - D\gamma_\mu + \tilde{\Lambda}_\mu)$

$D = -1 + z_1$

$\mathcal{L}_c = -(z_1 - 1) e \bar{\psi} A \psi$

$\mathcal{L}_{int} = -e \bar{\psi} A \psi$

$\mathcal{L}_{int} + \mathcal{L}_c = -z_1 e \bar{\psi} A \psi$

$= -z_1 e \frac{1}{z_2 z_3^{1/2}} \bar{\psi}_0 A_0 \psi_0 = -e_0 \bar{\psi}_0 A_0 \psi_0$

$\boxed{e_0 = e \frac{z_1}{z_2 z_3^{1/2}}}$ $z_1 = z_2$ ward identity

Summary

$\mathcal{L} = \bar{\psi}(i\partial\!\!\!/ - m)\psi - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - e \bar{\psi} A \psi$

くりこみ条件を満す様に

$\mathcal{L}_c = -(1 - z_2)\bar{\psi}(i\partial\!\!\!/ - m)\psi + \frac{1}{4}(1 - z_3) F_{\mu\nu} F^{\mu\nu}$

$+ z_2 \delta m \bar{\psi}\psi \qquad + (1 - z_1) e \bar{\psi} A \psi$

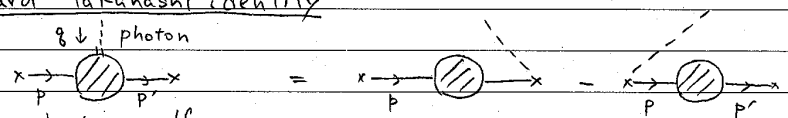
$\mathcal{L} + \mathcal{L}_c = z_2 \bar{\psi}(i\partial\!\!\!/ - m)\psi - z_1 e \bar{\psi} A \psi - \frac{z_3}{4} F_{\mu\nu} F^{\mu\nu}$

$+ z_2 \delta m \bar{\psi}\psi$

$= \bar{\psi}_0 (i\partial\!\!\!/ - m_0)\psi_0 - \frac{1}{4} F^0_{\mu\nu} F^{0\mu\nu} - ie_0 \bar{\psi}_0 A_0 \psi_0$

$= \underbrace{\frac{1}{z_2} \bar{\psi}_0 (i\partial\!\!\!/ - m_0)\psi_0 - \frac{1}{z_3} \frac{1}{4} F^0_{\mu\nu} F^{0\mu\nu}}_{\text{free}} + \frac{z_2 - 1}{z_2} \psi_0 (i\partial\!\!\!/ - m_0)\psi_0 - \frac{z_3 - 1}{z_3} \frac{1}{4} F^0_{\mu\nu} F^{0\mu\nu} - e_0 \bar{\psi}_0 A_0 \psi_0$

Ward - Takahashi identity



$S_F(p')\, i\partial_\mu \Gamma^\mu(p'p) S_F'(p) = S_F'(p) - S_F'(p')$

$i\partial_\mu \Gamma^\mu(p'p) = S_F'(p')^{-1} - S_F(p)^{-1}$

$(p'-p)_\mu \Gamma^\mu(p'p) = S_F'(p')^{-1} - S_F(p)^{-1}$

$(p'-p)_\mu \gamma^\mu = \not{p}' - m - (\not{p} - m) \qquad S_F^{-1}(p)\, u(p) = (\not{p} - m) u(p) = 0$

$u(p)\, q_\mu \Gamma^\mu(p+q, p)\, u(p) = u(p)\, S_F'(p+q)^{-1} u(p)$

$q_\mu u(p) \underbrace{\Gamma^\mu(p+q, p)}_{\hookrightarrow \gamma^\mu} u(p) \Big|_{q \to 0} = u(p)(\not{p} + \not{q} - m) u(p)$

unrenormalization

$q_\mu \bar{u}(p) \frac{\gamma^\mu}{z_1} u(p) = \bar{u}(p) \frac{+\not{q}}{z_2} - u(p)$

$\frac{\partial}{\partial q}\Big|_{q \to 0} \qquad z_1 = z_2$!

exercise)

$$\partial_\mu \langle 0|T(j^\mu(x)\psi(x_1)\cdots\bar\psi(y_1)\cdots A(z_1)\cdots)|0\rangle = \langle 0|T(\partial_\mu j^\mu(x)\cdots)|0\rangle$$

$$\langle 0|T(\psi(x_1)\cdots\bar\psi(y_1)\cdots A(z_1)\cdots)\sum_{i=1}^{m}[-\delta^4(x-x_i)+\delta^4(y-y_i)]$$

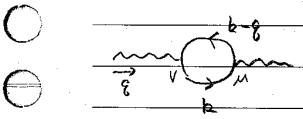
Noether $\partial_\mu j^\mu = 0$

Ward identity $m=1,\ n=0$

$$[j^0(x),\psi(x')] = -\psi(x)\delta^3(x-x')$$

polarization

photon self energy → vacuum pol'n tensor



$$ie^2\Pi^{(2)}_{\mu\nu}(q) = (-)(-ie)^2_{i^2}\int\frac{d^4k}{(2\pi)^4}\mathrm{Tr}\left(\gamma_\mu\frac{1}{\not k - m}\gamma_\nu\frac{1}{\not k - \not q - m}\right)$$

Pauli-Villars →

$$(-)(-ie)^2_{i^2}\int\frac{d^4k}{(2\pi)^4}\sum_i c_i\,\mathrm{Tr}\left(\gamma_\mu\frac{1}{\not k - M_i}\gamma_\nu\frac{1}{\not k - \not q - M_i}\right)$$

$$= (-ie)^2\int\frac{d^4k}{(2\pi)^4}\cdot\frac{\mathrm{Tr}(\gamma_\mu(\not k+m)\gamma_\nu(\not k-\not q+m))}{(k^2-m^2)((k-q)^2-m^2)}$$

Feynmann's trick

$$\frac{1}{AB} = \int_0^1 dx\,\frac{1}{[xA+(1-x)B]^2}$$

$$= (-ie)^2\int\frac{d^4k}{(2\pi)^4}\int_0^1 dx\,\frac{c_i N^i_{\mu\nu}}{(k^2-2kqx+q^2x-M_i^2+i\varepsilon)^2}$$

$$= (-ie)^2\int\cdots\int dx\sum_i\frac{N^i_{\mu\nu}}{[\underbrace{(k-qx)^2}_{k'^2}+q^2x(1-x)-M_i^2]^2}$$

$k' = k - qx$

$$N_{\mu\nu} = \mathrm{Tr}\{\gamma_\mu(\not k'+\not q x+M)\gamma_\nu(\not k'-\not q(1-x)+M_i)\} \quad \leftarrow O(k^{n\geq 2})$$

$$= 4\{(k'+qx)_\mu(k'-q(1-x))_\nu + (k'+qx)_\nu(k'-q(1-x))_\mu - (k'-qx)(k'-q(1-x))\eta_{\mu\nu} + M^2\eta_{\mu\nu}\}$$

$$= 8[k'_\mu k'_\nu - q_\mu q_\nu x(1-x)] - 4[k'^2 - q^2x(1-x) - M^2]$$

1より積分 $k_\mu k_\nu \Rightarrow \frac{1}{4}\eta_{\mu\nu}k^2$

$$N_{\mu\nu} = (-\eta_{\mu\nu}q^2 + q_\mu q_\nu)[-q(1-x)] - \eta_{\mu\nu}[k^2 + 2q^2x(1-x) - 2M^2]$$

$$\Pi_{\mu\nu} = (-\eta_{\mu\nu}q^2 + q_\mu q_\nu)\Pi(q^2) + \eta_{\mu\nu}\rho(q^2)$$

$$\rho(q^2) = i\int\frac{d^4k}{(2\pi)^4}\int_0^1 dx\sum_i c_i\frac{k^2-2\Delta_i^2}{(k^2-\Delta_i^2)^2} \qquad \Delta_i^2 = M_i^2 - q^2x(1-x)$$

$$\int d^4k\sum_i c_i\frac{k^2-2\Delta_i^2}{(k^2-\Delta_i^2)^2} = \frac{1}{k^2-\Delta_i^2} - \frac{\Delta_i^2}{(k^2-\Delta_i^2)^2} \qquad \int d^4k_E = 2\pi^2\int k_E^3\,dk = 2\pi^2\int k_E^3\frac{dk_E^2}{2}$$

$k^0 = ik^4 \quad \vec k = k \quad k_E$

$$i\int d^4k_E\sum_i c_i\left[\frac{1}{k_E^2+\Delta_i^2} - \Delta_i^2\frac{1}{(k_E^2+\Delta_i^2)^2}\right] = i\sum_i c_i\int\frac{dk_E^2}{2}\left[-1+\frac{\Delta_i^4}{(k_E^2+\Delta_i^2)^2}\right]$$

$\sum c_i = 0$

$= i\sum c_i\pi^2\Delta_i^2$

$\Delta_i^2 = M_i^2 - q^2x(1-x)$

$\sum c_i M_i^2 = 0$

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$N_{\mu\nu} = (-\eta_{\mu\nu} q^2 + q_\mu q_\nu)(-8x(1-x)) - \eta_{\mu\nu} 2\left[k^2 + 2q^2 x(1-x) - 2M^2\right]$

$\Pi_{\mu\nu} = (-\eta_{\mu\nu} q^2 + q_\mu q_\nu)\Pi(q^2) + \eta_{\mu\nu}\rho(q^2) \qquad \rho = 0$

$\Pi(q^2) = -8i \int \frac{d^4k}{(2\pi)^4} \sum c_i \int_0^1 dx \frac{x(1-x)}{(k^2 - \Delta_i^2 + i\varepsilon)^2} = -8i \int_0^1 dx\, x(1-x) \sum c_i \frac{-i\pi^2}{(2\pi)^4} \ln \underbrace{\Delta_i^2}_{M_i^2 - q^2 x(1-x)}$

$\begin{cases} c_1 = 1 & M_1 = m \\ c_2 = -1-\varepsilon & M_2 = m/\sqrt{2} \equiv M \\ c_3 = \varepsilon & M_3 = m/\Sigma \end{cases}$

$= -\frac{1}{2\pi^2} \int dx\, x(1-x) \sum c_i \ln(M_i^2 - q^2 x(1-x))$

$\ln \frac{m^2 - q^2 x(1-x)}{M^2 - q^2 x(1-x)}$

$\Pi(q^2) = \frac{1}{12\pi^2}\left[\ln\frac{M^2}{m^2} - 6\underbrace{\int_0^1 dx\, x(1-x) \ln\left[1 - \frac{q^2}{m^2} x(1-x)\right]}_{\tilde{\Pi}(q^2)}\right]$

$\Pi(0) = \frac{1}{12\pi^2} \ln\frac{M^2}{m^2}$

$e^2\Pi(0) = \frac{1}{3\pi}\ln\frac{M^2}{m^2} = 1 - Z_3 \qquad Z_3 = 1 - \frac{1}{3\pi}\ln\frac{M^2}{m^2} \qquad \tilde{\Pi}(q^2) = \frac{-1}{2\pi^2}\int_0^1 dx\, x(1-x) \ln\left[1 - \frac{q^2}{m^2} x(1-x)\right]$

$\Pi^{(2)} = \frac{1}{12\pi^2}\left[\ln\frac{M^2}{m^2} - 6x(1-x)\ln\left[1 - \frac{q^2}{m^2}x(1-x)\right]\right]$ ▷Jul 14

$M \to \infty$

$\Pi(0) = \frac{1}{12\pi^2}\ln\frac{M^2}{m^2}$

$Z_3 = 1 - e^2\Pi(0) = 1 - \frac{\alpha}{3\pi}\ln\frac{M^2}{m^2}$

$\tilde{\Pi} = -\frac{1}{2\pi^2}\int dx\, x(1-x)\ln\left[1 - \frac{q^2}{m^2}x(1-x)\right]$

$\tilde{\Pi}(q^2) = \Pi(q^2) - \Pi(0) \qquad e^2\tilde{\Pi}(q^2) = \frac{\alpha}{15\pi}\left(\frac{q^2}{m^2}\right) + \frac{\alpha}{140\pi}\left(\frac{q^2}{m^2}\right)^2 + \cdots \quad q^2 \text{小}$

$\simeq -\frac{\alpha}{3\pi}\ln\left(-\frac{q^2}{m^2}\right) + \frac{5\alpha}{9\pi} + O\left(-\frac{m^2}{q^2}\right) \quad q^2 \text{大}$

ex) $e^2\tilde{\Pi}(q^2) = \frac{\alpha}{3\pi}\left[\frac{5}{3} - \frac{1}{\rho} - \left(1 - \frac{1}{2\rho}\right)\sqrt{1+\frac{1}{\rho}}\ln\frac{\sqrt{1+1/\rho}+1}{\sqrt{1+1/\rho}-1}\right]$

$\rho = -\frac{q^2}{4m^2}$

N-dim $\rho^{(2)} = 0$ ← exersise

$\Pi^{(2)} = -8i\int_0^1 dx\, x(1-x)\int \frac{d^Nk}{(2\pi)^N}\frac{1}{(k^2-\Delta^2)^2} \to -8i\theta^2\int_0^1 dx\, x(1-x)\, i\frac{(\Delta^2)^{-2+\frac{N}{2}}}{(4\pi)^{\frac{N}{2}}}\cdot\frac{\Gamma(N/2)\Gamma(2-N/2)}{\Gamma(N/2)}$

$e^2(m^2)^{\frac{4-N}{2}}$

$[e^2] = \frac{4-N}{2} \qquad \varepsilon = \frac{4-N}{2} \to 0$

$\Gamma(\varepsilon) = \frac{\Gamma(1+\varepsilon)}{\varepsilon} = \frac{1}{\varepsilon} - \gamma$

$\tilde{\Pi}(q^2) = \frac{8e^2}{(4\pi)^2}\int_0^1 dx\, x(1-x)\left[\frac{1}{\varepsilon} - \gamma - \ln\frac{\Delta^2}{m^2} + \ln 4\pi\right]$

$a = \Delta^2/m^2$

$= \frac{e^2}{12\pi^2}\left(\frac{1}{\varepsilon} - \gamma + \ln 4\pi\right) - \frac{e^2}{2\pi^2}\int_0^1 dx\, x(1-x)\ln\frac{\Delta^2}{m^2}$

$a^{-\varepsilon}\Gamma(\varepsilon) = (1 - \varepsilon\ln a)\left(\frac{1}{\varepsilon} - \gamma\right) = \frac{1}{\varepsilon} - \ln a - \gamma$

MS scheme $1/\varepsilon$ を除去

(minimal sub..) $\tilde{\Pi} = \frac{e^2}{12\pi^2}(-\gamma + \ln 4\pi) - \frac{e^2}{2\pi^2}\int \cdots$

$\overline{MS}$, $1/\varepsilon - \gamma + \ln 4\pi$ を除去 $\tilde{\Pi}_{\overline{MS}} = -\frac{e^2}{2\pi^2}\int \cdots$ Dyson 流に一致

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$$\mathrm{Im}\,\Pi(q^2) = -\frac{1}{2\pi^2}\int_0^1 x(1-x)(-i\pi)\,\theta\left(-1+\frac{q^2}{m^2}x(1-x)\right)$$

$$\tilde{\Pi}(q^2) = -\frac{1}{2\pi^2}\int_0^1 dx\, x(1-x)\ln\left[1-\frac{q^2}{m^2}x(1-x)\right]$$

$$\ln\left(m^2 - q^2 x(1-x)\right)$$

$-i\varepsilon$ $\quad$ 0

$$= -\frac{1}{2\pi^2}\int_{x_-}^{x_+} x(1-x)\,dx$$

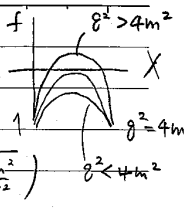
$$= \frac{1}{12\pi}\left(1+2\frac{m^2}{q^2}\right)\sqrt{1-\frac{4m^2}{q^2}}\;\theta(q^2-4m^2)$$

$$f = -1 + \frac{q^2}{m^2}x(1-x)$$

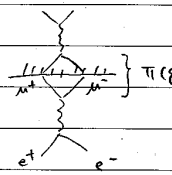
$$= -\frac{q^2}{m^2}\left[x^2 - x + \frac{m^2}{q^2}\right]$$

$$= -\frac{q^2}{m^2}(x_+ - x)(x_- - x)$$

$$x_\pm = \frac{1}{2}\left(1 \pm \sqrt{1-\frac{m^2}{q^2}}\right)$$



$$\sigma(e^+e^- \to \mu^+\mu^-) = \frac{16\pi}{q^2}\alpha^2\,\mathrm{Im}\,\tilde{\Pi}(q^2)$$



$$\tilde{\Pi}(q^2) = -\frac{1}{2\pi^2}\int_0^1 dx\, x(1-x)\ln\left[1-\frac{q^2}{m^2}x(1-x)\right]$$

$$x = \frac{1-u}{2}$$

$$= -\frac{1}{2\pi^2}\int_0^1 \frac{du}{2}\,\frac{1-u^2}{4}\ln\left[1-\frac{q^2}{4m^2}(1-u^2)\right]$$

$$= +\frac{1}{2\pi^2}\int_0^1 \frac{du}{2}\,\frac{u}{4}\left[1-\frac{u^2}{3}\right]\frac{\frac{q^2}{2m^2}u}{1-\frac{q^2}{4m^2}(1-u^2)}$$

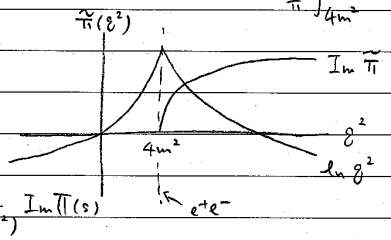
$$\frac{1}{q^2} \longrightarrow \frac{1}{q^2} + \frac{1}{q^2}e^2\tilde{\Pi}(q^2) = \frac{1}{q^2} + \frac{\alpha}{\pi}\int_0^1 du\, u^2\,\frac{1-\frac{u^2}{3}}{1-u^2}\,\frac{1}{q^2-\frac{4m^2}{1-u^2}}$$

$-\lambda^2$ $\quad$ $-\lambda^2$ $\quad$ $-\lambda^2$

$$* \quad u^2 = 1-\frac{4m^2}{s} \qquad u\,du = \frac{2m^2}{s}ds \qquad \tilde{\Pi}(q^2) = \frac{q^2}{12\pi^2}\int_{4m^2}^{\infty} ds'\,\frac{1}{s'}\,\frac{1}{s'-q^2}\left(1+\frac{2m^2}{s'}\right)\sqrt{1-\frac{4m^2}{s'}}$$

$$= \frac{1}{\pi}\int_{4m^2}^{\infty}\frac{ds}{s'(s'-q^2)}\,\mathrm{Im}\,\Pi(q^2) \qquad \text{dispersion relation}$$

$$\Pi(q^2) = \int ds\,\frac{\mathrm{Im}\,\Pi(s)}{s-q^2}$$



$$\Pi(q) - \Pi(0) = \int ds\,\frac{q^2}{s(s-q^2)}\,\mathrm{Im}\,\Pi(s)$$

一回引き算した分散式

$\Leftrightarrow$ 次

$\Leftrightarrow$ 次定数 $\Pi(0)$

$$V(q) = \frac{e^2}{q^2} \to \frac{e^2}{q^2}\,\frac{1}{1+e^2\tilde{\Pi}(q^2)} = \frac{e^2}{q^2}\left[1-\frac{\alpha}{15\pi}\frac{q^2}{m^2}+\cdots\right]$$

$q^2 = 0$ Coulomb interaction

$$V(r) = \frac{1}{(2\pi)^3}\int V(q^2)e^{i\vec{q}\vec{x}}\,d^3x = e^2\left(\frac{1}{4\pi}\frac{1}{r} - \frac{\alpha}{15\pi}\frac{1}{m^2}\delta^3(r)+\cdots\right)$$

H atom a level shift

$$R_y = \frac{m}{2}\alpha^2$$

$$\Delta E_{n\ell} = \langle n\ell\,|\,\delta V(r)\,|\,n\ell\rangle = -\frac{4}{15}\alpha^2\frac{1}{m^2}|\Psi(0)|^2 \qquad |\Psi(0)|^2 = \frac{1}{\pi n^3}(m\alpha)^3$$

$$= -\frac{8}{15}R_e\,\frac{1}{\pi n^3}\alpha^2\delta_{\ell n}$$

Vacuum polarization

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$\lambda = \frac{c}{\nu} = \frac{3\cdot 10^{10}}{26.9\times 10^{6}} = \frac{3\times 10^{4}}{26.9} \sim 10^{3}$ cm

$\nu(2S) = \Delta E_{nl}/2\pi\hbar = \frac{-Ry}{15\pi}\alpha^2 = -26.9\,MHz$

q^2 大きいと

$e^2 \to \frac{e^2}{1-\frac{\alpha}{3\pi}\ln(-\frac{q^2}{m^2})}$ e^2↗ as q^2↗

$\sim$○ e m μ τ $\ln(-\frac{q^2}{m_e^2}) + \ln(-\frac{q^2}{m_\mu^2}) + \ln(-\frac{q^2}{m_\tau^2})$

u d s c b ⋯ $+3\frac{4}{9}\ln(-\frac{q^2}{m_u^2}) + 3\frac{1}{9}\ln(-\frac{q^2}{m_d^2}) \cdots$

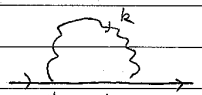
$q^2 = m_Z^2 = (90\,GeV)^2$ $e^2(q^2=m_Z^2) = \frac{1}{128}$ $e^2(q^2=0) = \frac{1}{137}$

7%大きい

ex) unren pert'n theory $e^2 \to \frac{e^2}{1+\alpha^2\Pi}$ このとき charge の繰り込みを考えた finite part ?

electron self energy

1PI $-i\Sigma^{(2)} = (-ie)^2\int\frac{d^4k}{(2\pi)^4}\gamma^\mu\frac{i}{\not p-\not k+m-i\varepsilon}\gamma^\nu\frac{i}{k^2-\lambda^2}\left[-\eta_{\mu\nu}+(1-a)\frac{k_\mu k_\nu}{k^2-a\lambda^2}\right]$



$\Sigma(p) = -ie^2\int\frac{d^4k}{(2\pi)^4}\gamma^\mu\frac{\not p-\not k+m}{(p-k)^2-m^2}\gamma_\mu\frac{1}{k^2-\lambda^2}$

$= +ie^2\int\frac{d^4k}{(2\pi)^4}\int_0^1 dx\,\frac{2(\not p-\not k)-4m}{[(k-px)^2-\Delta^2]}$ $\Delta^2 = -p^2x(1-x)+m^2x+\lambda^2(1-x)$

PV regulator

$\Sigma(p) = ie^2\int\frac{d^4k}{(2\pi)^4}\int_0^1 dx\,\sum c_i\frac{2(\not p-\not k)-4m}{[(k-px)^2-\Delta_i]^2}$

$k' = k-px$ $\int\frac{d^4k}{(2\pi)^4}\sum c_i\frac{2\not p(1-x)-4m}{(k'^2-\Delta_i^2)^2}$

$= \frac{ie^2}{(2\pi)^4}\int_0^1 dx\,2[\not p(1-x)-2m](-i\pi^2)\sum c_i\ln\Delta_i^2 = \frac{\alpha}{2\pi}\int_0^1 dx\,[2m-\not p(1-x)]\ln\frac{\Lambda^2(1-x)}{m^2x+\lambda^2(1-x)-p^2x(1-x)}$ ①

$\begin{cases} c_1 = 1 & \Lambda_1 = \lambda \\ c_2 = -1-\varepsilon & \Lambda_2 = \frac{\lambda}{\sqrt{\varepsilon}} = \Lambda \\ c_3 = \varepsilon & \Lambda_3 = \frac{\lambda}{\varepsilon} \end{cases}$

$\Sigma = A + B(-\not p+m) + \tilde\Sigma$ $\tilde\Sigma \sim O[(-\not p+m)^2]$ $\int dx\,\ln\frac{1-x}{x^2} = 1$

$A = \Sigma(p)\big|_{\not p=m} = \frac{\alpha}{2\pi}\int_0^1 dx\,m(1+x)\ln\frac{\Lambda^2(1-x)}{m^2x^2+\lambda^2(1-x)}$ $\int dx\,x\ln\frac{1-x}{x^2} = -\frac{1}{4}$

$\lambda\to 0$

IR finite

$$= \frac{\alpha m}{2\pi}\int_0^1 dx\left[(1+x)\ln\frac{\Lambda^2}{m^2} + (1+x)\ln\frac{1-x}{x^2}\right]$$

$$A = \frac{3}{4\pi}\alpha m\left(\ln\frac{\Lambda^2}{m^2} + \frac{1}{2}\right) = \delta m$$

$$B = -\left.\frac{\partial\Sigma}{\partial \not p}\right|_{\not p = m} = \frac{\alpha}{2\pi}\int_0^1 dx\left\{(1-x)\ln\frac{\Lambda^2(1-x)}{m^2x^2+\lambda^2(1-x)} + \left[2m - \not p(1-x)\right]\frac{-2\not p x(1-x)}{m^2x+\lambda^2(1-x)-p^2x(1-x)}\right|_{\not p = m}$$

$$= \frac{\alpha}{2\pi}\int_0^1 dx\left\{(1-x)\ln\frac{\Lambda^2(1-x)}{m^2x^2+\lambda^2(1-x)}\right. \quad \nearrow \text{no. IR} \quad \lambda = 0$$

$$\left. -(1+x)\frac{2m^2x(1-x)}{m^2x^2+\lambda^2(1-x)}\right\} \qquad \varepsilon = \frac{\lambda^2}{m^2} \qquad x(1-x)$$

$$\int_0^1 dx\,\frac{x}{x^2+\varepsilon(1-x)} = -\frac{1}{2}\ln\varepsilon \quad (\varepsilon \to 0)$$

$$B = \frac{\alpha}{2\pi}\left(\frac{1}{2}\ln\frac{\Lambda^2}{m^2} + \frac{9}{4} - \ln\frac{m^2}{\lambda^2}\right) = 1 - Z_2$$

$$Z_2 = 1 - \frac{\alpha}{4\pi}\left(\ln\frac{\Lambda^2}{m^2} + \frac{9}{2} - \frac{1}{2}\ln\frac{m^2}{\lambda^2}\right)$$

finite part $\quad \tilde{\Sigma}(p) = \Sigma(p) - \left[A + B(-\not p + m)\right]$

$$\Sigma(p) = (G + H(\not p - m))F(p) \qquad \tilde{\Sigma} = \left[G + H(\not p - m)\right](F(p) - F(m))$$

$$\tilde{\tilde{\Sigma}}(p) = \Sigma(p) - \left[A + B(-\not p + m)\right]$$

$$= \frac{\alpha}{2\pi}\int_0^1 dx\left\{\left[2m - \not p(1-x)\right]\left\{\ln\frac{\Lambda^2(1-x)}{m^2x^2+\lambda^2(1-x)-p^2x(1-x)} - \ln\frac{\Lambda^2(1-x)}{m^2x^2+\lambda^2(1-x)}\right\}\right.$$

$$\left. + (\not p - m)\,m(1+x)\frac{2mx(1-x)}{m^2x^2+\lambda^2(1-x)}\right]$$

$$\ln\frac{A}{A+B} = -\int_0^1 dz\,\frac{B}{A+Bz}$$

$$\Sigma(p) = \frac{\alpha}{2\pi}\int_0^1 dx\left[2m - \not p(1-x)\right]\ln\frac{m^2x^2+\lambda^2(1-x)}{\underbrace{m^2x^2+\lambda^2(1-x)}_{A} - \underbrace{(p^2-m^2)x(1-x)}_{B}} + \cdots$$

$$= \frac{\alpha}{2\pi}\int_0^1 dx\int_0^1 dz\,\left(2m - \not p(1-x)\right)\frac{(p^2-m^2)x(1-x)}{m^2x^2+\lambda^2(1-x)-(p^2-m^2)x(1-x)z}$$

$$(-\not p + m)^2 = (p^2-m^2) + 2m(-\not p + m)$$

$$\tilde{\Sigma} = \frac{\alpha}{2\pi}\int_0^1 dx\int_0^1 dz\,\frac{x(1-x)N}{m^2x^2+\lambda^2(1-x)-(p^2-m^2)x(1-x)z}$$

$$= (-\not p + m)^2\left[(\not p + m)(1-x)\left[1 - \frac{2x(1+x)}{x^2+\varepsilon(1-x)}z\right] - m(1+x)\right]$$

$$\left[2m - \not p(1-x)\right](p^2+m^2) - 2(-\not p + m)m^2(1+x)\,\frac{x^2+\varepsilon(1-x)+\frac{m^2p^2}{m^2}x}{x^2+\varepsilon(1-x)}\Big/\ln$$

$$\widehat{\Sigma} = (-\not p + m)^2 \frac{\alpha}{2\pi_1} \int_0^1 dx \int_0^1 dz \frac{-x(1-x)}{m^2x^2 + \lambda^2(1-x)} \cdots$$

$$\Sigma_f = \frac{\alpha}{2\pi_1} \int_0^1 dx \int_0^1 dz \frac{\omega(z,x)\,[\not p + M(z,x)]}{p^2 - m^2 r(z,x) + i\varepsilon}$$

$$r(z,x) = 1 + \frac{x^2 + \frac{\lambda^2}{m^2}(1-x)}{x(1-x)z}$$

$$\omega(z,x) = \frac{1-x}{z} s \qquad s = 1 - \frac{2x\,z(1+x)}{x^2 + \frac{\lambda^2}{m^2}(1-x)}$$

ミスプリ ??

$$M(z,x) = m\left[s - \frac{1+x}{1-x}\right]$$

Alpelgust-Brodsky 表示 PR A2, 2293 (1970)

$$\rho = \frac{m^2 - p^2}{m^2}$$

$$\Sigma_f = \frac{\alpha}{2\pi_1} \frac{1}{m}\left[\frac{1}{2(1-\rho)}\left(1 - \frac{2-3\rho}{1-\rho}\ln\rho\right) - \frac{\not p + m}{m}\left\{\frac{1}{2\rho(1-\rho)}\left(2 - \rho + \frac{\rho^2+4\rho-4}{1-\rho}\ln\rho\right) + \frac{1}{\rho} + \frac{1}{\rho}\ln\frac{\lambda^2}{m^2}\right\}\right]$$

$\Sigma_f(\rho=1) = \text{finite}$

gauge term の寄与

$$-i\Sigma_{gauge}^2 = (-ie)^2 \int \frac{d^4k}{(2\pi)^4} \gamma^\mu \frac{1}{\not p - \not k - m + i\varepsilon} \gamma^\nu \frac{i(1-a)}{k^2 - \lambda^2} \frac{k_\mu k_\nu}{k^2 - a\lambda^2}$$

$$\not k = -(\not p - \not k - m) \qquad \not k \frac{1}{\not p - \not k - m + i\varepsilon} \not k = -\not k - (\not p - m) + (\not p - m)\frac{1}{\not p - \not k - m}(\not p - m)$$

↓ Sym integer ≥ 0

$$= -\not k - (\not p - m)\frac{1}{\not p - \not k - m + i\varepsilon}\left[-\not p - \not k - m - (\not p - m)\right]$$

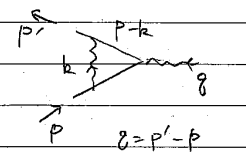
$$B_{gauge} = ie^2(1-a)\int \frac{d^4k}{(2\pi)^4} \frac{1}{k^2 - \lambda^2} \frac{1}{k^2 - a\lambda^2} = -\frac{\alpha}{4\pi_1}(1-\alpha)\ln\frac{\Lambda^2}{\lambda^2}$$

$$Z_2 = 1 - \frac{\alpha}{4\pi_1}\left[a\ln\frac{\Lambda^2}{m^2} + \frac{9}{2} - (3-\alpha)\ln\frac{m^2}{\Lambda^2}\right]$$

Landau gauge $a=0$ no UV

Fennie... gauge $a=3$ no IR

vertex part



$q = p' - p$

$$-ie\Gamma_\mu = -ie(\gamma_\mu + \Lambda_\mu(p'p))$$

$$\Lambda_\mu(p'p) = (-ie)^2 \int \frac{d^4k}{(2\pi)^4} \gamma^\nu \frac{i}{\not p' - \not k - m} \gamma_\mu \frac{i}{\not p - \not k - m} \gamma^\lambda i \frac{1}{k^2 - \lambda^2}\left(-\eta_{\mu\nu} + (1-a)\frac{k_\mu k_\nu}{k^2 - a\lambda^2}\right)$$

Feynman gauge $a=1$

$$= -ie^2 \int \frac{d^4k}{(2\pi)^4} \cdot \frac{f(k)}{\left[(p'-k)^2 - m^2\right]\left[(p-k)^2 - m^2\right]k^2}$$

$$f(k) = \gamma^\nu(\not p' - \not k + m)\gamma_\mu(\not p - \not k + m)\gamma_\nu$$

$$\frac{1}{ABC} = 2! \int dz_1 dz_2 dz_3\, \delta(1 - z_1 - z_2 - z_3) \frac{1}{(Az_1 + Bz_2 + Cz_3)^3}$$

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$$= -ie^2\, 2! \int_0^1 \pi\, dz_i\, \delta(1-\Sigma z_i) \int \frac{d^4k}{(2\pi)^4} \frac{f}{(k'^2-\Delta^2+i\varepsilon)^2}$$

denominator $= k^2 - 2kp'z_1 - 2kpz_2 - \lambda^2 z_3 - (p'^2-m^2)z_1 - (p^2-m^2)z_2$

$= \underbrace{(k-p'z_1-pz_2)}_{k'}{}^2 - (p'z_1+pz_2)^2 - (p'^2-m^2)z_1 - (p^2-m^2)z_2 - \lambda^2 z_3$

$\Delta' = -(p'^2-m^2)z_1(1-z_1) - (p^2-m^2)z_2(1-z_2) + (p^2+p'^2-2m^2)z_1z_2 - g^2 z_1 z_2 + m^2(1-z_3)^2 + \lambda^2 z_3$

$f(k'+p'z_1+pz_2)$ $\qquad \gamma^\nu \not k \not k \gamma_\nu = -2\not k \not k \not\alpha$

$= +\gamma^\nu(-\not k' + \not p'(1-z_1) - \not p z_2 + m)\gamma_\mu(-\not k' + \not k(1-z_3) - \not p' z_2 + m)\gamma_\nu = -2\not k \gamma_\mu \not k + N$

$N = -2[\not p'(1-z_1-z_2) - \not p(1-z_2)]\gamma_\mu[\not k(1-z_1-z_2) + \not q(1-z_1)]$
$\quad - 2m^2\gamma_\mu + 4m[p'(1-2z_1) + p(1-2z_2)]_\mu$

p, p' on-shell $\not p = m$ $\not p' = m$.

$$\Lambda_\mu(\text{on shell}) = -z_i e^2 \int_0^1 \pi\, dz_i\, \delta(1-\Sigma z_i) \frac{1}{(2\pi)^4}\left[\int d^4k \frac{-2k\gamma^\mu k}{(k^2-\Delta^2+i\varepsilon)^3} + \int d^4k \frac{N}{(k^2-\Delta^2+i\varepsilon)^3}\right]$$

$\not k \gamma^\mu \not k = \underbrace{k_\alpha k_\beta}_{-\frac{k^2}{4}\eta_{\alpha\beta}} \gamma^\alpha \gamma^\mu \gamma^\beta = -\frac{k^2}{2}\gamma_\mu$

等一项 $= \gamma_\mu \int d^4k \frac{k'}{(k^2-\Delta^2+i\varepsilon)^3} = \gamma_\mu \int d^4k \sum c_i \frac{k^2}{(k^2-\Delta_i^2+i\varepsilon)^3} = -i\pi^2\gamma_\mu \Sigma c_i \ln \Delta_i^2$

$= -i\pi^2\gamma_\mu \ln \frac{m^2(1-z_3)^2 + \lambda^2 z_3 - g^2 z_1 z_3}{m^2(1-z_3)^2 + \lambda^2 z_3 - g^2 z_1 z_2}$

$= i\pi^2\gamma_\mu \left(\ln\frac{\Lambda^2}{m^2} + \ln\frac{z_3}{(1-z_3)^2} + \ln\frac{m^2(1-z_3)^2}{m^2(1-z_3)^2 + \lambda^2 z_3 - g^2 z_1 z_2}\right)$

等二项 $= \int d^4k \frac{N}{(k^2-\Delta^2+i\varepsilon)^3} = \frac{N}{2i\Delta}$

$\bar u(p')N u(p) = \bar u(p')[-2\{mz_3 - g(1-z_2)\}\gamma_\mu\{mz_3 + g(1-z_1)\not\alpha$
$\quad - 2m^2\gamma_\mu + 4m[p'(1-2z_1) + p(1-2z_2)]_\mu u(p)$

Gordon 分解

$$\bar u(p')\gamma_\mu u(p) = \bar u(p')\left[\frac{(p+p')_\mu}{2m} + \frac{i\sigma_{\mu\nu}}{2m}q^\nu\right]u(p)$$

$\bar u N u = \bar u(p')[-\gamma_\mu\{2m^2(1-4z_3+z_3^2) + 2g^2(1-z_1)(1-z_2)\} - 2mz_3[\not q, \gamma_\mu](z_1+z_2)$
$\quad - mz_3\{\not q, \gamma_\mu\}(z_2-z_1) + 4m g_\mu(z_2-z_1)]u(p)$

分母 z_1, z_2 对称 symmetric

$$\int dz_1 dz_2 dz_3\, \delta(1-z_1-z_2-z_3) f(z_1 z_2 z_3) = \int dz_1 dz_2 \delta(z_1+z_2+1-z_1-z_2)$$

$z_1+z_2 \le 1$

$0 \le z_1, z_2$ $\qquad z_1+z_2 = x$

$$\int_0^1 dx \int_{-\frac{x}{2}}^{x/2} dy\, f\left(\frac{x}{2}+y, \frac{x}{2}-y, 1-x\right)$$

$\frac{z_1-z_2}{2} = y$

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$$\bar{u}(p')\Lambda_\mu u(p) = \frac{\alpha}{4\pi}\gamma_\mu\left[\ln\frac{\Lambda^2}{m^2} - \frac{1}{2} + 2\int_0^1 \frac{3}{\pi}\prod_{i=1} dz_i\,\delta(1-\Sigma z_i)\ln\frac{m^2(1-z_3)^2}{m^2(1-z_3)^2+\lambda^2 z_3 - q^2 z_1 z_2}\right]$$

$$-\frac{\alpha}{4\pi}\int \pi\, dz_i\,\delta(1-\Sigma z_i)\,\frac{-\gamma_\mu\{2m^2(1-4z_3+z_3^2)+2q^2(1-z_1)(1-z_2)\} - m z_3(1-z_3)[\not{q},\gamma_\mu]}{m^2(1-z_3)^2+\lambda^2 z_3 - q^2 z_1 z_2}$$

$$① = \int \pi dz_i\,\delta(1-\Sigma z_i)\ln\frac{m^2(1-z_3)^2}{m^2(1-z_3)^2+\lambda^2 z_3 - q^2 z_1 z_2} = \int_0^1 d\lambda \int_{-\frac{x}{2}}^{\frac{x}{2}} dy\,\ln\frac{m^2x^2}{m^2x^2+\lambda^2(1-x)-q^2(\frac{x^2}{4}-y^2)} \quad \begin{cases} z_1+z_2 = x \\ \frac{z_1-z_2}{2} = y \end{cases}$$

$y \to z = \frac{y+\frac{x}{2}}{x}$, $J = x$

$$= \int_0^1 dx \int_0^1 dz\; x\ln\frac{m^2x^2}{m^2x^2+\lambda^2(1-x)-q^2 z(1-z)x^2}$$

$$= \int_0^1 dx\int_0^1 dz\; z x\,\frac{-q^2x^2(1-2z)}{m^2x^2+\lambda^2(1-x)-q^2z(1-z)x^2} \sim O(q^2)$$

$$② = 2\int_0^1 dx\int_0^1 dz\; x\,\frac{x^2+2x-2+\frac{q^2}{m^2}(1-x+x^2z(1-z))}{x^2\left[1-\frac{q^2}{m^2}z(1-z)\right]+\frac{\lambda^2}{m^2}(1-x)} = 2\int_0^1 dx\int_0^1 dz\; x\,\frac{x^2+2x-2}{x^2+\frac{\lambda^2}{m^2}(1-x)} + O(q^2)$$

$$③ = -\frac{1}{m^2}\int_0^1 dx\int_{-\frac{x}{2}}^{\frac{x}{2}} dz\,\frac{x^2(1-x)}{x^2\left[1-\frac{q^2}{m^2}z(1-z)\right]+\frac{\lambda^2}{m^2}(1-x)}$$ IR finite ($\lambda \to 0$)

$$\bar{u}(p')\Lambda_\mu u(p) = \frac{\alpha}{4\pi}\gamma_\mu\left[\ln\frac{\Lambda^2}{m^2} + \frac{q}{2} - 2\ln\frac{m^2}{\lambda^2}\right] + \frac{\alpha}{2\pi}\gamma_\mu\int_0^1 dx\int_0^1 dz\left\{\frac{\frac{q^2}{m^2}(1-x+x^2z^2)-(2-2x-x^2)}{x^2\left[1-\frac{q^2}{m^2}z(1-z)+\frac{\lambda^2}{m^2}(1-x)\right]} + \frac{2-2x-x^2}{x^2+\frac{\lambda^2}{m^2}(1-x)}\right\}x$$

$$+\frac{\alpha}{4\pi}[\not{q},\gamma_\mu]\frac{1}{m}\int_0^1 dx\int_0^1 dz\,\frac{1-x}{1-\frac{q^2}{m^2}z(1-z)}$$

$1-z_1$

$$[\not{q},\gamma_\mu] = 2i\sigma_{\mu\nu}q^\nu$$

$$z_1 = 1 - \frac{\alpha}{4\pi}\left[\ln\frac{\Lambda^2}{m^2} + \frac{q}{2} - 2\ln\frac{m^2}{\lambda^2}\right]$$

$$\bar{u}(p')\Gamma_\mu u(p) = u(p')\left[F_1(q^2)\gamma_\mu + \frac{i\sigma_{\mu\nu}}{2m}q^\nu F_2(q^2)\right]u(p)$$

$F_1(0) = 1$

$$F_2(q^2) = \frac{\alpha}{\pi}\int_0^1 dx\int_0^1 dz\,\frac{1-x}{1-\frac{q^2}{m^2}z(1-z)}$$

$F_2(0) = \frac{1}{2}\frac{\alpha}{\pi}$ Schwinger's magnetic moment

$= 1.00116\,14$

exp) 1.00115 96 Kusch-Foley

$$\vec{\mu} = \frac{e}{2m}\left[1+F_2(0)\right]\vec{\sigma}$$ (magnetic moment)

Lande g-factor $\to g/2$

gauge dependent part $-(\not{p}-\not{k}-m)+(\not{p}-m)$ $\qquad -(\not{p}-\not{k}-m)+(\not{p}-m)$

$$\Lambda_\mu(p'p) = (-ie)^2\int\frac{d^4k}{(2\pi)^4}\not{k}\,\frac{i}{\not{p}-\not{k}-m-i\varepsilon}\gamma_\mu\frac{i}{\not{p}-\not{k}-m-i\varepsilon}\not{k}\cdot\frac{i(1-a)}{k^2-\lambda^2+i\varepsilon}\,\frac{1}{k^2-a\lambda^2+i\varepsilon}$$

$$= (-ie)^2\int\frac{d^4k}{(2\pi)^4}\,i\gamma_\mu i\,\frac{i(1-a)}{(k^2-\lambda^2+i\varepsilon)(k^2-a\lambda^2+i\varepsilon)} = \frac{\alpha}{4\pi}(1-a)\ln\frac{\lambda^2}{\Lambda^2} = 1 - z_1^{\text{gauge}}$$

$$z_1 = 1 - \frac{\alpha}{4\pi}\left[\ln\frac{\Lambda^2}{m^2} + \frac{q}{2} - 2\ln\frac{m^2}{\lambda^2}\right]$$

$z_1 = z_2$

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$$F_1(q^2) = 1 + \underbrace{\frac{\alpha}{3\pi}\frac{q^2}{m^2}\left[\ln\frac{m}{\lambda} - \frac{3}{8}\right]}_{\frac{1}{2}q^2\langle r^2\rangle} + O(q^4)$$

charge — charge radius

$$F_2(q^2) = \frac{\alpha}{\pi}\frac{m^2}{q^2}\frac{1}{\sqrt{1-\frac{4m^2}{q^2}}}\ln\frac{\sqrt{1-\frac{4m^2}{q^2}}-1}{\sqrt{1-\frac{4m^2}{q^2}}+1} \qquad x = \frac{1-\sqrt{\ }}{1+\sqrt{\ }}$$

$$= \frac{\alpha}{\pi}\frac{x}{1-x^2}\ln(-x)$$

$$F_1(q^2) = \frac{\alpha}{\pi}\left[\ln\frac{m}{\lambda}\left(1+\frac{1+x^2}{1-x^2}\ln(-x)\right) - 1 - \frac{3x^2-2x+3}{4(1-x^2)}\ln(-x)\right.$$

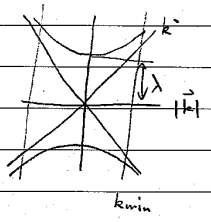
$$\left. + \frac{1+x^2}{1-x^2}\right\} - \frac{1}{4}\ln^2(-x) + \frac{1}{2}\zeta(2) + Li_2(x) + \ln(-x)\ln(1-x)\Big\}\Big]$$

Li_2 Euler dialog (Spence form)

$$Li_2(x) = -\int_0^x dt\,\frac{\ln(1-t)}{t}$$

IR div の取り扱い $|k| > k_{min}$

$$\int_{|k|>k_{min}} = \underbrace{\int_0^\infty dk}_{iD_F} - \underbrace{\int_0^{k_{min}} dk}_{\delta}$$



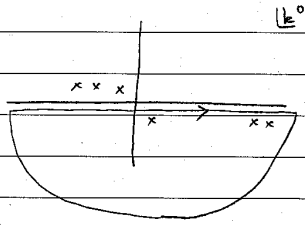
$$k^2 = p^\circ \pm \sqrt{(p-k)^2+\lambda^2} \mp i\varepsilon$$

$$= \pm\sqrt{\vec{k}^2+\lambda^2} \mp i\varepsilon$$

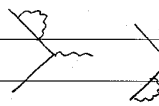
$$\delta\Lambda_\mu = \Lambda_\mu(p',p) - \Lambda_\mu(p',p,|\vec{k}|>k_{min})$$

$$= -ie^2\int_{|k|<k_{min}}\frac{d^3k}{(2\pi)^3}\int\frac{dk^\circ}{(2\pi)}\left[\gamma^\nu\frac{1}{\not p - \not k - m + i\varepsilon}\gamma_\mu\frac{1}{\not p' - \not k - m + i\varepsilon}\gamma_\nu\frac{1}{k^2-\lambda^2+i\varepsilon}\right]$$

$\frac{1}{(p-k)^2-m^2}$ → $-2\pi i\frac{1}{2\sqrt{\vec{k}^2+\lambda^2}}$



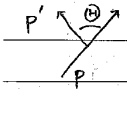
$$\bar{u}\delta\Lambda_\mu u = -e^2\gamma_\mu\int\frac{d^3k}{(2\pi)^3}\frac{1}{2\sqrt{k^2+\lambda^2}}\left\{\frac{(p'p)}{(p'k)(pk)} - \frac{1}{2}\frac{m^2}{(p'k)^2} - \frac{1}{2}\frac{m^2}{(pk)^2}\right\}$$



$$\int_0^{k_{min}}\frac{d^3k}{\sqrt{k^2+\lambda^2}}\frac{m^2}{(pk)^2} = 4\pi\ln\left[\ln\frac{2k_{min}}{\lambda} - \frac{1}{2\beta}\ln\frac{1+\beta}{1-\beta}\right]$$

$\beta = \frac{|\vec{p}|}{p^\circ}$ $\quad k = (\sqrt{k^2+\lambda^2}, \vec{k})$ $\quad p = (p^\circ, \vec{p})$ $\quad \hat{p}k = \theta$

$$\int_0^{k_{min}} d^3k\,\frac{(pp')}{(pk)(p'k)} = \int_0^{k_{min}} dk\,\frac{k^2}{\sqrt{\ }}\int d\Omega\,\frac{p^\circ p'^\circ(1-\vec{\beta}\cdot\vec{\beta}')}{p^\circ p'^\circ(\sqrt{k^2+\lambda^2}-\vec{\beta}'\vec{k})(\sqrt{k^2+\lambda^2}-\vec{\beta}\vec{k})}$$



$$= (1-\beta\beta')\int\frac{k^2dk}{\sqrt{\ }}\,2\pi\int_0^1 dx\int_{-1}^1 dz\,\frac{1}{\left[k^\circ - k\underbrace{\left|\vec{\beta}x + \vec{\beta}'(1-x)\right|}_{B}z\right]^2}$$

$$= 4\pi(1-\beta\beta')\int_0^1 dx\,\frac{1}{1-B^2}\left[\ln\frac{k_{min}}{\lambda} - \frac{1}{2B}\ln\frac{1+B}{1-B}\right]$$

static 近似

$k \sim 0$ $\quad p' \sim p$ $\quad \beta' \sim \beta$

$$= 4\pi\left[1 + \frac{2}{3}\beta^2(1-\cos\theta)\ln\frac{2k_{min}}{\lambda} - \left(1+\frac{8}{9}\beta^2 - \frac{5}{9}\beta^2\cos\theta\right)\right]$$

$q^2 = -2m^2\beta^2(1-\cos\theta)$

$$\bar{u}\delta\Lambda u = -e^2\gamma_\mu\frac{4\pi}{2(2\pi)^3}\left[\left\{1+\frac{2}{3}\beta^2(1-\cos\theta)-1\right\}\ln\frac{2k_{min}}{\lambda} - \left(1+\frac{8}{9}\beta^2 - \frac{5}{9}\beta^2\cos\theta + 1 + \frac{\beta^2}{3}\right)\right.$$

$$= \frac{\alpha}{\pi}\bar{u}\gamma_\mu u\frac{q^2}{3m^2}\left(\ln\frac{2k_{min}}{\lambda} - \frac{5}{6}\right)$$

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$\ln\frac{m}{\lambda} - \frac{3}{8}$

$\bar{u}\Gamma_\mu(p',p,|\vec{k}|>k_{min})u = \bar{u}(\Gamma_\mu - \delta\Gamma_\mu)u = \bar{u}\gamma_\mu\left\{\left[1+\frac{\alpha}{3\pi}\frac{g^2}{m^2}\left(\ln\frac{m}{2k_{min}} + \frac{5}{6} - \frac{3}{8}\right)\right] + \frac{i\sigma_{\mu\nu}}{2m}g^\nu\frac{\alpha}{2\pi}\right\}u$

$\ln\frac{2k_{min}}{\lambda} - \frac{5}{6}$

energy shift

$e\bar{u}(p')\gamma_\mu u(p)A^\mu_{ext} \qquad eA^\mu_{ext} = \left(-\frac{Ze^2}{|\vec{q}|^2}, \vec{0}\right)$

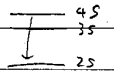
$\gamma_\mu \to P_\mu$

$e\bar{u}Pu = -\frac{Ze^2}{|\vec{q}|^2}u^\dagger(p')\left[1 - \frac{\alpha}{3\pi}\frac{|\vec{q}|^2}{m^2}\left(\ln\frac{m}{2k_{min}} - \frac{3}{8} + \frac{5}{6}\right)\right] \to V(r)$

$V(q)$

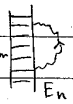
$V(r) = -Z\alpha\left[\frac{1}{r} - \frac{4\alpha}{3m^2}\delta^3(r)\left[\ln\frac{m}{2k_{min}} - \frac{3}{8} + \frac{5}{6}\right]\right]$

$\Delta E_{nl} = \frac{4\alpha}{3m^2}Z\alpha|\Psi(0)|^2\left(\ln\frac{m}{2k_{min}} - \frac{3}{5} + \frac{5}{6}\right)$ $\quad 2S_{1/2}$



$|\vec{k}| < k_{min}$ の項は

$E^{bound} = e^2\int\frac{d^3k}{2k(2\pi)^3}\sum_\lambda\sum_m\frac{\langle n|\vec{\xi}^\lambda|m\rangle\langle m|\vec{\xi}^\lambda|n\rangle}{E_n - (E_m+k)}$ $\quad E_m$, E_n



$\gamma^\mu \sim \frac{P^\mu}{m} = V_\mu$

$= e^2\int\sum_\lambda\frac{''}{''}\frac{\langle n|V_i|m\rangle\langle m|V_j|n\rangle}{E_n - E_m - k}\xi_i\xi_j$ $\qquad \sum_\lambda V_i V_j \xi_i^\lambda \xi_j^\lambda = (1-\cos^2\theta)|\vec{V}|^2$

$= \frac{e^2}{2(2\pi)^2}\frac{4}{3}\int kdk\sum_m\frac{|V_{nm}|^2}{E_n - E_m - k}$ $\qquad E^{free} = \frac{e^2}{2(2\pi)^2}\frac{4}{3}\int kdk\frac{V^2}{-k} \to -\int kdk\frac{(V^2)_{av}}{-k}$

$\Delta E = E^{bd} - E^{free} = \frac{e^2}{2(2\pi)^2}\frac{4}{3}\int dk\sum_m\frac{(E_n - E_m)|V_{nm}|^2}{E_n - E_m - k} = \frac{-2\alpha}{3\pi}\sum_m|V_{nm}|^2(E_n - E_m)\ln\frac{K}{|E_n - E_m|}$

$\langle m|[\vec{p}[\vec{p},H]]|n\rangle = -\nabla^2 V = -4\pi Z\alpha\delta^3(r) = 2\sum_m\langle n|\vec{p}|m\rangle\langle m|p'|n\rangle(E_n - E_m)$

$\vec{p} \to m\vec{r}$

$= \frac{2\alpha}{3\pi}\frac{2\pi Z\alpha}{m^2}|\Psi(0)|^2\ln\frac{K}{\langle E_n - E_m\rangle_{Av}}$

$\Delta E = \frac{4\alpha}{3m^2}Z\alpha|\Psi_{nl}(0)|^2\left[\left(\ln\frac{m}{\langle E_m - E_n\rangle_{Av}} - \frac{3}{8} + \frac{5}{6}\right) - \frac{1}{5} + \frac{1}{2}\right]$

Bethe $\quad \tilde{P}^2 \quad \tilde{\pi}^2(g^2) \quad F_2(g^2)$

$F_1'(g^2)$

Bethe log

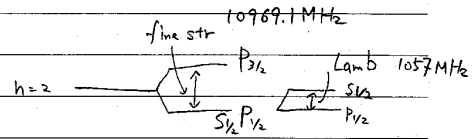
	E/h
Bethe + $\tilde{P}^2$	1010 MHz
$\tilde{\pi}^2$	−27.1 MHz
2nd order magnetic	+68
2nd order binding	+17
	1058
expt)	1057.84

$\ln\frac{m}{\lambda} \to \ln\frac{m}{k_{min}} \to \ln\frac{m}{\langle E_m - E_n\rangle}$

$F(q^2) = 1 + \frac{1}{6}q^2\langle r^2\rangle - \cdots$

Rutherford 1947

$2S_{1/2} - 2P_{1/2} \approx 1000$ MHz



electron anomalous magnetic moment

$a_e - 2$ $\qquad$ analytic

$F_2(0) = \frac{\alpha}{2\pi} - 0.328478965\left(\frac{\alpha}{\pi}\right)^2 + 1.181241456\left(\frac{\alpha}{\pi}\right)^3$

$- 1.4092(384)\left(\frac{\alpha}{\pi}\right)^4$

Laporta Remiddi PL 379B 283(1996)

$+\frac{1}{10}a^{(5)}$ 他の寄与

expt) 1057.841 (12)

845 (9)

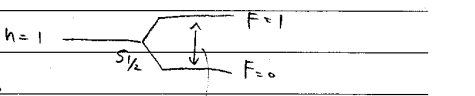
851 (2)

$\langle r_p\rangle$

th) .821 (6) ← 0.805 (11) ← 4th order

.840 (1) ← 0.862 (11)

proton radius (strong interaction)



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$\lesssim 100$ GeV

- QEDの現状

NO. 61

e^- $\frac{g_e-2}{2}$ = 1159652188.4(4.3) $\times 10^{-12}$ 3.7 ppb

e^+ 187.9 (4.3) Dehmelt Penning Trap

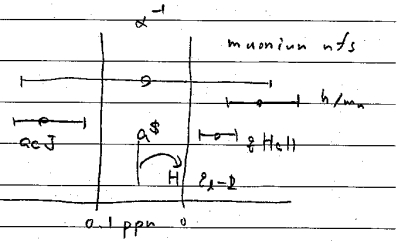
α^{-1} (g-Hall) = 137.0360037 (27) 0.020 ppm

(ac J vNp) — .0359770 (77) 0.059

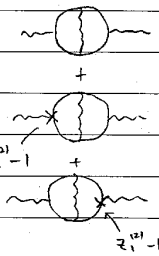
(h/m_n) — .03601087 (524) 0.039

$\frac{g_e-2}{2}$ — .3599993 (52) 0.00038 (sys error) ~5 (8次)

$\sim \frac{1}{10}$ (6次)



Higher order



Lia

Nielsen polylog

$S_{np}(x) = \frac{(-1)^{n+p-1}}{(n-1)!\,p!}\int_0^1 \frac{ln^{n-1}t\, l_n^p(1-yt)}{t}dt$

BPHZ

くりこみ群

$m_e = 0$

$D_u(q^2) = \frac{1}{q^2}\left[1 - \frac{\alpha}{3\pi} ln\frac{\Lambda^2}{q^2} + const\right] = \frac{1}{q^2}\left[1 - \frac{\alpha}{3\pi} ln\frac{\Lambda^2}{-\mu^2} + \frac{\alpha}{3\pi} ln\frac{q^2}{-\mu^2} + \cdots\right]$

$= \frac{1}{q^2}\left[1 - \frac{\alpha}{3\pi} ln\frac{\Lambda^2}{-\mu^2}\right]\left[1 + \frac{\alpha}{3\pi} ln\frac{\Lambda^2}{-\mu^2}\right] = Z_3(\Lambda^2/m^2, \alpha_0)\, D(q^2, \mu^2, \alpha)$

$D(\mu^2, \mu^2, \alpha) = 1$

$e_0^2 Z_3 = e^2$

$e_0^2 D_u = e^2 D_u$

$\mu^2 \to \mu^2 - \delta\mu^2$

$D(q^2, \mu^2+\delta\mu^2, \alpha) = \frac{1}{q^2}\left[1 + \frac{\alpha}{3\pi} ln\frac{q^2}{-(\mu^2+\delta\mu^2)}\right]$

$= \frac{1}{q^2}\left[1 + \frac{\alpha}{3\pi} ln\frac{q^2}{-\mu^2}\right]\left(1 - \frac{\alpha}{3\pi}\frac{\delta\mu^2}{\mu^2}\right]$

$Z_3(\mu^2+\delta\mu^2) = Z_3(\mu^2)\left[1 + \frac{\alpha}{3\pi}\frac{\delta\mu^2}{\mu^2}\right]$

$e(\mu^2+\delta\mu^2) = e(\mu^2)\left[1 + \frac{\alpha}{3\pi}\frac{\delta\mu^2}{\mu^2}\right]^{1/2}$ $\quad e(\mu^2+\delta\mu^2) = e(\mu^2) + \delta e$ $\quad \mu\frac{\partial e}{\partial\mu} = 2\mu^2\frac{\partial e}{\partial\mu^2} = \frac{\alpha}{3\pi}e = \beta(e)$

くりこみ群のβ関数

$\frac{e^3}{12\pi^2} = be^3$

$\frac{\delta\mu}{\mu} = \frac{\delta e}{\beta(e)} \to ln\,\mu = -\frac{1}{2b}\left(\frac{1}{e^2} + const\right)$

$ln\,\mu_0 = -\frac{1}{2b}\left(\frac{1}{e_0^2} + const\right)$

$e^2 = \frac{e_0^2}{1 - 2be_0^2 ln\frac{\mu}{\mu_0}}$ $\quad \alpha = \frac{\alpha_0}{1 - \frac{\alpha}{3\pi} ln\left(\frac{\mu}{\mu_0}\right)^2}$

Green関数の心

$G(q^2, \mu^2, \alpha) = \frac{G_u(q^2, \Lambda^2, \alpha)}{Z(\mu^2/\Lambda^2, \alpha_0)}$

$G(q^2, \mu^2+\delta\mu^2, \alpha+\delta\alpha) = \frac{G_u(\ ,\ ,\ )}{Z(\mu^2+\delta\mu^2/\Lambda^2, \alpha_0)} = \frac{Z(\frac{\mu^2}{\Lambda^2}, \alpha_0)}{Z(\frac{\mu^2+\delta\mu^2}{\Lambda^2}, \alpha_0)} G(q^2, \mu^2, \alpha)$

$1 - 2\gamma\frac{\delta\mu^2}{2\mu^2}$ $\quad \gamma = \frac{\mu^2}{Z}\frac{\partial Z}{\partial\mu^2}$

$\delta G(q^2, \mu^2, e) = \frac{\delta G}{\delta\mu^2}\delta\mu^2 + \frac{\delta G}{\delta e}\frac{\delta e}{\delta\mu^2}\delta\mu^2$ $\quad \left(2\mu^2\frac{\partial G}{\partial\mu^2} + \beta\frac{\partial G}{\partial e} + 2\gamma G\right) = 0$

$= -2\mu\frac{\delta\mu^2}{2\mu^2}G$ $\quad \beta(e)/2\mu^2$ $\quad \left(\mu\frac{\partial}{\partial\mu} + \beta\frac{\partial}{\partial e} + 2\gamma\right)$

$\mu = \mu_0 e^{\frac{3\pi}{2\alpha}}$ Landau pole

500

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コーナー reduce ver.
600 page → 200 page

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$G^{(n)} = \langle \phi \cdots \phi \rangle = \frac{1}{Z_\phi^{n/2}} \langle \phi_0 \cdots \phi_0 \rangle$

$n=2$

$\left(\mu \frac{\partial}{\partial \mu} + \beta \frac{\partial}{\partial e} + n\gamma \right) G^{(n)} = 0$

1PI $\Gamma = \dfrac{\langle \phi \cdots \phi \rangle}{\underbrace{\langle \phi\phi \rangle \cdots \langle \phi\phi \rangle}_{n}} = Z_\phi^{1/2} P_u$ $\quad \left(\mu \frac{\partial}{\partial \mu} + \beta \frac{\partial}{\partial e} - n\gamma \right) \Gamma^n(\mu, e) = 0$

formal に行け

$P_u(P_i, \Lambda, e_0)$

$P(P_i, \mu, e)$ $\quad P_u = Z^{-n/2} P$

$\mu \dfrac{\partial P_u}{\partial \mu}\Big|_{\Lambda, e_0} = 0 = \mu \dfrac{\partial P}{\partial \mu} + \beta \dfrac{\partial P}{\partial e} - \dfrac{n}{2} \dfrac{1}{Z} \dfrac{\partial Z}{\partial \mu} P$ $\quad \mu \dfrac{\partial \Gamma}{\partial \mu} + \beta \dfrac{\partial \Gamma}{\partial e} - n\gamma P = 0$ $\quad \gamma = \frac{1}{2} \mu \frac{\partial}{\partial \mu} \ln Z$

ヤラナカ…月事

1) γ^5 の取扱い
2) mass singularity
3) bound state e^+e^-
4) non linear σ model O(3) Heisenberg model のくりこみ → 臨界現象 Kogut Willson Phys. Report
5) Callan - Symanzik